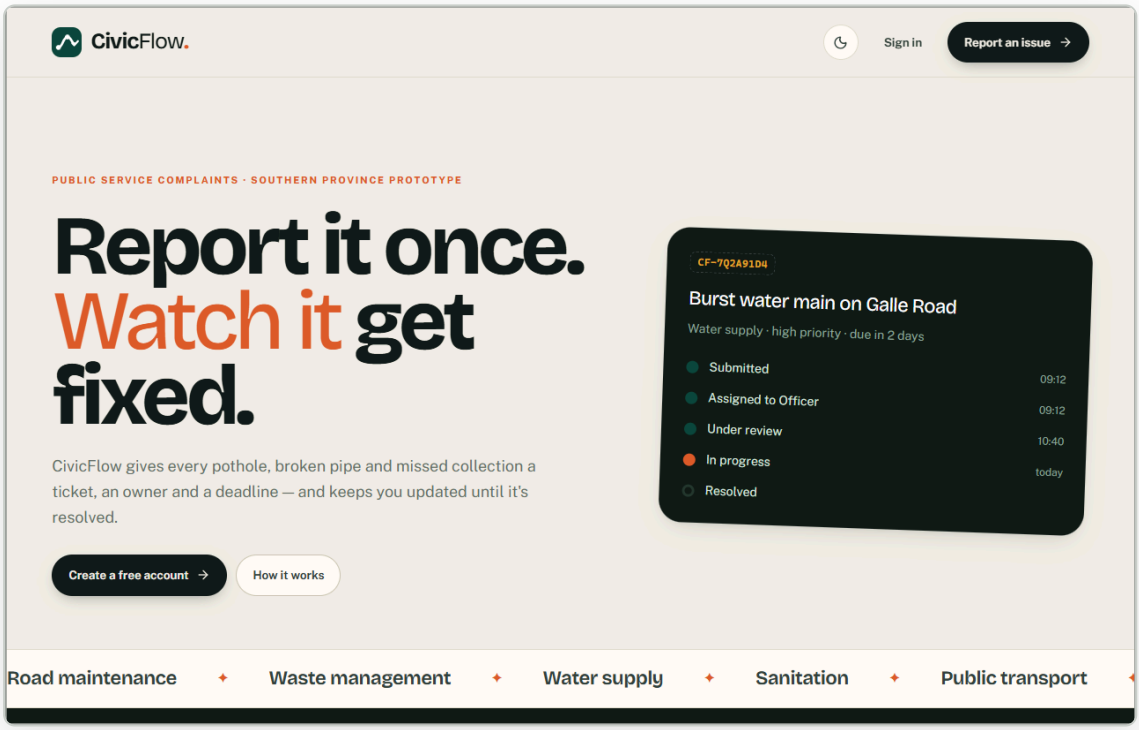
UNIVERSITY OF RUHUNA – FACULTY OF TECHNOLOGY

ICT2232 Software Engineering · Mini Project

# CivicFlow

## Public Service Complaint & Resolution System

Final Project Report · SDLC documentation · Group 15




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# Declaration

We declare that this report and the CivicFlow software are the work of Group 15 for ICT2232 Software Engineering. Material from other sources is acknowledged in the References. All names, complaints and locations shown in this report, in the screenshots and in the demonstration are fictional.

Use of AI tools: as allowed by Section 4 of the project guidelines, we used AI-assisted tools to help with parts of the coding and documentation. The requirements, design decisions, priorities and trade-offs were decided by the team. We reviewed everything the tools produced, checked it with the 283 automated tests that run on every commit, and documented it. Each member can explain the parts of the system they worked on.

# Executive summary

In Sri Lanka, people report problems with roads, waste collection, water supply, sanitation and public transport by telephone, by visiting an office or through people they know. These complaints are often lost, the person who complained cannot see what is happening, and the organisation cannot easily tell who is responsible or which complaints are overdue.

CivicFlow is a web application that handles each complaint from submission to resolution in one place. A Citizen submits a complaint with a category, a description, a location on the map and, if needed, photos or documents. The system sends it to the department responsible for that category and assigns it to the Officer there with the fewest open complaints. The Officer reviews it, sets a priority, works on it and resolves it. Each step is saved in a history that cannot be edited, and the Citizen is told about it in an Updates inbox. After resolution the Citizen can rate the service. Administrators manage users, invite Officers, maintain the service categories, move complaints that are stuck in departments without staff, and follow performance through analytics.

We followed Scrum over five sprints. The final system covers all 25 functional requirements (every Must and Should item, the one Could item and three requirements added through change request CR-006) and 11 measurable non-functional requirements. Data is stored in PostgreSQL with Row Level Security, every change goes through checked functions on the server, and evidence files are kept in private storage.

We wrote 283 automated tests (77 unit, 147 database integration and 59 system tests), and all of them pass. We also ran an automated layout check at four screen sizes. 19 defects were recorded and fixed. Eight security controls cover the eleven threats we assessed. The system is live at https://civic-flow-rho.vercel.app.

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# 1 Introduction and Problem Domain

Road maintenance, waste collection, water supply, sanitation and public transport affect people every day. When something goes wrong, such as a pothole, an overflowing bin, a burst water main or a bus that does not stop at a halt, people need an easy way to report it and to know that someone is dealing with it.

Today most complaints are made by telephone, by visiting a council or divisional office, by letter or e-mail, or through personal contacts. None of these channels keeps a shared record, so complaints get lost or are reported twice. The person who complained cannot see any progress and has to call again to follow up. The service organisations, on the other hand, have no reliable way to record, categorise, prioritise, assign and monitor complaints, or to show how quickly they responded.

Our group (Group 15) was given the domain "Public Service Complaint & Resolution System: a system for citizens to submit service complaints, track progress and receive resolution updates". We named the product CivicFlow. This report describes how we took it through the ten stages of the Software Development Life Cycle (SDLC) and how the output of each stage was used in the next.

## 1.1 Report structure

The chapters follow the structure given in Section 16 of the project guidelines. Chapters 2 to 5 cover problem definition and process planning (Milestone 1), Chapter 6 covers Agile planning (Milestone 2) and Chapter 7 covers requirements engineering (Milestone 3). Chapter 8 describes the design (Milestone 4), Chapters 9 and 10 the implementation and testing (Milestone 5), and Chapter 11 the deployment (Milestone 6). Chapters 12 and 13 deal with maintenance and security (Milestone 7), and Chapter 14 with project management and each member's contribution.

## 1.2 How to read the reference codes

Every requirement, story, use case, diagram, test, defect, change and threat in this report has a short code made of a type and a number, for example UC-08. The letters give the type and the number is its position in that list; numbers are given in order and never reused. The same codes are used in the project repository, so any item can be followed from requirement to design, code and tests (Section 7.7).

**Table 1.1:** Reference codes used in this report

| Code | Meaning | Example | Listed in |
|---|---|---|---|
| CF-xx | Functional requirement | CF-19: the Citizen rates a resolved complaint once | Section 7.2 |
| NFR-xx | Non-functional (quality) requirement | NFR-05: users see only the data their role allows | Section 7.3 |
| US-xx | User story | US-18: "As a citizen, I want to rate … my resolved complaint" | Section 6.2 |
| AC1, AC2 … | Acceptance criterion of a user story | AC1 of US-18: rating 1 to 5, submitted once | Section 6.3 |

| Code | Meaning | Example | Listed in |
|---|---|---|---|
| UC-xx | Use case | UC-08: Provide feedback | Sections 7.4 and 7.5 |
| SD-x | Sequence diagram | SD-3: Citizen gives feedback | Section 8.4 |
| TC-AREA-xx | Browser (system) test case | TC-FB-01: the feedback form used through the real screens | Appendix B |
| TC-DB-AREA-xx | Database integration test case | TC-DB-FB-03: feedback refused before resolution and after withdrawal | Appendix B |
| TC-U-AREA-xx | Unit test case | TC-U-FB-01: invalid ratings and over-long comments rejected | Appendix B |
| TC-INT-xx | HTTP integration test case | TC-INT-01: owner, department and role checks with real sign-in tokens | Appendix B |
| DEF-xx | Defect found and fixed | DEF-13: complaints stuck in departments without Officers | Section 10.5 |
| CR-xxx | Change request | CR-006: every complaint must reach a responsible Officer | Chapter 12 |
| O1 … O10 | Project objective | O3: Officers can manage, assign, prioritise and resolve complaints | Section 3.2 |
| T1 … T11 | Security threat | T3: skipping workflow steps by calling the database directly | Section 13.1 |
| SR-xx | Security requirement | SR-03: Citizens read only their own complaints | Section 13.2 |
| C1 … C8 | Implemented security control | C1: Row Level Security | Section 13.3 |
| S1 … S12 | Smoke-test check on the live site | S1: opening the site signed out leads to the landing or sign-in page | Section 11.5 |
| 001 … 015 | Database migration file | 015: automatic assignment and dispatch | Section 7.7 |

Test areas: AUTH sign-up and login, SUB submit complaint, LOC location, EV evidence files, TRK tracking, OFF Officer processing, NTF updates inbox, FB feedback, WD withdrawal, CAT categories, ADM user administration, MON monitoring, DSP dispatch, UX screen sizes and layout, DEMO demonstration data. A range such as TC-DB-FB-01…05 means tests 01 to 05. For example, the traceability row for CF-19 reads US-18, UC-08, SD-3, migration 012 and tests TC-FB-01…02, TC-U-FB-01 and TC-DB-FB-01…05: the requirement, its story, its use case, its design, where it was built and how it was tested.

# 2 Problem Statement

**Problem statement**

Citizens find it difficult to report public service issues and to follow up on them, because complaint information is handled through separate, manual channels. This leads to poor visibility of complaint status, slow communication, unclear responsibility and difficulty in monitoring unresolved complaints. Public service organisations also lack an organised way to record, categorise, prioritise, assign and monitor complaints. A centralised, role-based Public Service Complaint & Resolution System is therefore required to improve accessibility, transparency, accountability and efficiency.

**Table 2.1:** Problem analysis

| Aspect | Description |
|---|---|
| Who is affected | Citizens, front-line public service Officers, department managers and public service organisations |
| Current situation | Phone, walk-in, e-mail and informal reports; no shared record; no status visibility; manual follow-up |
| Impact | Lost or duplicated complaints, no audit trail, unclear accountability, slow resolution and low public trust |
| Root causes | No single system of record; no defined workflow or ownership; no notification of progress |
| Proposed solution | CivicFlow: a web application with a defined complaint lifecycle (Submitted → Under review → In progress → Resolved, or Withdrawn by the owner), automatic assignment, history, updates and feedback |
| Success measure | Every complaint has a reference, a responsible Officer, a service target and a complete history visible to the Citizen |

# 3 Stakeholders and Objectives

## 3.1 Stakeholders and their needs

We identified the stakeholders from the problem domain and refined the list while writing the proposal. Three of them use the system directly, as the roles Citizen, Officer and Administrator. Department managers are stakeholders too, but we decided not to give them a separate login role; their needs are covered by the department queues and the Administrator's analytics.

**Table 3.1:** Stakeholders, needs and how CivicFlow addresses them

| Stakeholder | Type | Needs | How CivicFlow addresses them |
|---|---|---|---|
| Citizen | Primary user (application role) | Easy submission, tracking, status visibility, updates and feedback | Complaint form with map, My Complaints, timeline, Updates inbox, feedback (CF-04…07, 13, 18, 19) |
| Public Service Officer | Primary user (application role) | Centralised complaints, search/filter, assignment, prioritisation and status management | Department queue, Assigned to me, filters, protected workflow actions (CF-08…12, 17) |
| Administrator | Primary user (application role) | User and category management, monitoring and secure administration | User management, Officer invitations, Service categories, monitoring overview (CF-14, 20, 21) |
| Department manager | Secondary stakeholder (not an application role) | Visibility of department complaints, workload and unresolved cases | Served by department-scoped queues and admin monitoring. Not a fourth login role (decision recorded in requirements). |
| Public service organisations | Indirect stakeholder | Accountability, organised records, monitoring | Immutable history, audit events, data-driven categories and departments |
| Community | Indirect stakeholder | Transparent, responsive handling | Status and history visible to the complainant; resolution date recorded |
| Lecturer (ICT2232) | Project stakeholder / reviewer | Evidence of each SDLC stage | This evidence pack |

## 3.2 Project objectives

We kept the ten objectives from the proposal. The table below gives the evidence for each one.

**Table 3.2:** Objectives and evidence of achievement

| # | Objective | Evidence of achievement |
|---|---|---|
| O1 | Convenient digital complaint submission | UC-03 implemented; browser test TC-SUB-01 |
| O2 | Citizens can track status and progress | UC-04, timeline, Updates inbox; TC-TRK-01, TC-NTF-01 |
| O3 | Officers can manage, assign, prioritise and resolve | UC-05…07; TC-OFF-01 |
| O4 | Accurate, centralised records | Database constraints, atomic history and rollback; TC-DB-SUB-05, TC-DB-OFF-08 |

| # | Objective | Evidence of achievement |
|---|---|---|
| O5 | Timely progress and resolution updates | Durable notifications created in the same transaction as every change; TC-NTF-* |
| O6 | Feedback after resolution | UC-08; TC-FB-* |
| O7 | Administrators manage users and categories | UC-09, UC-10; TC-ADM-, TC-CAT- |
| O8 | Transparency, accountability, efficiency | Append-only history, admin audit log, department queues |
| O9 | Authentication and role-based authorisation | RLS and protected RPCs; security |
| O10 | Extensible to more public services | Categories and departments are data. CF-20 lets an Administrator add services without a code change. |

# 4 Scope, Assumptions and Constraints

## 4.1 In scope

- Citizen registration, login, logout and profile management; role-based access for Citizen, Officer and Administrator.
- Complaint submission with category, description, map location (or typed coordinates) and private evidence files.
- Routing to the category's department, automatic workload-balanced assignment, and Administrator dispatch of stuck complaints.
- Officer processing: review, assign, prioritise, transfer, start, progress remarks and resolution, with service-level targets.
- Complaint tracking, an append-only history, an Updates inbox with read state, citizen withdrawal and post-resolution feedback.
- Administration: user management, Officer invitations, service-category management, monitoring and analytics with CSV export.

## 4.2 Out of scope

- Online payments; real-time GPS tracking of field Officers; AI-based complaint classification.
- Automatic legal case management; emergency response and disaster management; automated field dispatch of crews.
- Integration with every government information system; SMS, e-mail or push notifications for complaint updates (updates are delivered in-system).

## 4.3 Assumptions

**Table 4.1:** Assumptions and their validation status

| Assumption | Status |
|---|---|
| Users have an internet-connected device and browser | Accepted; responsive UI tested at 375 px and desktop widths |
| Citizens provide truthful, sufficient information | Accepted. The system validates form and bounds but can't verify truth. Officers review each complaint. |
| Officers have access and update complaints | Accepted; Officer accounts are invited by an Administrator |
| Categories and departments are system data | Implemented as database tables; six prototype categories chosen by the team |
| Evidence follows supported types and size | Enforced: JPEG/PNG/PDF, ≤ 5 MiB, ≤ 5 files per complaint |
| Demonstrations use fictional data | Enforced for all automated tests and screenshots |
| No external government integration | Accepted |

## 4.4 Constraints

- Limited academic duration and a five-member student team.
- No access to real government databases or systems; demonstrations use fictional data only.
- Free-tier hosting (Supabase and Vercel), so no paid services, custom servers or SMS gateways.
- Security and privacy must still be designed in, even though this is a prototype.
- The system cannot represent every organisation's workflow; categories and departments are therefore configuration data.

# 5 Software Process Model and Justification

## 5.1 Selected model: Agile with Scrum

We used an Agile approach based on Scrum. We kept a prioritised product backlog and refined it during the project. The work was split into five time-boxed sprints. Each sprint ended with a working increment that we could demonstrate, followed by a sprint review and a retrospective, and what we learned in these meetings went back into the backlog.

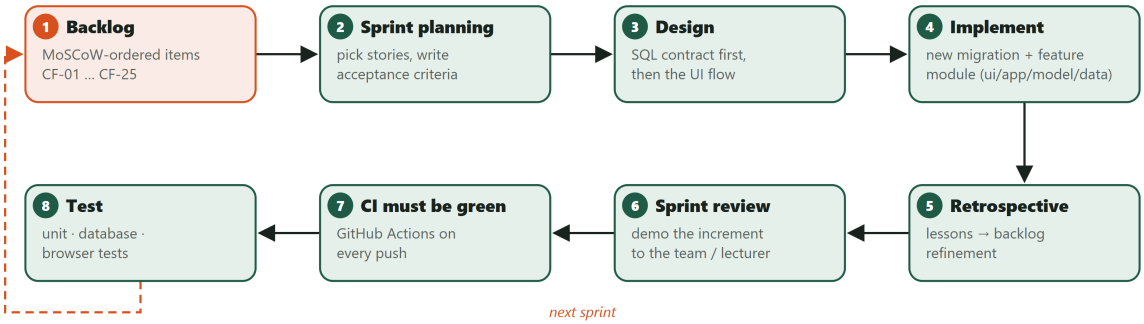



**Figure 5.1:** Scrum cycle applied in every sprint

## 5.2 Justification

**Table 5.1:** Why Scrum fits this project better than a plan-driven model

| Project characteristic | Why Scrum fits | Why a plan-driven model (Waterfall/V-model) fits worse |
|---|---|---|
| Requirements refined through lecturer and stakeholder feedback | Backlog refinement at the end of every sprint absorbs change. Three changes (CF-22 withdrawal, CF-04 location, CF-14 invitations) were absorbed this way. | Changes after the requirements phase are expensive and need re-baselining |
| Five-member student team with a fixed deadline | Time-boxed sprints give regular checkpoints and visible progress | One big delivery at the end gives risky feedback very late |
| Security-sensitive, role-based system | Each sprint adds tests with its features, so defects are found early. The PT409 defect, for example, was found and fixed within Sprint 3. | Testing at the end finds design-level defects late |
| Needs early demonstrations (Milestones 1–3) | Working increments can be shown at every milestone | Working software appears only near the end |

We did not use prototyping on its own because the module asks for documented requirements, design and testing. We also decided against the Spiral model, because its formal risk analysis is too heavy for a team of five working for one semester.

## 5.3 Development approach and Definition of Done

Every sprint followed the same steps. First we agreed the database side of a feature: the tables, security rules, function signatures and error codes. Then we designed the screens. The implementation added a new SQL migration (applied migrations are never edited) and a feature module with four layers. We wrote tests at three levels and ran the full set of checks locally and in continuous integration before demonstrating the increment.

**Table 5.2:** Definition of Done used for every backlog item

| # | A backlog item is done only when… |
|---|---|
| 1 | its acceptance criteria are demonstrably met; |
| 2 | authorisation is enforced in the database (grants, Row Level Security and function checks), not only in the user interface; |
| 3 | input limits are identical in the client schema and in SQL; |
| 4 | allow and deny tests exist (unit, database and a browser journey where there is UI); |
| 5 | loading, empty and error states are accessible; |
| 6 | lint (0 warnings), strict typecheck, all tests and the build pass locally and in CI; |
| 7 | documentation and traceability are updated. |

## 5.4 Project plan and tooling

**Table 5.3:** Project plan by SDLC phase

| Phase | Main activities | Output | Status |
|---|---|---|---|
| Project definition | Problem, stakeholders, objectives, scope | Approved proposal (Milestone 1) | Done |
| Process planning | Select and justify Scrum; plan | Chapter 5 | Done |
| Agile planning | Backlog, user stories, acceptance criteria, sprint plan | Chapter 6 (Milestone 2) | Done, refined |
| Requirements | FRs, NFRs, use cases, prioritisation, traceability | Chapter 7 (Milestone 3) | Done, refined by CRs |
| Design | Architecture, UML, database, UI | Chapter 8 (Milestone 4) | Done |
| Implementation & testing | Sprints 1–5, unit / database / system tests | Working software, Chapter 9–10 | Done |
| Deployment | Vercel + Supabase, configuration, smoke test | Live URL, Chapter 11 | Done |
| Maintenance & security | Change requests, threat model, controls | Chapters 12–13 | Done; lecturer CR pending |

**Table 5.4:** Tooling and configuration management

| Concern | Tool / practice |
|---|---|
| Version control | Git and GitHub (https://github.com/razim2003/civic-flow); feature branches merged to main |

| Concern | Tool / practice |
| --- | --- |
| Continuous integration | GitHub Actions runs install, lint, typecheck, unit tests, database tests, a type-drift check, the build and the Playwright tests on every push |
| Database change control | Ordered, timestamped SQL migrations; never edited after being applied; hosted rollout only after explicit approval with data checksums |
| Backlog tracking | Product backlog, roadmap and traceability matrix kept in the repository (docs/) |
| Code quality | ESLint (0 warnings), strict TypeScript, database types generated from the migrations |

# 6 Agile Planning

## 6.1 Product backlog

The product backlog lists every requirement as a backlog item with its MoSCoW priority, the sprint that delivered it and its final status. Items CF-22 to CF-25 were added during the project through change requests (Chapter 12).

**Table 6.1:** Product backlog

| CF ID | Backlog item | MoSCoW | Sprint | Status (2026-10-29) |
|---|---|---|---|---|
| CF-01 | User registration (Citizen) | Must | 1 | Done |
| CF-02 | Secure login/logout | Must | 1 | Done |
| CF-03 | Role-based access control | Must | 1 | Done |
| CF-04 | Submit complaint (description, location) | Must | 2 | Done |
| CF-05 | Complaint category selection | Must | 2 | Done |
| CF-06 | Unique complaint ID, initial Submitted status | Must | 2 | Done |
| CF-07 | View submitted complaints / track status | Must | 2 | Done |
| CF-08 | Officer review and complaint queue | Must | 3 | Done |
| CF-09 | Assign complaint (officer/department) | Must | 3 | Done |
| CF-10 | Set complaint priority | Must | 3 | Done |
| CF-11 | Update status and progress remarks | Must | 3 | Done |
| CF-12 | Resolve complaint | Must | 4 | Done |
| CF-13 | Complaint notifications/updates | Must | 4 | Done (inbox and read state added 2026-10-29) |
| CF-14 | Admin user management | Must | 1 (+3) | Done |
| CF-15 | Citizen profile management | Should | 1 | Done (display name) |
| CF-16 | Upload supporting evidence | Should | 2 | Done |
| CF-17 | Search and filter complaints | Should | 2–3 | Done |
| CF-18 | Complaint history | Should | 2–3 | Done |
| CF-19 | Citizen feedback | Should | 4 | Done (2026-10-29) |
| CF-20 | Complaint category management | Should | 2→4 | Done (2026-10-29) |
| CF-21 | Dashboard/reports | Could | 4 | Basic monitoring done; enhanced reports not planned |
| CF-22 | Citizen withdrawal (approved scope extension) | Should (team-assigned) | 2 | Done |

| CF ID | Backlog item | MoSCoW | Sprint | Status (2026-10-29) |
|---|---|---|---|---|
| CF-23 | Automatic workload-balanced assignment (CR-006) | Must | 5 | Done |
| CF-24 | Administrator dispatch centre (CR-006) | Must | 5 | Done |
| CF-25 | Service-level targets and overdue flags (CR-006) | Should | 5 | Done |
| — | Integration/system testing, security, deployment | Sprint 5 scope | 5 | In progress |

## 6.2 User stories

We expanded the story titles from the proposal into 24 user stories written as "As a ‹role›, I want ‹capability›, so that ‹benefit›". Story points are our own relative estimates on a scale of 1 to 8.

**Table 6.2:** User stories

| ID | User story | CF | MoSCoW | Pts |
|---|---|---|---|---|
| US-01 | As a citizen, I want to create an account with my name, email and a password, so that I can submit and follow my complaints. | CF-01 | Must | 3 |
| US-02 | As a registered user, I want to log in and log out securely, so that only I can use my account. | CF-02 | Must | 3 |
| US-03 | As any user, I want to see only the pages and data my role allows, so that private information stays protected. | CF-03 | Must | 5 |
| US-04 | As a citizen, I want to describe a public service problem and mark its location on a map or by coordinates, so that the service team can find it. | CF-04 | Must | 8 |
| US-05 | As a citizen, I want to choose a service category, so that my complaint reaches the right department. | CF-05 | Must | 2 |
| US-06 | As a citizen, I want a unique reference and confirmation after submitting, so that I know my complaint was recorded. | CF-06 | Must | 2 |
| US-07 | As a citizen, I want to list, search and open my complaints and see their status, so that I know what is happening. | CF-07, CF-17 | Must | 5 |
| US-08 | As an officer, I want a queue of my department's complaints with filters, so that I can see and pick up work. | CF-08, CF-17 | Must | 5 |
| US-09 | As an officer, I want to assign a complaint to an officer in my department, so that responsibility is clear. | CF-09 | Must | 3 |
| US-10 | As an officer, I want to set a complaint's priority, so that urgent problems are handled first. | CF-10 | Must | 2 |
| US-11 | As an assigned officer, I want to start work and add progress remarks, so that the citizen sees what is being done. | CF-11 | Must | 5 |
| US-12 | As an assigned officer, I want to record resolution details and mark the complaint resolved, so that the case is closed with a record. | CF-12 | Must | 3 |
| US-13 | As a citizen, I want an Updates inbox that shows what changed and which updates I haven't read, so that I don't miss progress on my complaints. | CF-13 | Must | 5 |
| US-14 | As an administrator, I want to view users and change their role, department or active status, and invite officers, so that the team has the right access. | CF-14 | Must | 8 |
| US-15 | As a citizen, I want to change my display name, so that my profile is correct. | CF-15 | Should | 1 |

| ID | User story | CF | MoSCoW | Pts |
|---|---|---|---|---|
| US-16 | As a citizen, I want to attach photos or PDF documents privately, so that officers have evidence of the problem. | CF-16 | Should | 8 |
| US-17 | As a citizen or officer, I want to see the full history of a complaint, so that every change is transparent. | CF-18 | Should | 3 |
| US-18 | As a citizen, I want to rate and comment on how my resolved complaint was handled, so that the service can improve. | CF-19 | Should | 3 |
| US-19 | As an administrator, I want to add, rename, route and deactivate service categories, so that the system can grow to new services without code changes. | CF-20 | Should | 5 |
| US-20 | As a citizen, I want to withdraw my unresolved complaint with a reason, so that officers don't spend time on a problem that no longer needs action. | CF-22 | Should* | 5 |
| US-21 | As an administrator, I want summary counts of complaints by status and the average feedback rating, so that I can monitor service performance. | CF-21 | Could | 2 |
| US-22 | As a citizen, I want my complaint to be given to a responsible Officer as soon as I submit it, so that it never waits unnoticed. | CF-23 | Must | 5 |
| US-23 | As an administrator, I want to route and assign complaints that are stuck in departments without Officers, so that every complaint has an owner. | CF-24 | Must | 5 |
| US-24 | As an officer, I want to see which complaints are due soon or overdue, so that I handle urgent work first. | CF-25 | Should | 3 |

## 6.3 Acceptance criteria for the major user stories

The acceptance criteria are written in Given/When/Then form, and each one is linked to the automated test that checks it.

**Table 6.3:** Acceptance criteria (major stories)

| Story | Acceptance criteria | Verified by |
|---|---|---|
| US-04/05/06 Submit complaint | AC1 Given a signed-in Citizen, when a valid category, description (10–5,000 chars), location (1–300) and coordinates are submitted, then the complaint is saved with a unique ID and status Submitted, and a confirmation shows the reference. AC2 Missing or out-of-range input shows field errors and nothing is saved; the database rejects the same input sent directly. AC3 A retried lost request returns the same complaint (no duplicate). AC4 Non-citizens, inactive and anonymous callers are refused. | TC-SUB-01…03, TC-DB-SUB-01…04 |
| US-07 Track complaint | AC1 My Complaints shows only my own complaints with reference, category and status. AC2 Opening one shows progress steps, history and, if resolved, the resolution text and date. AC3 A reference that is not mine shows no details. | TC-TRK-01…03 |
| US-11 Update status | AC1 Only an Officer of the complaint's department can review it; only the assigned Officer can start, add progress or resolve. AC2 Only Submitted → Under review → In progress → Resolved is accepted. AC3 Each change writes the new status, one history event and one Citizen update atomically. AC4 A stale change is rejected and the Officer's input is kept. | TC-DB-OFF-01…05, TC-OFF-03, TC-OFF-05 |

| Story | Acceptance criteria | Verified by |
|---|---|---|
| US-12 Resolve complaint | AC1 Resolution needs 10–5,000 characters and confirmation; status becomes Resolved with date and Officer stored. AC2 The Citizen sees the resolution and receives an update. AC3 Feedback becomes available. AC4 Resolved and withdrawn complaints reject further processing. | TC-OFF-01, TC-DB-OFF-06…07, TC-FB-01 |
| US-13 Updates inbox | AC1 One item per change, newest first, unread items counted in the menu badge. AC2 Mark one or all as read changes only my notifications. AC3 I can never see another user's notifications. | TC-NTF-01, TC-DB-NTF-01…02 |
| US-14 Manage users | AC1 Only an active Administrator can manage users. AC2 Role, department and active state can change; every change is audited. AC3 The last active Administrator cannot be removed, even concurrently. AC4 A deactivated user loses access immediately. | TC-ADM-01…02, TC-DB-ADM-01…04 |
| US-18 Feedback | AC1 On a Resolved complaint the owner submits a 1–5 rating and optional comment once; it is then read-only. AC2 A different second submission is refused. AC3 Feedback on unresolved complaints or by others is refused. | TC-FB-01…02, TC-DB-FB-01…04 |
| US-22/23 Every complaint has an owner | AC1 If the department has active Officers, the least-loaded one is assigned and notified on submission. AC2 Otherwise the complaint is flagged in the Dispatch centre. AC3 An Administrator dispatch with a reason changes route and assignee in one versioned transaction with history and notifications. | TC-DB-DSP-01…07, TC-DSP-01 |

## 6.4 Sprint plan and sprint history

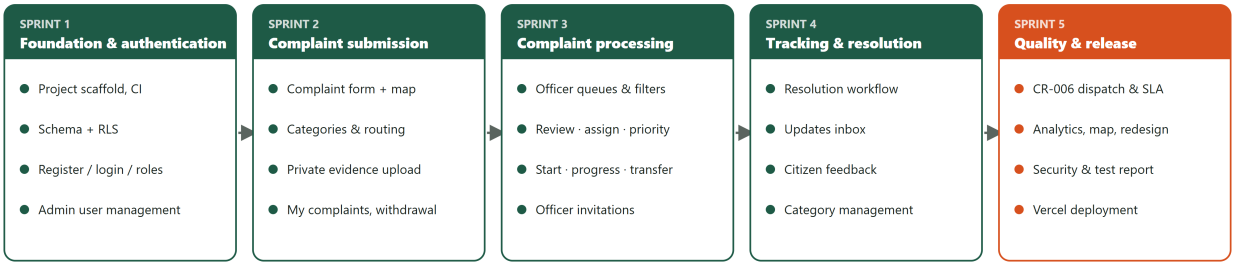



**Figure 6.1:** Five-sprint roadmap (from the proposal) with the delivered increments

**Table 6.4:** Sprint plan, delivered increments and review outcomes

| Sprint | Goal | Delivered increment | Review / retrospective outcome |
|---|---|---|---|
| 1 | Foundation & authentication | React/Vite scaffold; Supabase schema with Row Level Security; sign-up, login, recovery; role workspaces; admin user management; CI | Decided to keep sessions in memory only, choosing security over convenience |
| 2 | Complaint submission | Complaint form with map; six categories with routing; idempotent submission; private evidence; My Complaints; withdrawal (CR-001) | An empty category list in the demo led to a clear "missing setup" message and a database check before each demo |
| 3 | Complaint processing | Officer queues and filters; review, assign, priority, start, progress, transfer; Officer invitations (CR-003) | A concurrency test found an endless retry loop (DEF-01), fixed with a conflict code that is not retried (CR-002) |
| 4 | Tracking & resolution | Resolution; Updates inbox with read state; feedback; category management; monitoring | The inbox work was tracked as a separate item so that its progress was reported accurately |

| Sprint | Goal | Delivered increment | Review / retrospective outcome |
|---|---|---|---|
| 5 | Quality & release | CR-006 auto-assignment, dispatch centre and service targets; analytics; map; redesign; security and test report; deployment | Hosted use showed that 4 of 5 open complaints had no owner, so CR-006 was done before release; the layout audit found 4 layout defects, which were fixed |

## 6.5 Task allocation

We kept the task allocation from the proposal. Each member owned one module from start to finish and worked on its requirements, design, implementation and testing, as Section 13 of the guidelines requires. Chapter 14 gives the details.

**Table 6.5:** Task allocation by module

| Member | Main area | Backlog items owned |
|---|---|---|
| TG/2023/1731 H.M. Amadi Nethma | Requirements & Citizen module | CF-01, 02, 04, 07, 13 (UI), 15, 19 (UI), 22 |
| TG/2023/1710 M.F. Rishma | Complaint management module | CF-05, 06, 09, 10, 16, 25 |
| TG/2023/1757 N.M. Razim Mohamed | Officer module | CF-08, 11, 12, 17 (Officer), Officer map |
| TG/2023/1761 M.S.M. Sharfan | Administration & database | CF-03 (database), 14, 18, 20, 21, 23, 24 |
| TG/2023/1776 V. Kayaththiry | Quality, security & deployment | Test plan and suites, NFRs, security controls, deployment, maintenance CR |

# 7 Requirements Engineering

We gathered the requirements from the allocated domain, the stakeholder analysis and what we know about how local authorities in Sri Lanka handle complaints, and refined them later through sprint reviews and change requests. Every requirement has a fixed identifier (CF-xx for functional and NFR-xx for non-functional requirements), which is used in the backlog, the design, the code, the tests and this report.

## 7.1 Actors

**Table 7.1:** Actors

| Actor | Kind | Description |
|---|---|---|
| Citizen | Primary, application role | Registers, submits and tracks own complaints, receives updates, gives feedback, may withdraw an unresolved complaint |
| Public Service Officer | Primary, application role | Belongs to one department; reviews, assigns, prioritises and transfers its complaints; when assigned, starts work, adds progress and resolves |
| Administrator | Primary, application role | Manages users and Officer invitations, service categories, dispatch of stuck complaints, monitoring and analytics; does not process complaints |
| Supabase Auth | Secondary system actor | Stores credentials, verifies e-mail, issues sessions, sends invitation and recovery e-mails |
| OpenStreetMap | Secondary system actor | Supplies map tiles for location selection |

## 7.2 Functional requirements

The specification has 25 functional requirements (the guidelines ask for at least 12). Each one describes a single behaviour that can be tested, and the last column lists the test cases that check it.

**Table 7.2:** Functional requirements

| ID | Requirement ("The system shall …") | Priority | Source | Verified by |
|---|---|---|---|---|
| CF-01 | … allow a visitor to register a Citizen account with name, email and a password of at least 12 characters, and shall assign the Citizen role only. | Must | p. 11 | TC-AUTH-02, TC-DB-FND-01 |
| CF-02 | … allow registered users to log in and log out, and shall deny protected pages and data after logout. | Must | p. 11 | TC-AUTH-03, TC-AUTH-06 |
| CF-03 | … enforce role-based access to every protected function in the database, independently of the user interface. | Must | p. 11 | TC-AUTH-04, TC-RBAC-01, TC-DB-FND-02…04 |
| CF-04 | … allow a Citizen to submit a complaint with a description (10–5,000 characters), location text (1–300) and a latitude/longitude selected on a map or entered as numbers. | Must | p. 11 | TC-SUB-01, TC-SUB-02, TC-DB-SUB-03 |
| CF-05 | … require the Citizen to select one active service category and shall route the complaint to that category's department. | Must | p. 11 | TC-SUB-04, TC-DB-SUB-01, TC-DB-SUB-07 |

| ID | Requirement ("The system shall …") | Priority | Source | Verified by |
|---|---|---|---|---|
| CF-06 | … generate a unique identifier for each complaint, set its initial status to Submitted and show a confirmation with its reference. | Must | p. 11 | TC-SUB-01, TC-DB-SUB-01 |
| CF-07 | … allow a Citizen to list, open and track only their own complaints, including status, progress and resolution. | Must | p. 11 | TC-TRK-01…03, TC-DB-SUB-06 |
| CF-08 | … allow an active Officer to view and open the complaints of their own department, and complaints assigned to them. | Must | p. 11 | TC-OFF-02, TC-DB-SUB-06 |
| CF-09 | … allow an Officer to assign a department complaint to an active Officer of the same department, and to transfer an unstarted complaint to another department with a reason. | Must | p. 11 | TC-OFF-01, TC-OFF-04, TC-DB-OFF-04, TC-DB-OFF-09 |
| CF-10 | … allow an Officer to set complaint priority to Low, Normal, High or Urgent. | Must | p. 11 | TC-OFF-01, TC-OFF-08 |
| CF-11 | … allow only valid status transitions (Submitted → Under review → In progress → Resolved) and allow only the assigned Officer to start work and record progress remarks. | Must | p. 11 | TC-OFF-01, TC-OFF-05, TC-DB-OFF-05 |
| CF-12 | … allow the assigned Officer to resolve an In progress complaint with resolution details (10–5,000 characters), storing the resolution date and Officer. | Must | p. 11 | TC-OFF-01, TC-DB-OFF-06 |
| CF-13 | … record a durable update for the Citizen with every complaint change, and shall let the Citizen list updates and mark them read. | Must | p. 11 | TC-NTF-01, TC-DB-NTF-01…05 |
| CF-14 | … allow an Administrator to view users, change role, department and active state, and invite Officer accounts. It shall audit every change and protect the last active Administrator. | Must | p. 11 | TC-ADM-01…03, TC-DB-ADM-01…04 |
| CF-15 | … allow every user to change their own display name only. | Should | p. 6 | TC-AUTH-08 |
| CF-16 | … allow a Citizen to attach up to five private JPEG/PNG/PDF files (≤ 5 MiB each) to their Submitted complaint, downloadable only by authorised users. | Should | p. 11 | TC-SUB-01, TC-DB-EV-01…06 |
| CF-17 | … allow searching by reference or description and filtering by status (Citizen), and by status, category, priority and assignment (Officer). | Should | p. 6 | TC-TRK-01, TC-OFF-02, TC-OFF-09 |
| CF-18 | … maintain an append-only history of every complaint change, visible to authorised users. | Should | p. 11 | TC-SUB-01, TC-DB-OFF-01 |
| CF-19 | … allow the owning Citizen to submit exactly one rating (1–5) with an optional comment (≤ 2,000 characters) on a Resolved complaint. | Should | p. 11 | TC-FB-01, TC-FB-02, TC-DB-FB-01…05 |
| CF-20 | … allow an Administrator to create, edit, route and deactivate service categories without deleting them, auditing every change. | Should | p. 11 | TC-CAT-01, TC-CAT-02, TC-DB-CAT-01…06 |
| CF-21 | … provide Administrators with read-only complaint monitoring information (counts by status, average rating). | Could | p. 11 | TC-MON-01, TC-U-MON-01 |
| CF-22 | … allow the owning Citizen to withdraw an unresolved complaint with a 10–2,000 character reason, retaining its record, evidence and history. (approved scope extension) | Should | change record | TC-WD-01…04, TC-DB-WD-01…05 |
| CF-23 | … automatically assign each new complaint to the active Officer with the fewest open assignments in the routed department, and notify that Officer. (CR-006) | Must | CR-006 | TC-DB-DSP-01…03, TC-DSP-03 |

| ID | Requirement ("The system shall …") | Priority | Source | Verified by |
|---|---|---|---|---|
| CF-24 | … allow an active Administrator to dispatch an open complaint: route it to another active department/category and/or assign an active Officer of that department, with a recorded 10–2,000 character reason. (CR-006) | Must | CR-006 | TC-DSP-01, TC-DB-DSP-04…09 |
| CF-25 | … apply a service-level target per priority (urgent 1 day, high 3, normal 7, low 14), flag open complaints that are due soon or overdue, and report on-target resolution. (CR-006) | Should | CR-006 | TC-U-SLA-01, TC-DSP-02, TC-DSP-03 |

## 7.3 Non-functional requirements

The quality goals in the proposal were turned into eleven non-functional requirements, each with a measurable acceptance threshold. They cover performance, usability, security, reliability, integrity, availability, maintainability, scalability and privacy.

**Table 7.3:** Non-functional requirements

| ID | Category | Requirement | Measure / acceptance threshold | Verification method | Status |
|---|---|---|---|---|---|
| NFR-01 | Performance | List screens shall load a bounded page of results, never the whole table. | Citizen/Officer lists ≤ 10 rows per request; admin users ≤ 20; ownership, department and status columns indexed | Code review of range() calls; indexes in migration 001 | Verified |
| NFR-02 | Performance | Common screens should be usable quickly on the demo deployment. | Proposed: first meaningful content ≤ 3 s and page navigation ≤ 1.5 s on a typical broadband connection with demo data | Manual timing in browser DevTools on the deployed URL | Pending (deployment) |
| NFR-03 | Usability / accessibility | Forms shall be usable by keyboard and screen reader, and pages shall fit a phone screen. | Every input has a programmatic label; errors announced (role="alert"); skip link; no horizontal scroll at 375 px width | Playwright getByLabel/role queries; mobile overflow tests TC-UX-01, TC-TRK-01, TC-OFF-06 | Verified |
| NFR-04 | Security – authentication | Protected functions shall require an authenticated, active account. | 100% of RPCs deny the anonymous role; inactive accounts denied immediately | TC-DB-FND-03, TC-DB-ADM-04, security advisor | Verified |
| NFR-05 | Security – authorisation | Users shall access only data and actions permitted by role, ownership and department. | RLS enabled on 9/9 application tables; 0 direct browser INSERT/UPDATE/DELETE grants | TC-DB-FND-02, TC-DB-OFF-03, TC-DB-FB-05 | Verified |
| NFR-06 | Reliability / integrity | Every complaint change shall be atomic and concurrency-safe. | Each version change writes exactly 1 event and 1 notification, or nothing; stale writes rejected (HTTP 409); retries are idempotent | TC-DB-SUB-05, TC-DB-OFF-08, TC-DB-WD-05, TC-INT-04…06 | Verified |
| NFR-07 | Data integrity / validation | Invalid or incomplete records shall be rejected in both client and database. | Identical bounds in Zod schemas and SQL CHECK/RPC validation | Unit TC-U- and database TC-DB--bounds tests | Verified |

| ID | Category | Requirement | Measure / acceptance threshold | Verification method | Status |
|---|---|---|---|---|---|
| NFR-08 | Availability | The deployed system shall be reachable during the evaluation. | Smoke-test checklist (08 §8.6) passes within 24 h before the demo | Manual smoke test | Pending (deployment) |
| NFR-09 | Maintainability | Code shall be modular and consistently checked. | Feature modules with ui/application/model/data layers; ESLint 0 warnings; strict TypeScript; CI green on main | CI workflow | Verified |
| NFR-10 | Scalability / extensibility | New categories, departments and users shall be added without code changes. | Category added via the UI appears in the Citizen form | TC-CAT-01, TC-DB-CAT-02 | Verified |
| NFR-11 | Privacy | Complaint data and evidence shall be visible only to authorised users; demo data is fictional. | Evidence bucket private; no public complaint feed; tests use fictional identities | TC-DB-EV-01, TC-TRK-03, security review | Verified |

## 7.4 Use case diagram

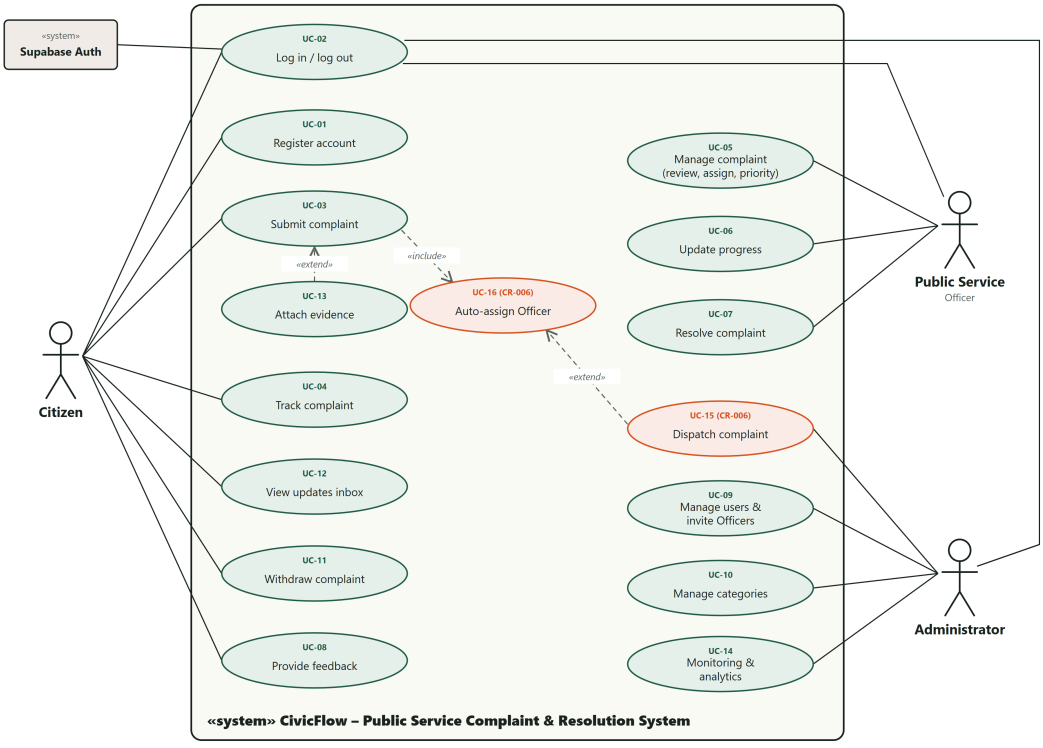



**Figure 7.1:** UML use case diagram (UC-15 and UC-16 were added by change request CR-006)

## 7.5 Use case descriptions

**Table 7.4:** UC-03 Submit complaint (CF-04, 05, 06, 16, 23)

| Element | Description |
|---|---|
| Primary actor | Citizen |
| Preconditions | Signed in with an active Citizen account; at least one active category exists |
| Main flow | 1) Open Make a complaint. 2) Select a service category. 3) Describe the issue. 4) Select the location on the map, type coordinates or use "Use my current location", and describe the location. 5) Submit. 6) The system validates the input, routes the complaint to the category's department, assigns the least-loaded active Officer, records history event 1 and updates, and shows the reference. 7) Optionally attach evidence (UC-13). |
| Alternative flows / exceptions | 3a/4a Invalid or missing fields: field errors, nothing saved. 5a Network failure: the same request key is retried and no duplicate is created. 6a No Officer in the department: the complaint is left unassigned and flagged for dispatch. 7a Upload fails: the complaint stays saved and the file can be retried. |
| Postcondition | The complaint is Submitted, visible only to its owner, the department's Officers and Administrators. |

**Table 7.5:** UC-04 Track complaint (CF-07, 17, 18)

| Element | Description |
|---|---|
| Primary actor | Citizen |
| Preconditions | Signed in |
| Main flow | 1) Open My complaints. 2) Search by reference or description and filter by status. 3) Open a complaint. 4) View the progress line, details, location, evidence, resolution and history. |
| Alternative flows / exceptions | The reference does not exist or belongs to someone else: "Complaint unavailable" with no details. |
| Postcondition | No data is changed. |

**Table 7.6:** UC-05 Manage complaint (CF-08, 09, 10)

| Element | Description |
|---|---|
| Primary actor | Officer |
| Preconditions | Active Officer of an active department; complaint in that department and not closed |
| Main flow | 1) Open the department queue and filter. 2) Open a complaint. 3) Begin review (Submitted → Under review). 4) Assign an Officer of the department. 5) Set the priority. Each step saves one history event and one Citizen update. |
| Alternative flows / exceptions | 4a Transfer to another department with a reason (Submitted or Under review only). Exceptions: complaint changed meanwhile (stale-version message, input kept); target Officer inactive or in another department (refused). |
| Postcondition | Assignment, priority and status are saved with history. |

**Table 7.7:** UC-06/07 Update and resolve complaint (CF-11, 12)

| Element | Description |
|---|---|
| Primary actor | Assigned Officer |

| Element | Description |
|---|---|
| Preconditions | Status Under review (to start) or In progress |
| Main flow | 1) Start work, optionally with a remark. 2) Add progress updates (1–2,000 characters). 3) Resolve with resolution details (10–5,000 characters) and confirmation. |
| Alternative flows / exceptions | Not the assignee, invalid transition, empty remark or short resolution: refused atomically with no partial update. |
| Postcondition | Status Resolved; date and Officer stored; the Citizen is notified and can give feedback. |

**Table 7.8:** UC-08 Provide feedback (CF-19)

| Element | Description |
|---|---|
| Primary actor | Citizen (owner) |
| Preconditions | Complaint Resolved with no feedback yet |
| Main flow | 1) Open the resolved complaint. 2) Choose a 1–5 rating. 3) Optionally add a comment. 4) Submit. |
| Alternative flows / exceptions | No rating: client error. Already submitted: "Feedback was already submitted". Not resolved: refused. |
| Postcondition | Feedback shown read-only to the owner, the department's Officers and Administrators. |

**Table 7.9:** UC-09 Manage users and invite Officers (CF-14)

| Element | Description |
|---|---|
| Primary actor | Administrator |
| Preconditions | Active Administrator |
| Main flow | 1) Open User management. 2) Search. 3) Edit role, department or active state, or Create Officer account (name, e-mail, department). 4) Save; the change is audited and the invitee sets their own password. |
| Alternative flows / exceptions | Removing the last active Administrator: refused. An Officer with open assignments cannot be moved or deactivated until the work is reassigned. |
| Postcondition | Access changed and audited. |

**Table 7.10:** UC-15 Dispatch complaint (CF-24)

| Element | Description |
|---|---|
| Primary actor | Administrator |
| Preconditions | Complaint is Submitted, Under review or In progress |
| Main flow | 1) Open the Dispatch centre; coverage cards show departments with no Officers and the Needs-attention filter lists unstaffed, unassigned or overdue complaints. 2) Choose Dispatch. 3) Select department, category and optionally an Officer. 4) Enter a reason and confirm. |
| Alternative flows / exceptions | Work in progress cannot change department or lose its assignee. Changed meanwhile: refresh. Closed complaints: refused. |
| Postcondition | Route and/or assignee changed; history records the Administrator's reason; Citizen and Officer notified. |

The other use cases (UC-01 Register, UC-02 Log in/out, UC-10 Manage categories, UC-11 Withdraw complaint, UC-12 View updates inbox, UC-13 Attach evidence and UC-14 Monitoring) use the same format and are described in the repository specification (docs/sdlc/04-requirements-specification.md).

## 7.6 Prioritisation (MoSCoW)

We used MoSCoW because it fits the sprint order well: within each sprint the Must items were scheduled before the Should items, and the Could item was done last.

**Table 7.11:** MoSCoW prioritisation

| Priority | Requirements | Rationale |
|---|---|---|
| Must have | CF-01 – CF-14, CF-23, CF-24 | Without these the core complaint lifecycle, its security or its ownership does not work |
| Should have | CF-15 – CF-20, CF-22, CF-25 | Important for usability, evidence, transparency and administration, but the lifecycle works without them |
| Could have | CF-21 (monitoring and analytics) | Valuable for management; delivered after all Must and Should items |
| Won't have (this release) | AI classification, GPS Officer tracking, payments, emergency response, broad government integration, SMS/e-mail alerts | Out of scope (Chapter 4) or beyond the free-tier constraints |

## 7.7 Requirements traceability matrix

The matrix links each requirement to its user story, use case, design artefact, implementation (database migration and frontend module) and tests. Migration numbers are shortened: 001 is 20261019000100_foundation.sql and 015 is 20261029000400_dispatch_and_coverage.sql.

**Table 7.12:** Requirements traceability matrix

| Req | Story | Use case | Design | Implementation (DB → frontend) | Tests | Status |
|---|---|---|---|---|---|---|
| CF-01 | US-01 | UC-01 | Auth flow, handle_new_user trigger | 001 trigger → features/auth | TC-AUTH-02, TC-U-AUTH-01, TC-DB-FND-01 | Done |
| CF-02 | US-02 | UC-02 | Memory-only session decision | Supabase Auth → features/auth/AuthProvider | TC-AUTH-03, TC-AUTH-06, TC-U-AUTH-02 | Done |
| CF-03 | US-03 | all | Access matrix (DB design) | 001–002 RLS, private.active_role → route guards | TC-AUTH-04, TC-RBAC-01, TC-DB-FND-02…04 | Done |
| CF-04 | US-04 | UC-03 | Seq. SD-1, shared/location | 004 submit_complaint → complaints/NewComplaintPage | TC-SUB-01…02, TC-LOC-01…02, TC-U-CMP-01, TC-DB-SUB-03 | Done |
| CF-05 | US-05 | UC-03 | Category routing | 004, 006 → category selector | TC-SUB-04…05, TC-DB-SUB-07 | Done |
| CF-06 | US-06 | UC-03 | SD-1 | 004 → submission success view | TC-SUB-01, TC-SUB-03, TC-DB-SUB-01…02 | Done |
| CF-07 | US-07 | UC-04 | Citizen screens | 004 RLS → MyComplaintsPage, ComplaintDetailPage | TC-TRK-01…03 | Done |
| CF-08 | US-08 | UC-05 | Officer screens | 004 RLS → officer/OfficerQueue | TC-OFF-02, TC-OFF-07 | Done |

| Req | Story | Use case | Design | Implementation (DB → frontend) | Tests | Status |
|---|---|---|---|---|---|---|
| CF-09 | US-09 | UC-05 | State diagram, SD-2 | 009 assign_complaint, transfer_complaint → OfficerActions | TC-OFF-01, TC-OFF-04, TC-DB-OFF-04, TC-DB-OFF-09 | Done |
| CF-10 | US-10 | UC-05 | — | 009 set_complaint_priority | TC-OFF-01, TC-OFF-08 | Done |
| CF-11 | US-11 | UC-06 | Activity diagram, state diagram | 009 review/start/add_progress | TC-OFF-01, TC-OFF-03, TC-OFF-05, TC-DB-OFF-01…05 | Done |
| CF-12 | US-12 | UC-07 | SD-2 | 010 resolve_complaint | TC-OFF-01, TC-DB-OFF-06…07 | Done |
| CF-13 | US-13 | UC-12 | notifications table | 004/009 writes, 013 read policy + mark_notifications_read → features/notifications | TC-NTF-01, TC-U-NTF-01, TC-DB-NTF-01…05 | Done |
| CF-14 | US-14 | UC-09 | Admin audit, Edge Function | 002 admin_update_user, provision-officer → features/admin | TC-ADM-01…03, TC-U-INV-01, TC-DB-ADM-01…04, TC-INT-08 | Done |
| CF-15 | US-15 | — | — | 002 update_my_profile → features/profiles | TC-AUTH-08 | Done |
| CF-16 | US-16 | UC-13 | Evidence storage design | 005 bucket, policies, reserve/finalize → features/evidence | TC-SUB-01, TC-U-EV-01, TC-DB-EV-01…06 | Done |
| CF-17 | US-07, US-08 | UC-04, UC-05 | — | Data adapters (bounded ilike, filters) | TC-TRK-01, TC-OFF-02, TC-OFF-09 | Done |
| CF-18 | US-17 | UC-04 | complaint_events | 004–011 event writes → timeline | TC-SUB-01, TC-DB-OFF-01 | Done |
| CF-19 | US-18 | UC-08 | SD-3, complaint_feedback | 012 submit_feedback → features/feedback | TC-FB-01…02, TC-U-FB-01, TC-DB-FB-01…05 | Done |
| CF-20 | US-19 | UC-10 | Category rules | 014 admin_upsert_category → features/categories | TC-CAT-01…02, TC-U-CAT-01, TC-DB-CAT-01…06 | Done |
| CF-21 | US-21 | UC-14 | Analytics view | 015 admin_analytics, community_impact → features/analytics, admin/AdminOverview | TC-MON-01, TC-DSP-02, TC-U-MON-01, TC-U-ANL-01, TC-DB-DSP-10…12 | Done (enhanced) |
| CF-23 | US-22 | UC-03 | SD-4 | 015 submit_complaint, private.pick_officer | TC-DB-DSP-01…03, TC-DSP-03 | Done |
| CF-24 | US-23 | UC-15 | SD-4 | 015 admin_dispatch_complaint → features/dispatch | TC-DSP-01, TC-U-DSP-01, TC-DB-DSP-04…09 | Done |
| CF-25 | US-24 | UC-05, UC-15 | SLA table | shared/sla.ts + private.sla_hours | TC-U-SLA-01, TC-DSP-02, TC-DSP-03 | Done |
| CF-22 | US-20 | UC-11 | State diagram | 007–008, 011 withdraw_complaint → WithdrawalPanel | TC-WD-01…04, TC-U-WD-01, TC-DB-WD-01…05 | Done |
| NFR-04/05 | US-03 | all | Security | Grants, RLS, definer RPCs | TC-DB-FND-, TC-INT- | Done |
| NFR-06 | — | — | Lock order, version checks | All workflow RPCs | TC-DB-*-rollback, TC-INT-04…06 | Done |

## 7.8 Requirements validation

**Table 7.13:** Requirements validation activities

| Activity | What was done | Outcome |
|---|---|---|
| Proposal review | The proposal (Milestones 1–3) was submitted as the requirements baseline | Accepted as the baseline for development |
| Stakeholder refinement | The project owner requested withdrawal, Administrator-created Officers and the location refinement; hosted testing revealed unowned complaints | Recorded as change requests CR-001 – CR-006 with impact analysis (Chapter 12) |
| Prototype walkthroughs | Each increment was demonstrated in the browser and screenshots were reviewed | Defects such as the empty category list were found and fixed |
| Consistency checks | Client and database input limits compared; tests assert both | No mismatches outstanding |
| Not yet done | Validation with real citizens or Officers | Labelled as an open assumption |

# 8 System Design

## 8.1 High-level system architecture

CivicFlow uses a three-tier client-server architecture. The presentation tier is a React single-page application served from Vercel's CDN. Supabase provides the application tier (authentication, a REST/RPC API and the business functions that run on the server) and the data tier (PostgreSQL and private file storage).

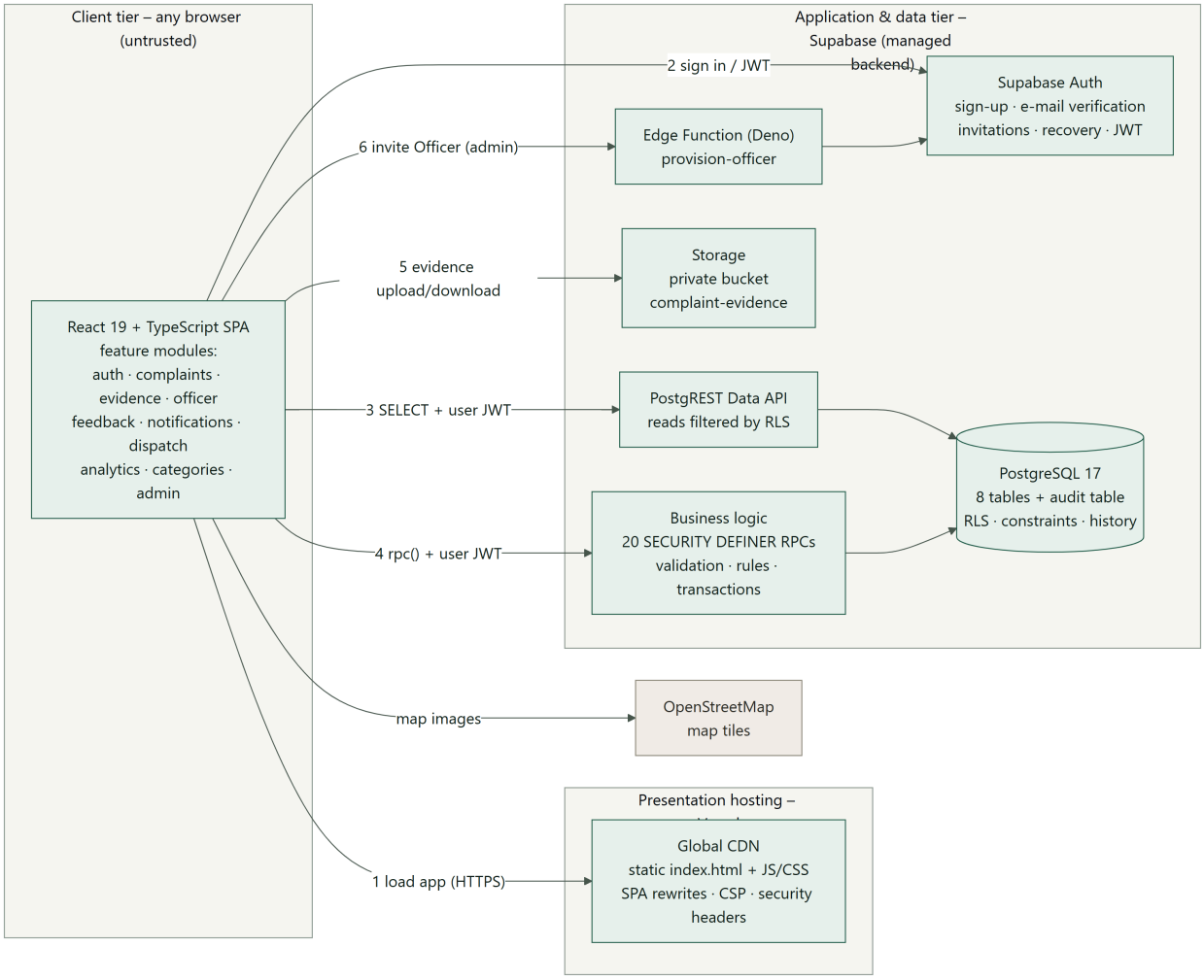



**Figure 8.1:** High-level system architecture

### Trust boundary

We treat the browser as untrusted, so every authorisation decision is made on the server by PostgreSQL grants, Row Level Security policies, checks inside SECURITY DEFINER functions and Storage policies. Hiding buttons and guarding routes in the interface only makes the application easier to use; it does not protect any data.

## 8.2 Frontend architecture

The frontend code is grouped by feature (auth, complaints, evidence, officer, feedback, notifications, dispatch, analytics, categories, admin and landing). Every feature has the same four layers, shown below. Screens never call the backend directly, and the business rules can be unit-tested without React or a network connection.

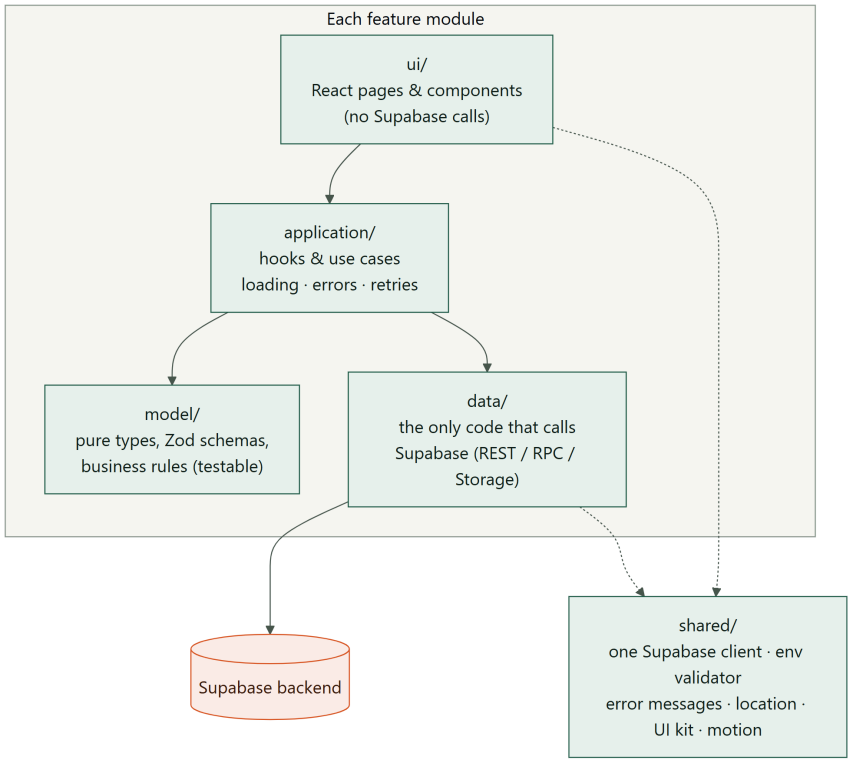



**Figure 8.2:** Layered structure of every feature module

## 8.3 Class / domain model

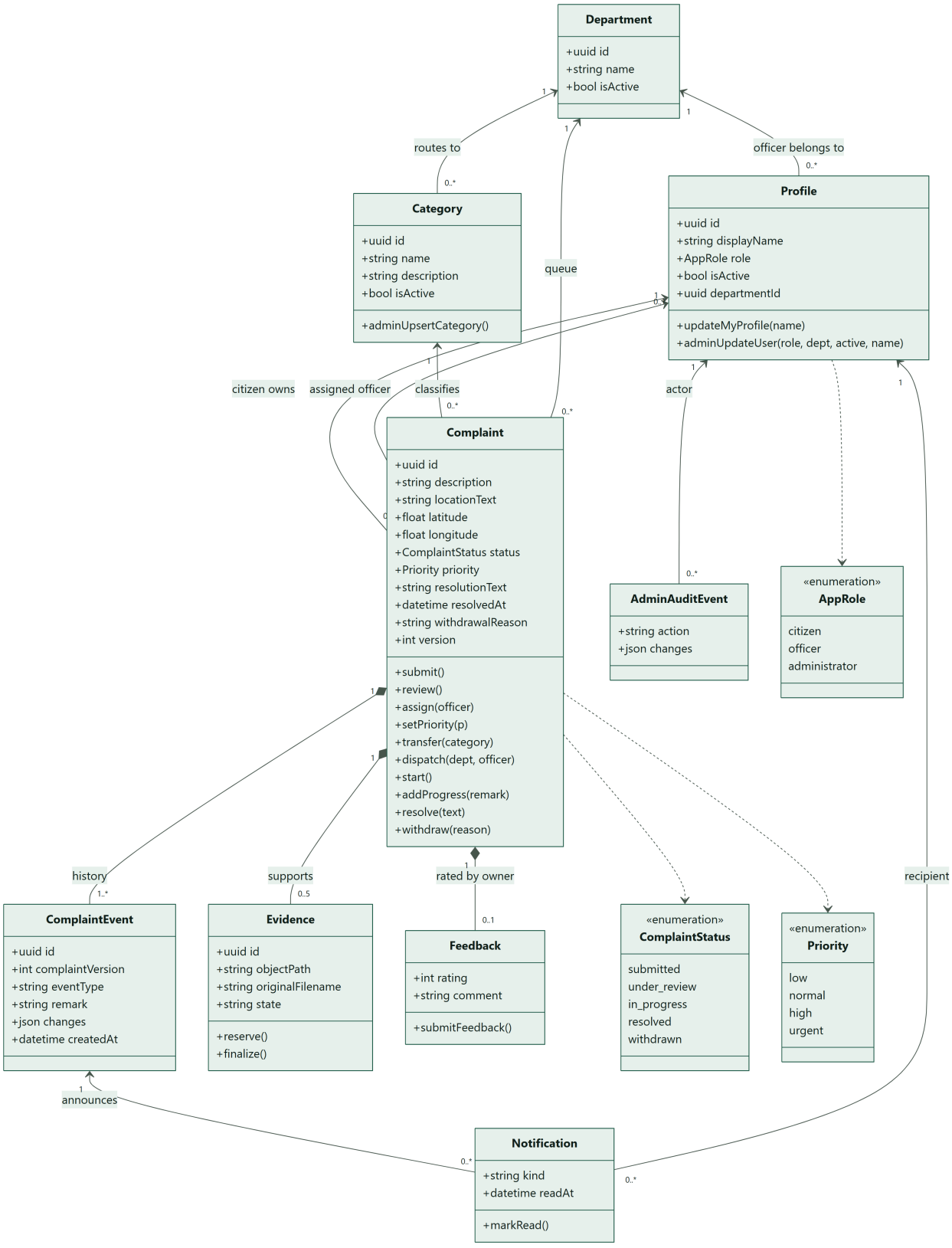



**Figure 8.3:** UML class diagram of the domain model (operations are the only way to change data)

Browsers have no direct insert, update or delete rights on any table. Every change goes through one of the operations shown in the diagram, each implemented as a protected database function.

## 8.4 Sequence diagrams

We drew sequence diagrams for the four most important interactions. SD-1 covers complaint submission, including validation, protection against duplicate submissions and automatic assignment. SD-2 shows an Officer resolving a complaint with row locking and version checks. SD-3 shows a Citizen giving feedback, and SD-4 shows automatic assignment and Administrator dispatch, which were added by CR-006.

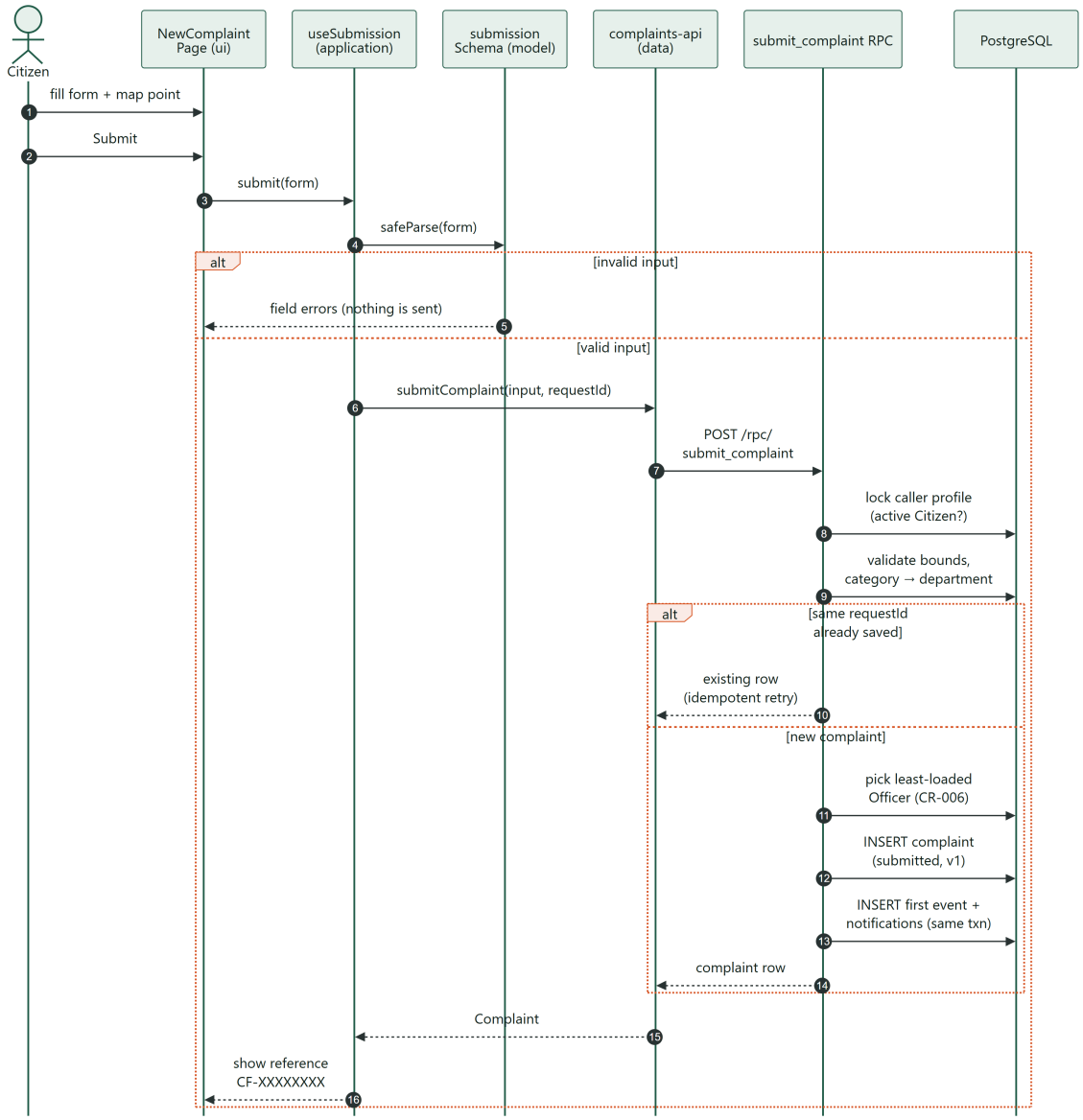



**Figure 8.4:** SD-1: Citizen submits a complaint (UC-03)

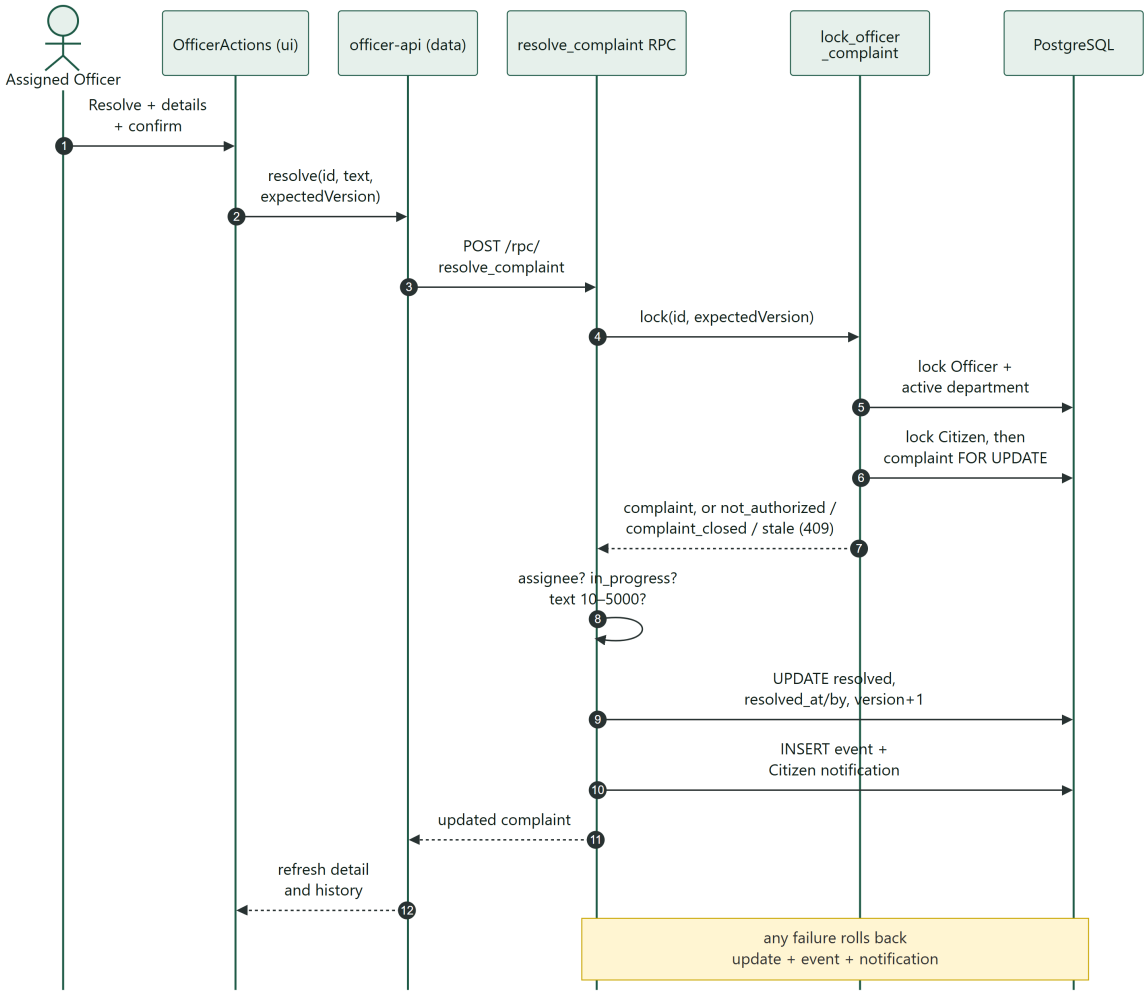



**Figure 8.5:** SD-2: Assigned Officer resolves a complaint (UC-07)

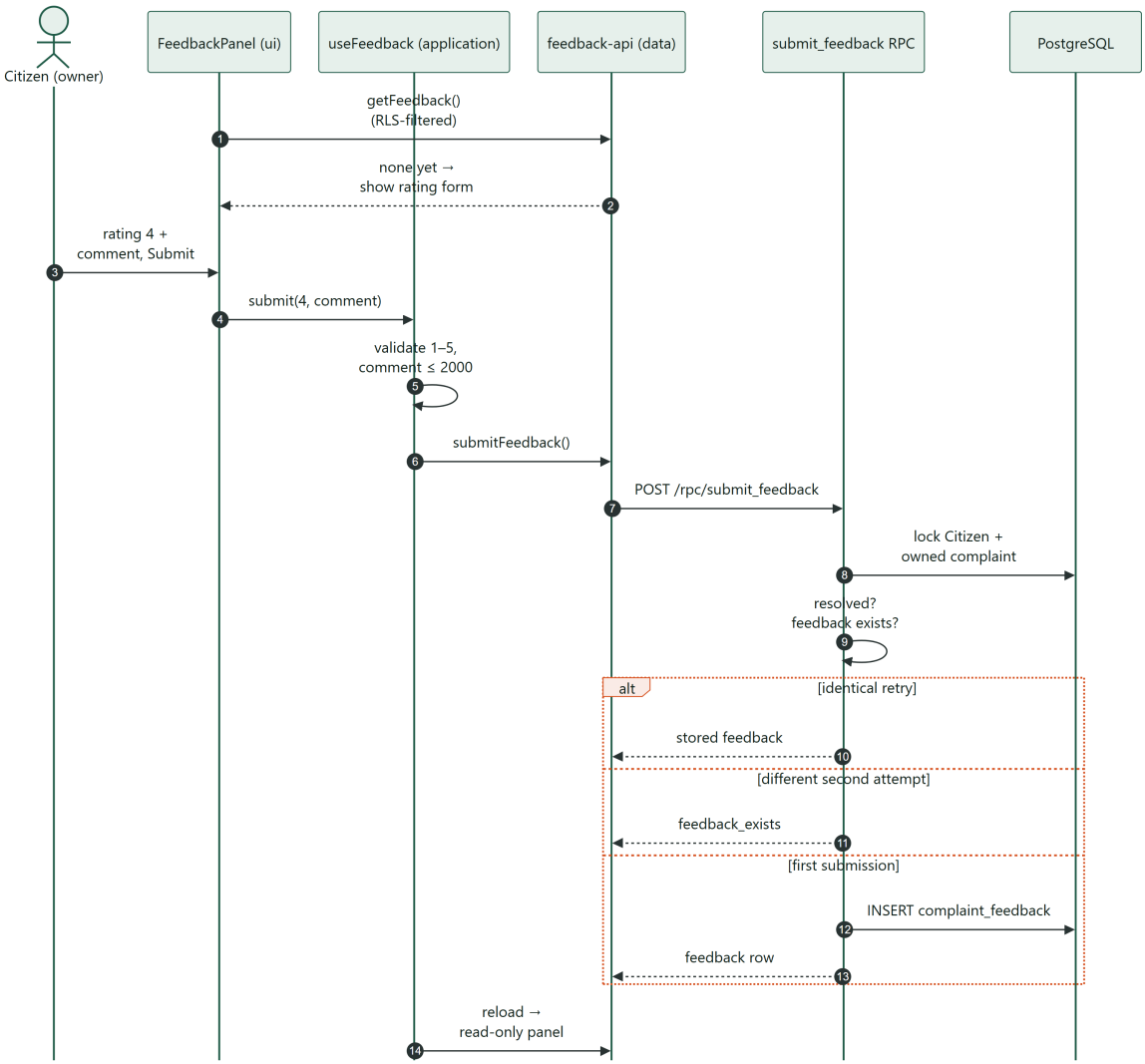



**Figure 8.6:** SD-3: Citizen gives feedback (UC-08)

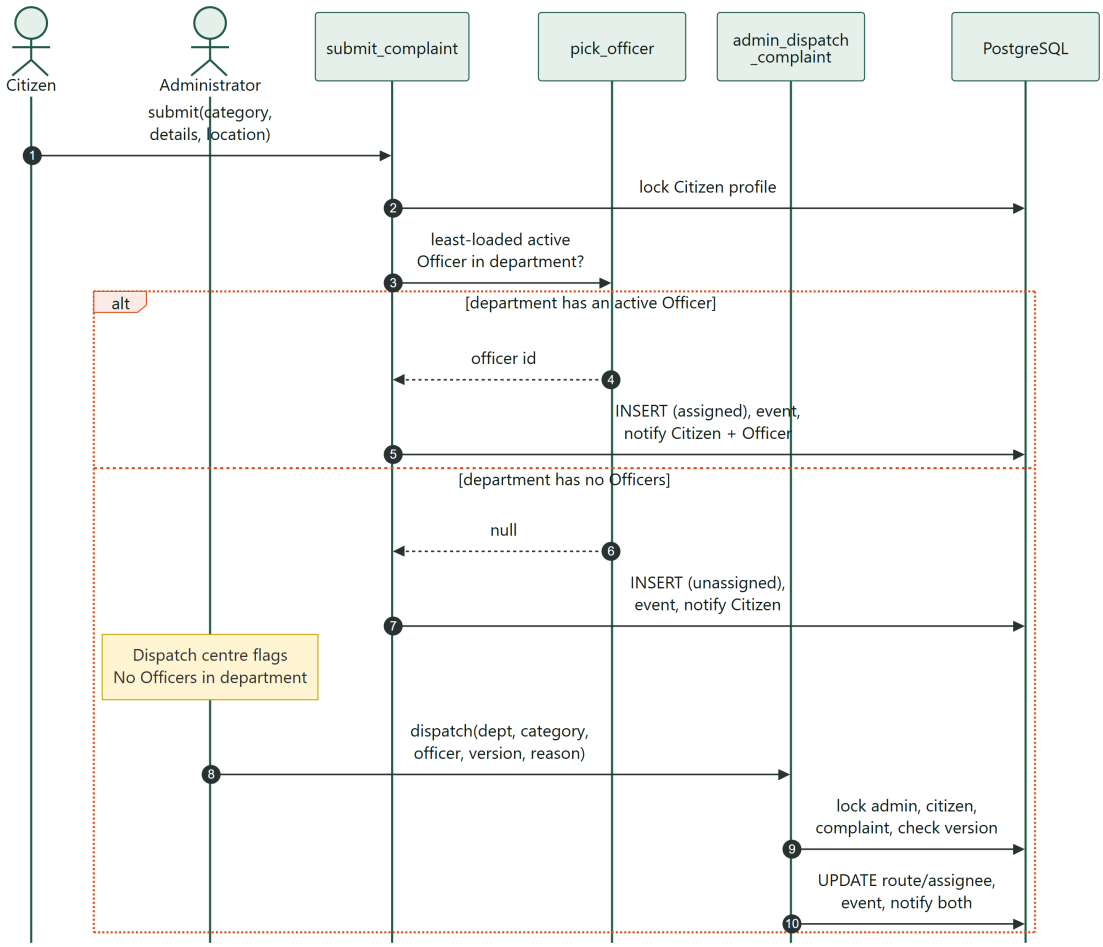



**Figure 8.7:** SD-4: Automatic assignment and Administrator dispatch (CR-006, UC-15, UC-16)

## 8.5 Activity diagram of the complaint workflow

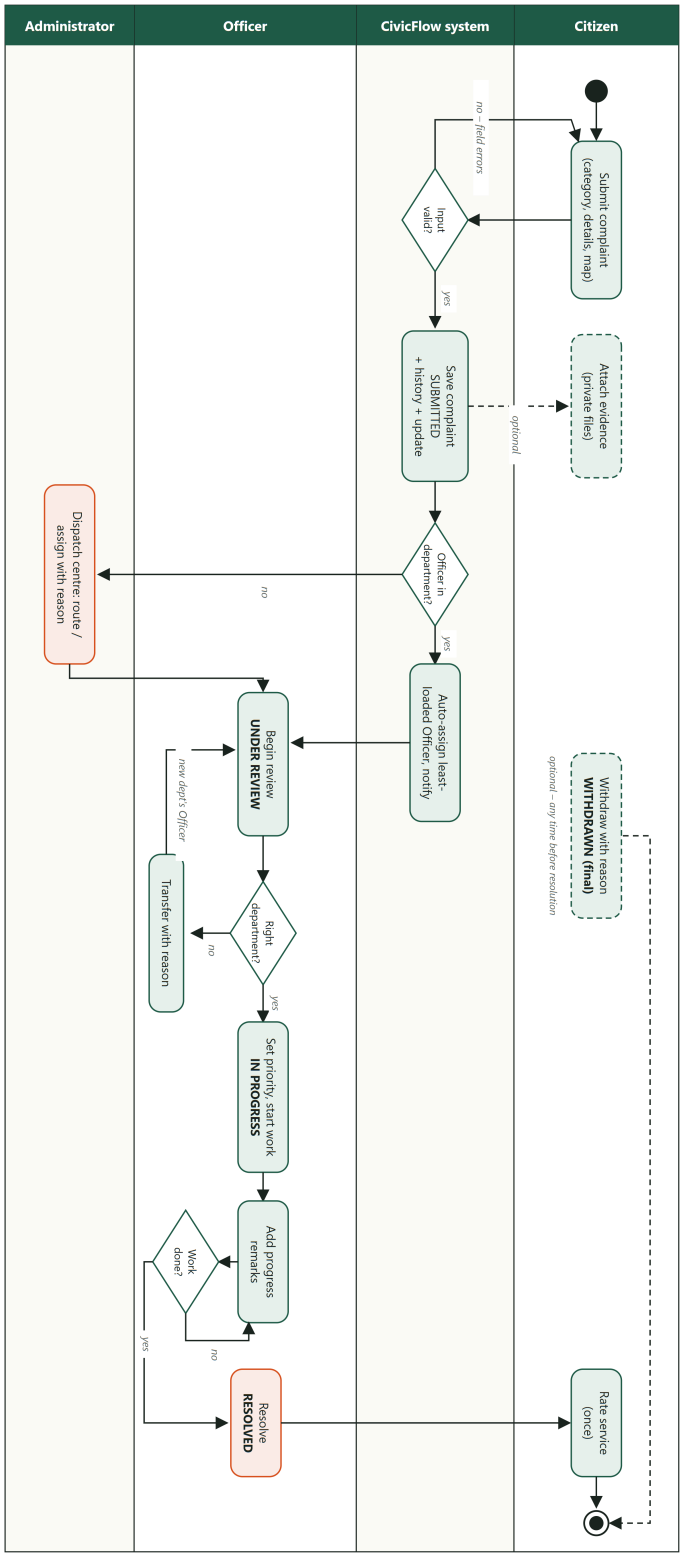



**Figure 8.8:** UML activity diagram with swimlanes for the complete complaint workflow (turned sideways to fit the page)

## 8.6 State machine of the complaint status

The status of a complaint follows the state machine below, which is enforced inside the database functions. Assigning an Officer, changing the priority and adding a progress remark leave the status unchanged but increase the version number, which we use for optimistic concurrency control.

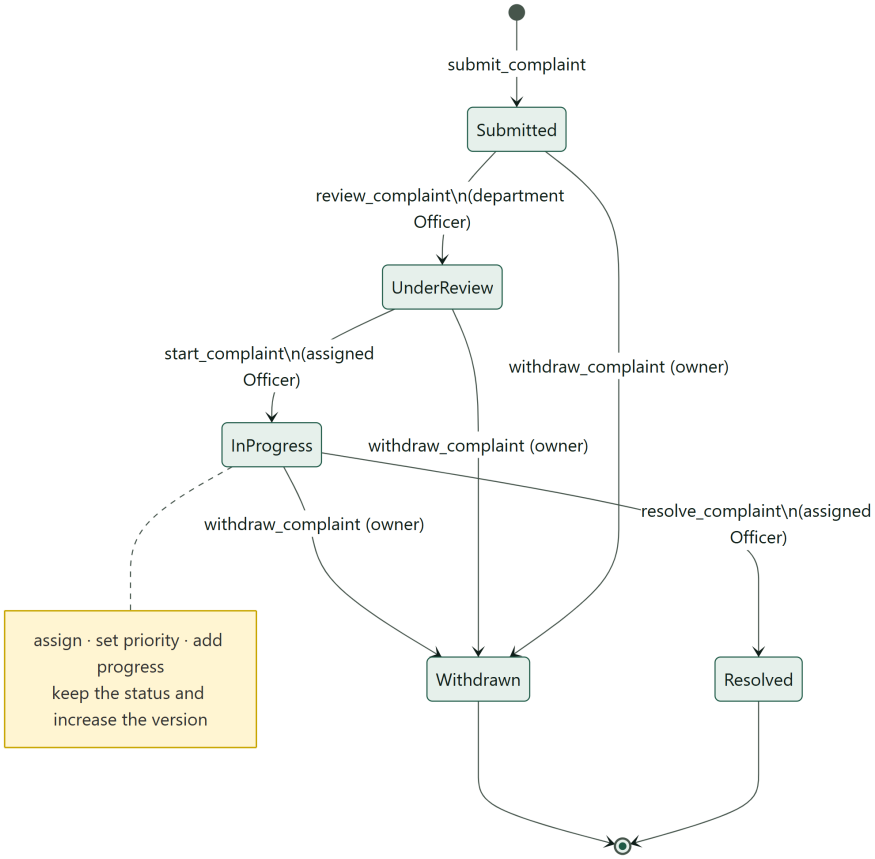



**Figure 8.9:** UML state machine of the complaint status

## 8.7 Database design

The database has eight application tables in the public schema and one audit table in a private schema. The entity-relationship diagram below shows their keys and relationships, and the table after it lists the main integrity rules.

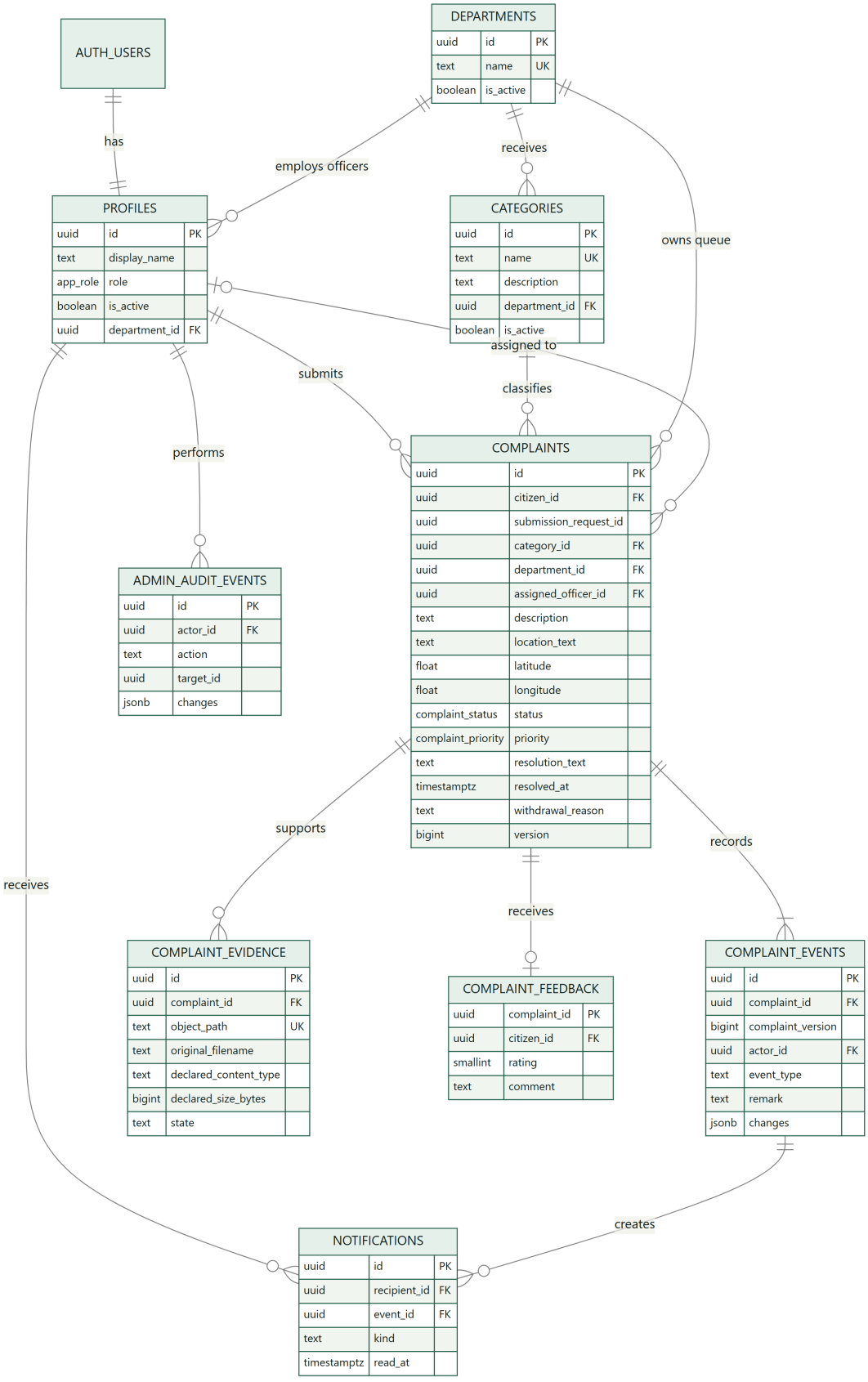



**Figure 8.10:** Entity-relationship diagram (PostgreSQL 17)

**Table 8.1:** Major tables and integrity rules

| Table | Purpose | Key rules |
|---|---|---|
| profiles | One row per Auth user: display name, role, active flag, department | Signup always creates a Citizen; an Officer must have a department and only Officers may have one |
| departments / categories | Service departments and the categories that route to them | Unique names; categories are deactivated, never deleted; a used category keeps its department |
| complaints | The complaint and its current state | Description 10–5,000; location 1–300; resolution fields set if and only if Resolved; In progress/Resolved require an assignee; version ≥ 1; (citizen, request id) unique for idempotency |
| complaint_events | Append-only history | Exactly one event per complaint version (unique complaint_id, complaint_version); progress events need a remark |
| complaint_evidence | Private file metadata | JPEG/PNG/PDF, 1 byte – 5 MiB, ≤ 5 files; server-reserved unique object path; pending → attached |
| complaint_feedback | One rating per complaint | Primary key = complaint; rating 1–5; comment ≤ 2,000 |
| notifications | Durable in-system updates | Unique (recipient, event); read_at for the inbox |
| private.admin_audit_events | Audit of administrative changes | Not exposed to browsers; before/after values in JSON |

All foreign keys use ON DELETE RESTRICT, so records are never deleted by the application. Row Level Security is switched on for every table and browsers have no INSERT, UPDATE or DELETE rights. Every change goes through one of twenty narrowly scoped functions.

## 8.8 User-interface prototype and wireframes

Before building the screens we drew low-fidelity wireframes of the main pages to agree on the layout. The working screens were then built and improved after each sprint review; Chapter 9 shows the final versions.

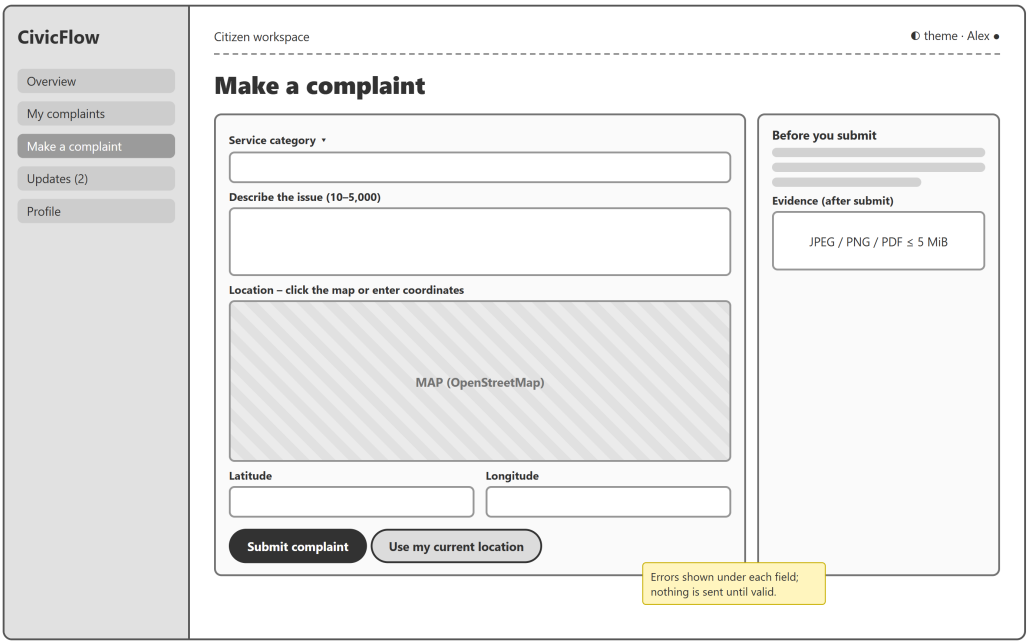



**Figure 8.11:** Wireframe of the Citizen "Make a complaint" screen

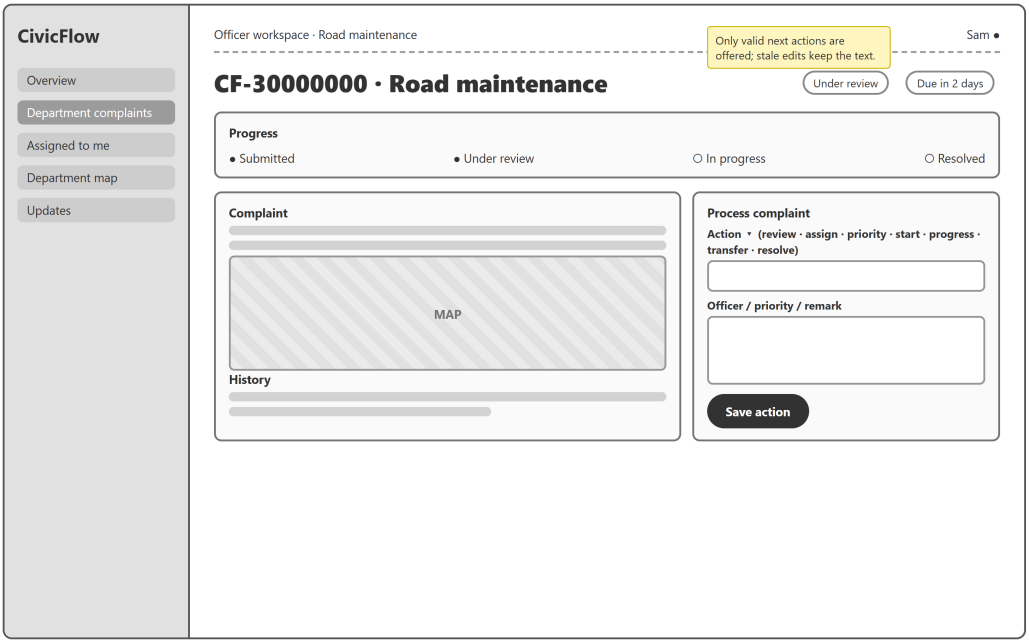



**Figure 8.12:** Wireframe of the Officer complaint processing screen

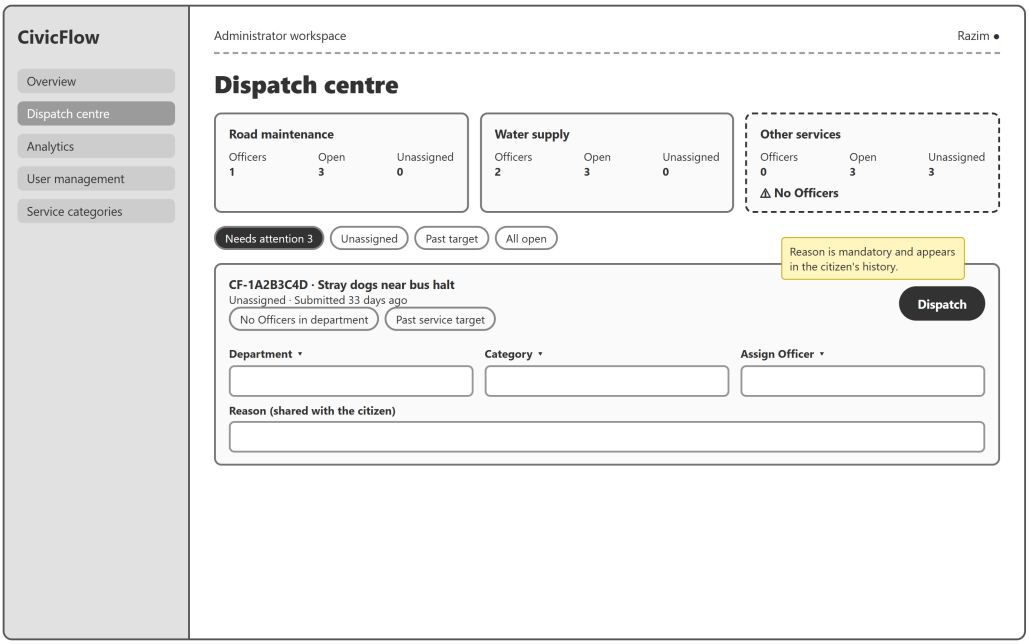



**Figure 8.13:** Wireframe of the Administrator dispatch centre (CR-006)

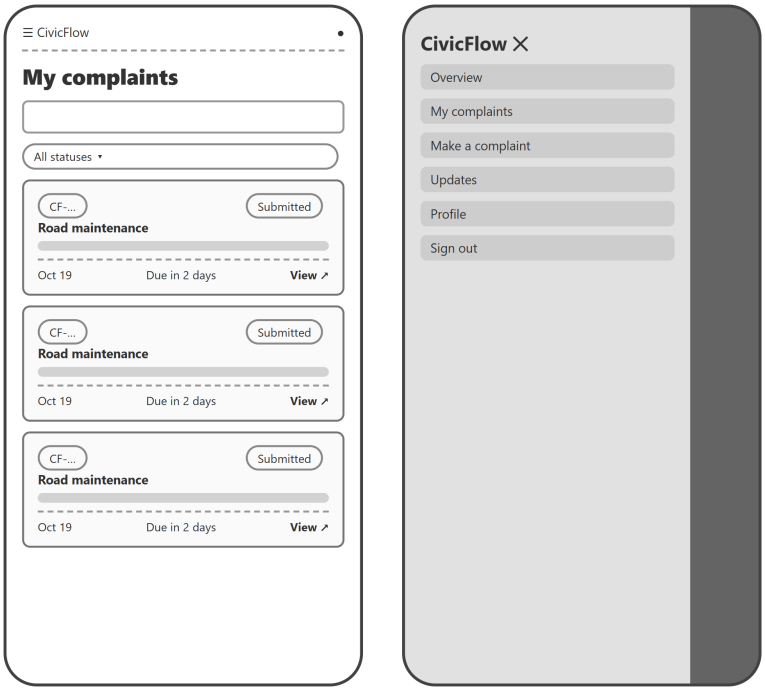



**Figure 8.14:** Wireframe of the mobile complaint list and slide-out menu

The interface follows a few simple rules. Each screen has one main action. Status names are written in plain language and shown with both text and colour. Every form field has a label, a hint and an error message. There is a skip link and a visible keyboard focus, a light and a dark theme, and animations are turned off when the device asks for reduced motion. Every page works on screens from 360 px phones up to desktops.

## 8.9 Design decisions and justification

**Table 8.2:** Important design decisions

| Decision | Alternatives considered | Reason |
|---|---|---|
| React + TypeScript + Vite SPA | Server-rendered (Next.js), plain JS | Typed domain models catch errors at build time. Vite gives fast builds. The team already knew React. An SPA is enough because there is no SEO requirement. |
| Supabase (PostgreSQL, Auth, Storage) instead of a custom API server | Node/Express + MySQL; Firebase | Relational integrity (FKs, CHECKs, transactions) suits a workflow with history. RLS puts authorisation next to the data. Managed Auth avoids storing passwords ourselves. The free tier fits the project. |
| Business writes only through SECURITY DEFINER RPCs; direct table DML revoked | Direct table updates with RLS UPDATE policies | A status change must update the complaint, history and notification atomically with transition rules and locks. Generic UPDATE policies can't express "only valid transitions". |
| Optimistic concurrency (expected_version, error PT409) | Last-write-wins | Two Officers editing the same complaint must not overwrite each other silently. PT409 maps to HTTP 409 without PostgREST retry loops (defect DEF-01). |
| Append-only history + durable notifications in PostgreSQL | Realtime push, email/SMS | Meets CF-13/18 within the scope constraint on external notifications. History is the audit trail. |

| Decision | Alternatives considered | Reason |
|---|---|---|
| Memory-only sessions (persistSession: false) | localStorage tokens | Avoids leaving bearer tokens in persistent storage on shared computers. Trade-off: a page refresh requires signing in again (documented). |
| Private evidence bucket with server-reserved paths | Public bucket, signed URLs | Complaint evidence may contain personal information. Paths can't be chosen by users, and downloads are authorised per request. |
| Department-scoped Officers; assigned-Officer-only progress/resolution | Any Officer can edit any complaint | Least privilege and clear accountability, as the problem statement asks ("identifying responsible officers"). |
| Categories deactivated, never deleted; referenced department immutable | Hard delete / free edit | Historic complaints must keep their original routing and meaning. |
| Leaflet + OpenStreetMap with numeric coordinate fallback | Google Maps | No API key or cost. Accessible fallback when tiles fail. The OSM attribution policy is respected. |
| Vercel static hosting | Netlify, GitHub Pages | Simple Vite deployment, SPA rewrites and security headers configured in vercel.json. |
| Auto-assignment + Administrator dispatch (CR-006) | Manual pick-up only; giving Administrators full workflow rights | Guarantees an owner whenever staff exist, and gives a narrow, audited escape hatch when they don't, without making Administrators process complaints. |
| motion library + CSS tokens | Hand-written CSS only; heavy UI kits | Declarative, interruptible animations that respect reduced motion. Design tokens keep light and dark themes consistent. |
| Vitest + PGlite + Playwright | Jest only, manual testing | Tests each layer: pure rules, real SQL/RLS inside PostgreSQL, and user journeys in a real browser. |

# 9 Implementation

## 9.1 Technology stack

**Table 9.1:** Technology stack as built

| Layer | Technology (pinned in package.json) |
|---|---|
| UI | React 19, React Router 7, Tailwind CSS 4, lucide-react icons |
| Language / build | TypeScript 6 (strict), Vite 8 (Rolldown) |
| Validation | Zod 4 (client); SQL CHECK constraints and RPC validation (server) |
| Maps | Leaflet 1.9 + React Leaflet 5, OpenStreetMap tiles |
| Backend | Supabase: PostgreSQL 17, Auth, Storage, one Deno Edge Function |
| Tests | Vitest 5 (unit), PGlite (embedded PostgreSQL for database tests), Playwright (browser) |
| CI / hosting | GitHub Actions; Vercel configuration (vercel.json) |

We chose these technologies because they suit the requirements. TypeScript finds many errors before the code runs. PostgreSQL gives us relational integrity and transactions, which a workflow with a full history needs. Row Level Security keeps the access rules next to the data. Because Supabase manages authentication, we never store passwords ourselves, and the free tiers fit the budget constraint. Section 8.9 lists the alternatives we considered.

## 9.2 Module structure

**Table 9.2:** Module map

| Module | Path | Responsibility | Requirements |
|---|---|---|---|
| App shell | src/app/App.tsx | Routes, role navigation, protected layout, unread badge | CF-02, CF-03 |
| Auth | src/features/auth | Register, login/logout, recovery, invitation password, session/profile loading | CF-01–03 |
| Profiles | src/features/profiles | Own display name | CF-15 |
| Complaints | src/features/complaints | Form, submission, list, detail, withdrawal | CF-04–07, 17, 18, 22 |
| Evidence | src/features/evidence | Private reserve → upload → finalize, download | CF-16 |
| Officer | src/features/officer | Queues, filters, review/assign/priority/start/progress/transfer/resolve | CF-08–12, 17 |
| Feedback | src/features/feedback | Post-resolution rating | CF-19 |
| Notifications | src/features/notifications | Updates inbox, read state, unread badge | CF-13 |
| Categories | src/features/categories | Admin category maintenance | CF-20 |

| Module | Path | Responsibility | Requirements |
|---|---|---|---|
| Admin | src/features/admin | User management, Officer invitation, monitoring overview | CF-14, 21 |
| Shared | src/shared | One Supabase client, environment validation, errors, location, UI primitives | — |
| Database | supabase/migrations (15 files) | Schema, RLS, RPCs, Storage policies | All |
| Edge Function | supabase/functions/provision-officer | Admin-only Officer invitation using the Auth Admin API | CF-14 |

## 9.3 Core workflows implemented end to end

Each core use case works end to end: from the screen, through the application hook and the data adapter, to a protected database function, with tests at every level. The screenshots below were taken from the running system using fictional data.

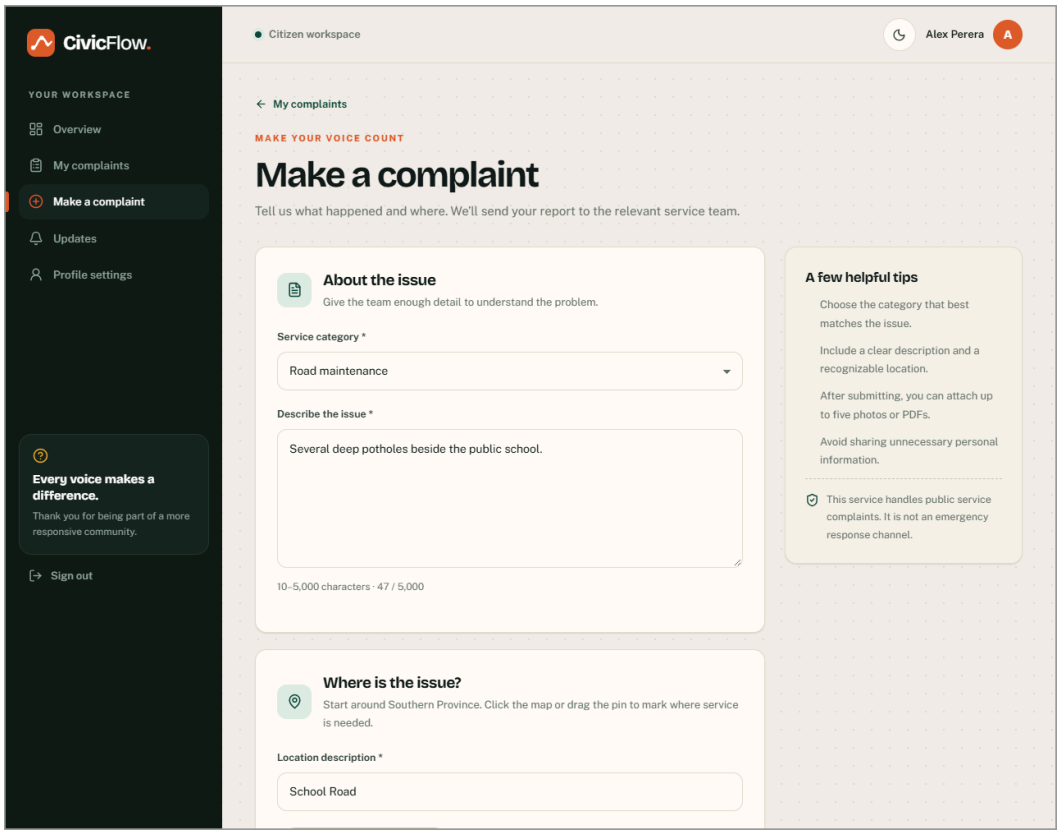



**Figure 9.1:** Make a complaint screen (Citizen): category, description, OpenStreetMap location and typed coordinates

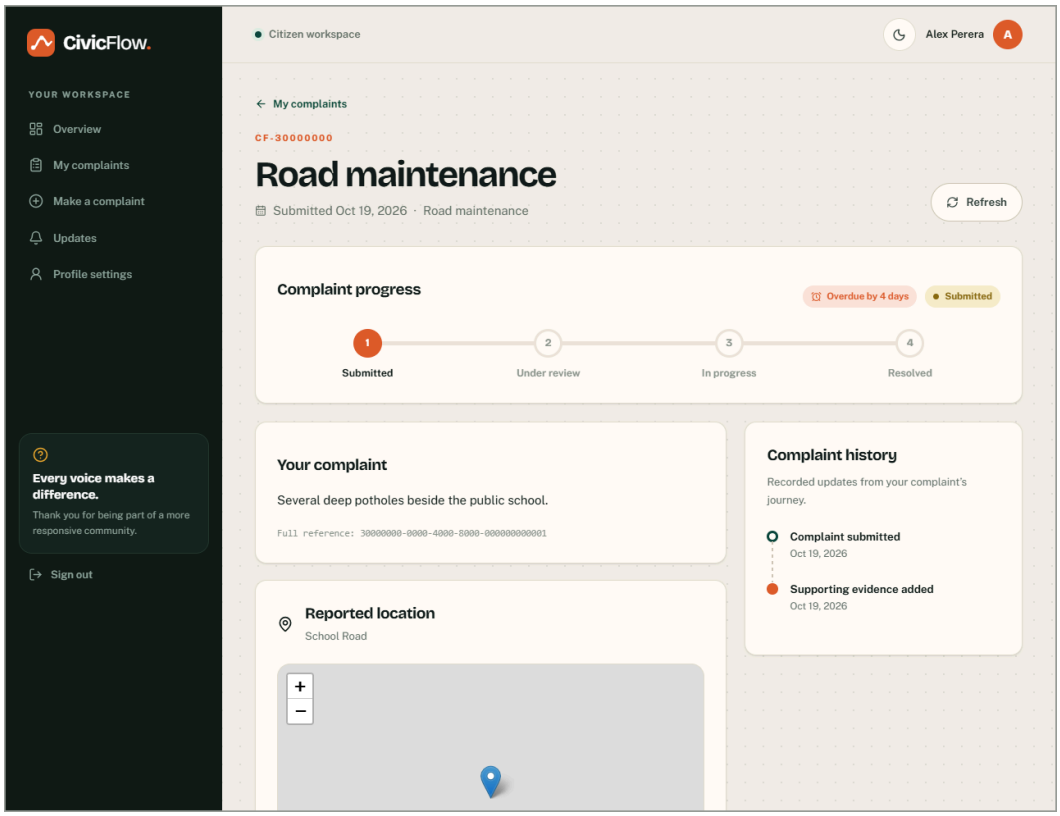



**Figure 9.2:** Complaint detail (Citizen) with the progress line, location, history and service-target badge

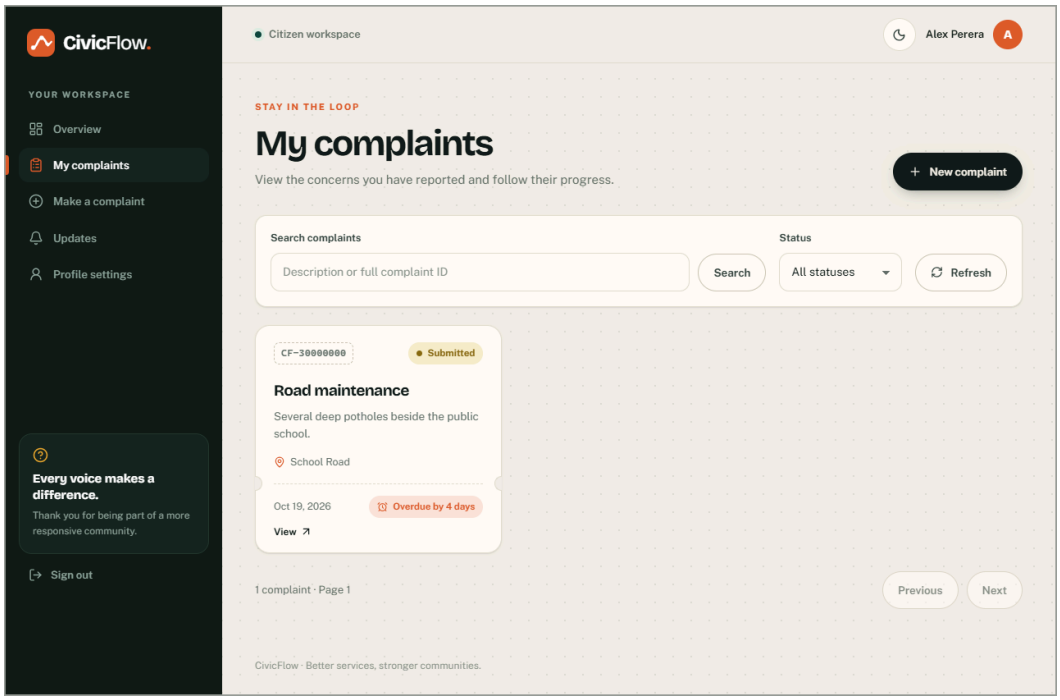



**Figure 9.3:** My complaints (Citizen): each complaint is shown as a ticket with its status and a due or overdue badge

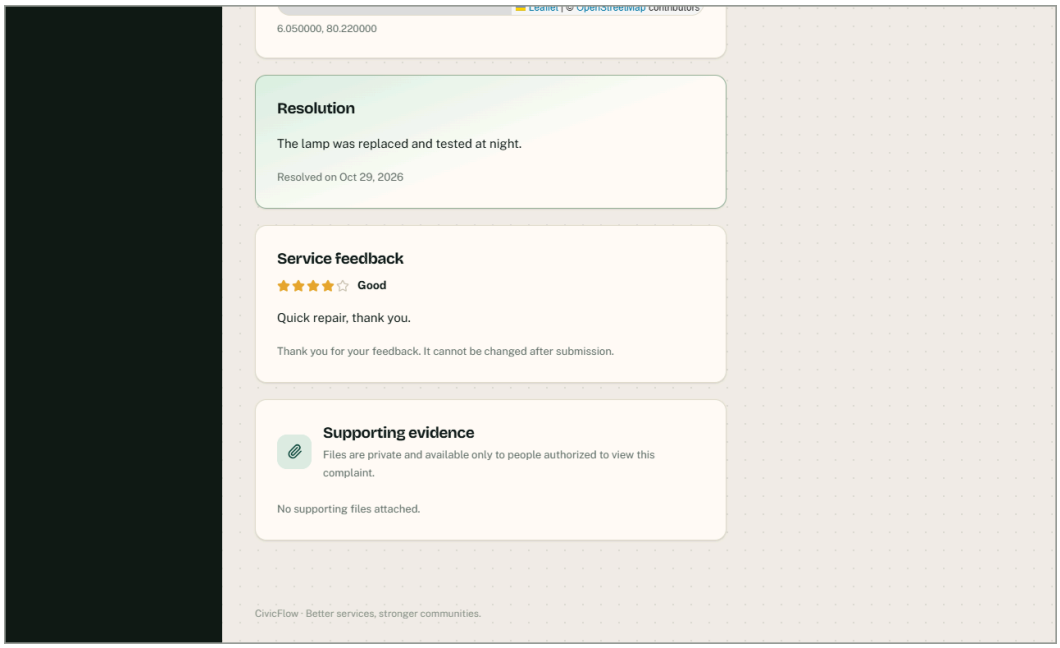



**Figure 9.4:** Resolution details and the one-time service feedback form (Citizen)

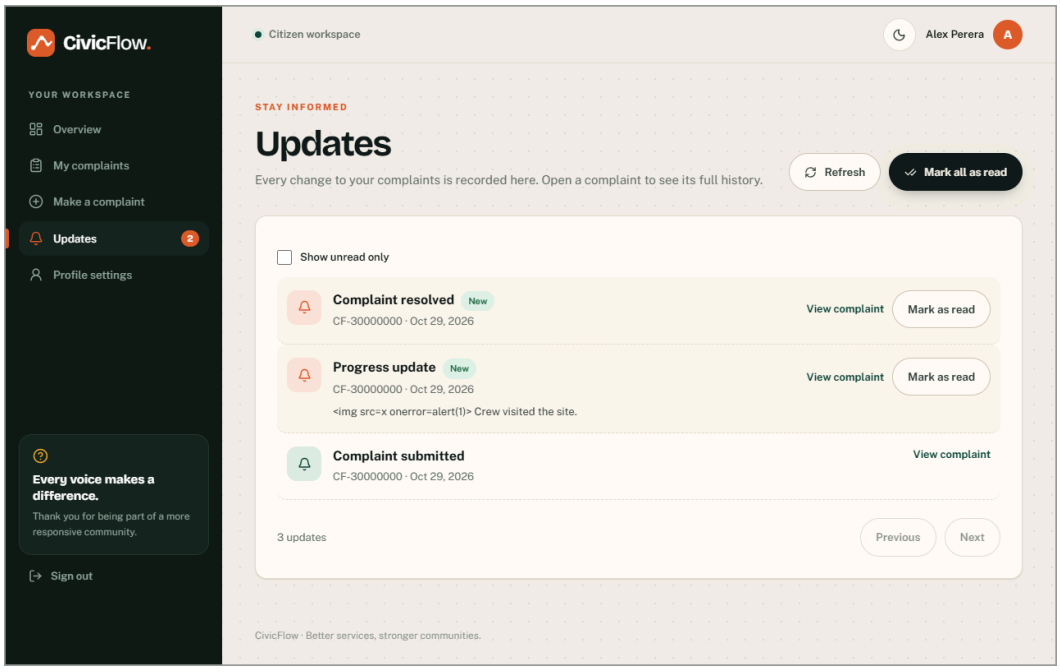



**Figure 9.5:** Updates inbox (Citizen) with the unread count and mark-as-read

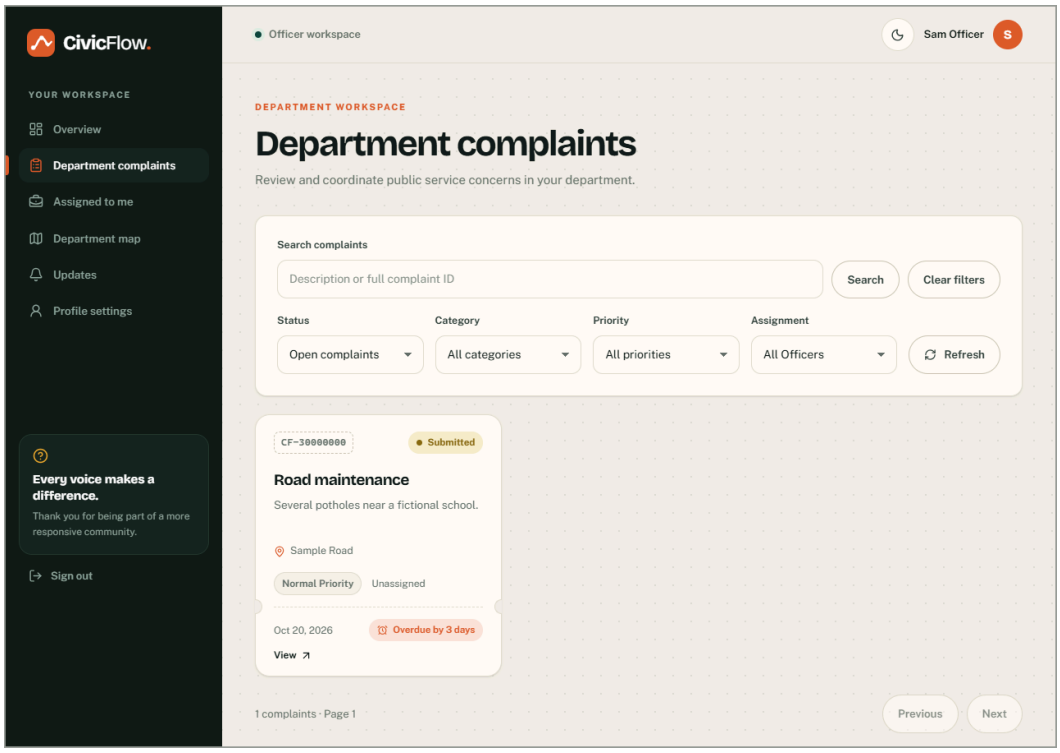



**Figure 9.6:** Department queue (Officer) with filters for status, category, priority and assignment

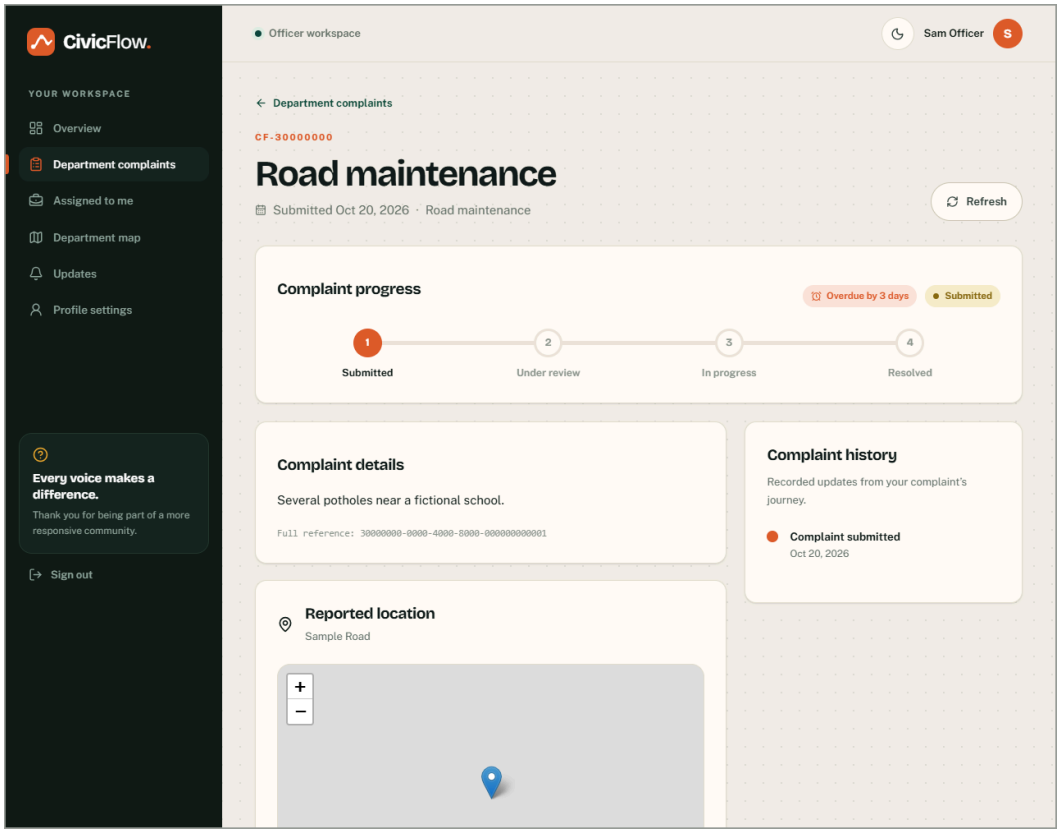



**Figure 9.7:** Processing a complaint (Officer): only the valid next actions are offered

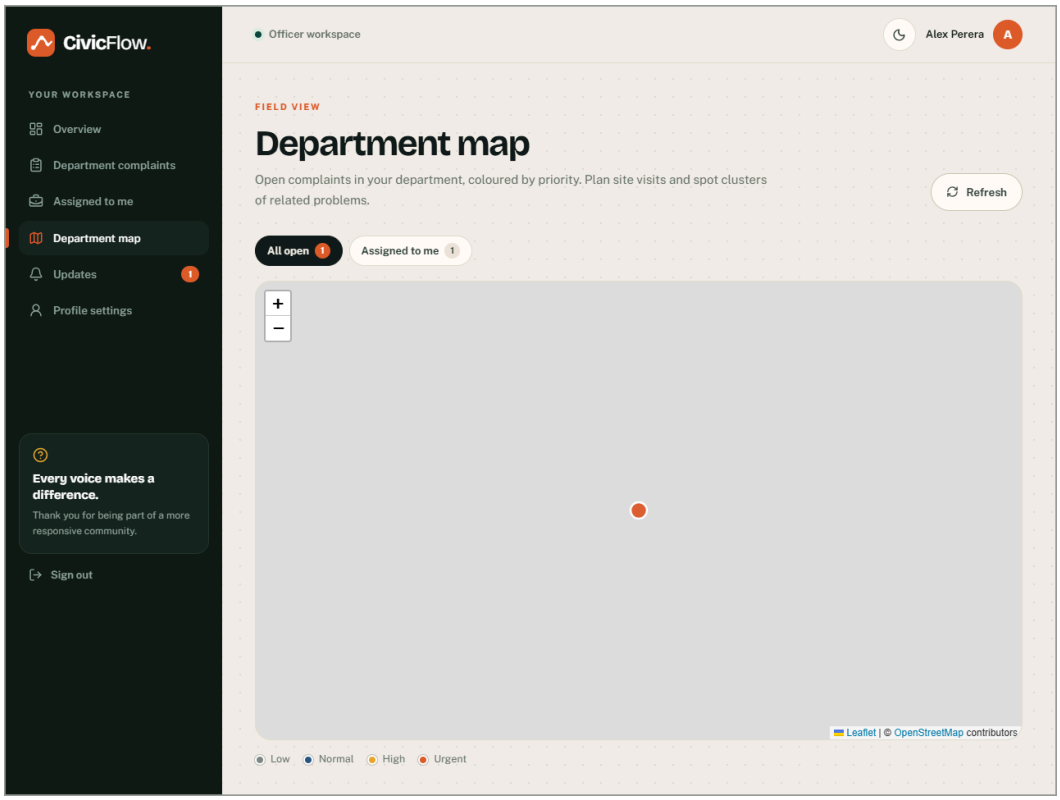



**Figure 9.8:** Department map of open complaints (Officer)

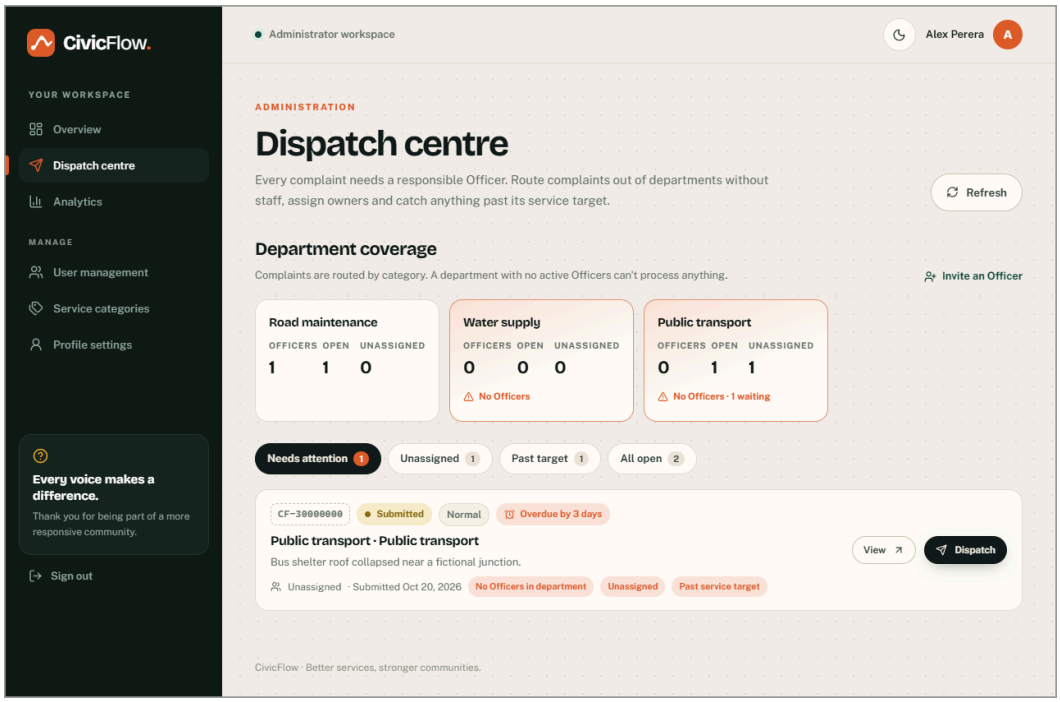



**Figure 9.9:** Dispatch centre (Administrator) showing department coverage and complaints that need attention

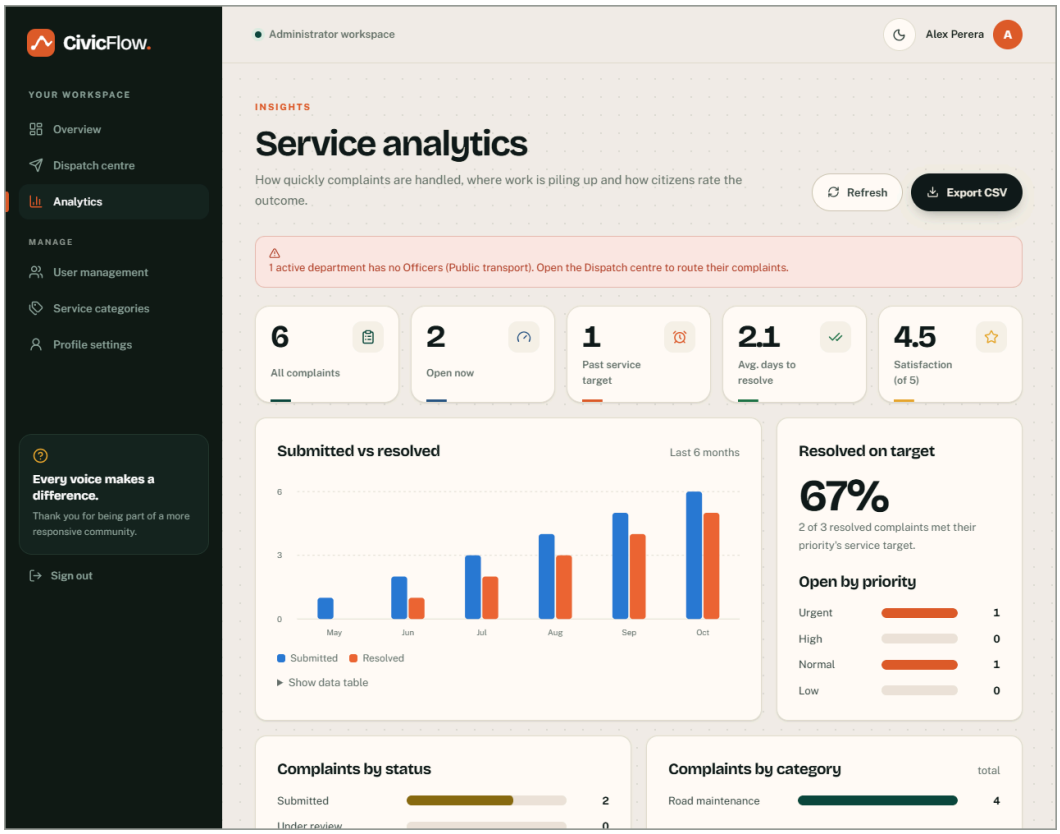



**Figure 9.10:** Service analytics (Administrator): targets, coverage, workload, satisfaction and the six-month trend, with CSV export

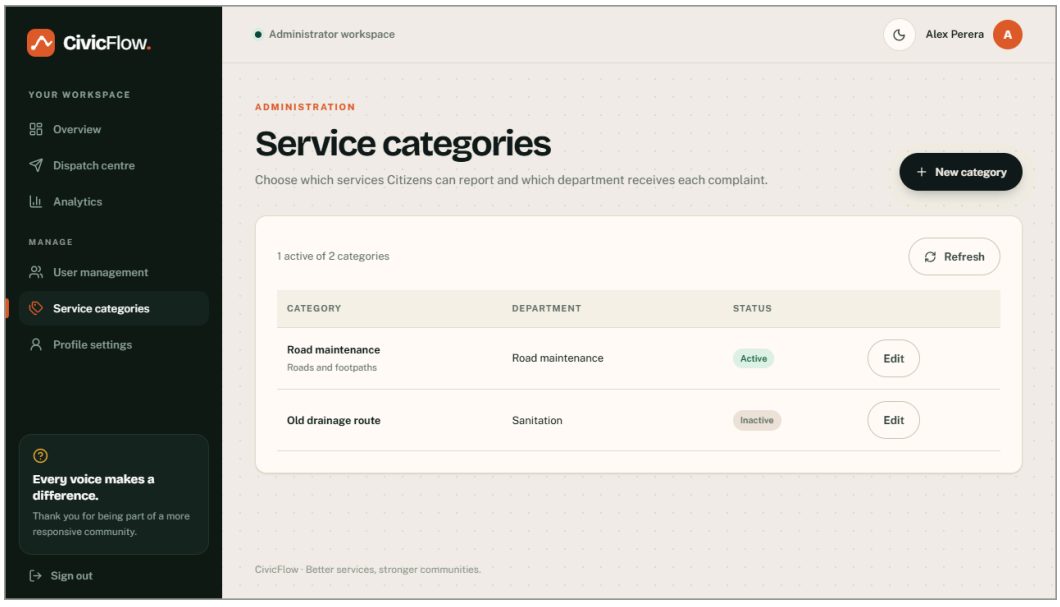



**Figure 9.11:** Service category management (Administrator)

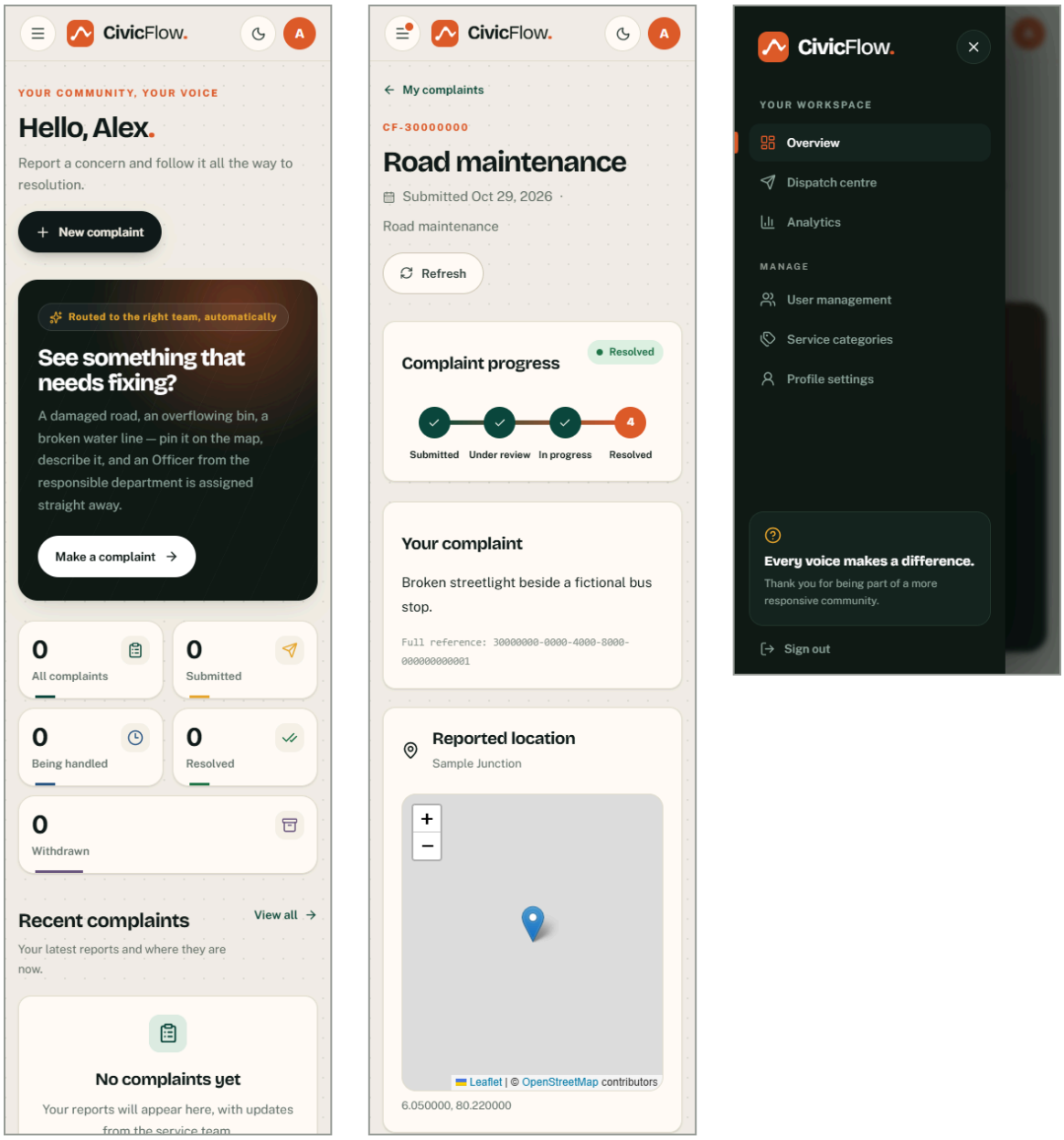



**Figure 9.12:** Citizen overview, feedback form and slide-out menu on a 360 px wide phone

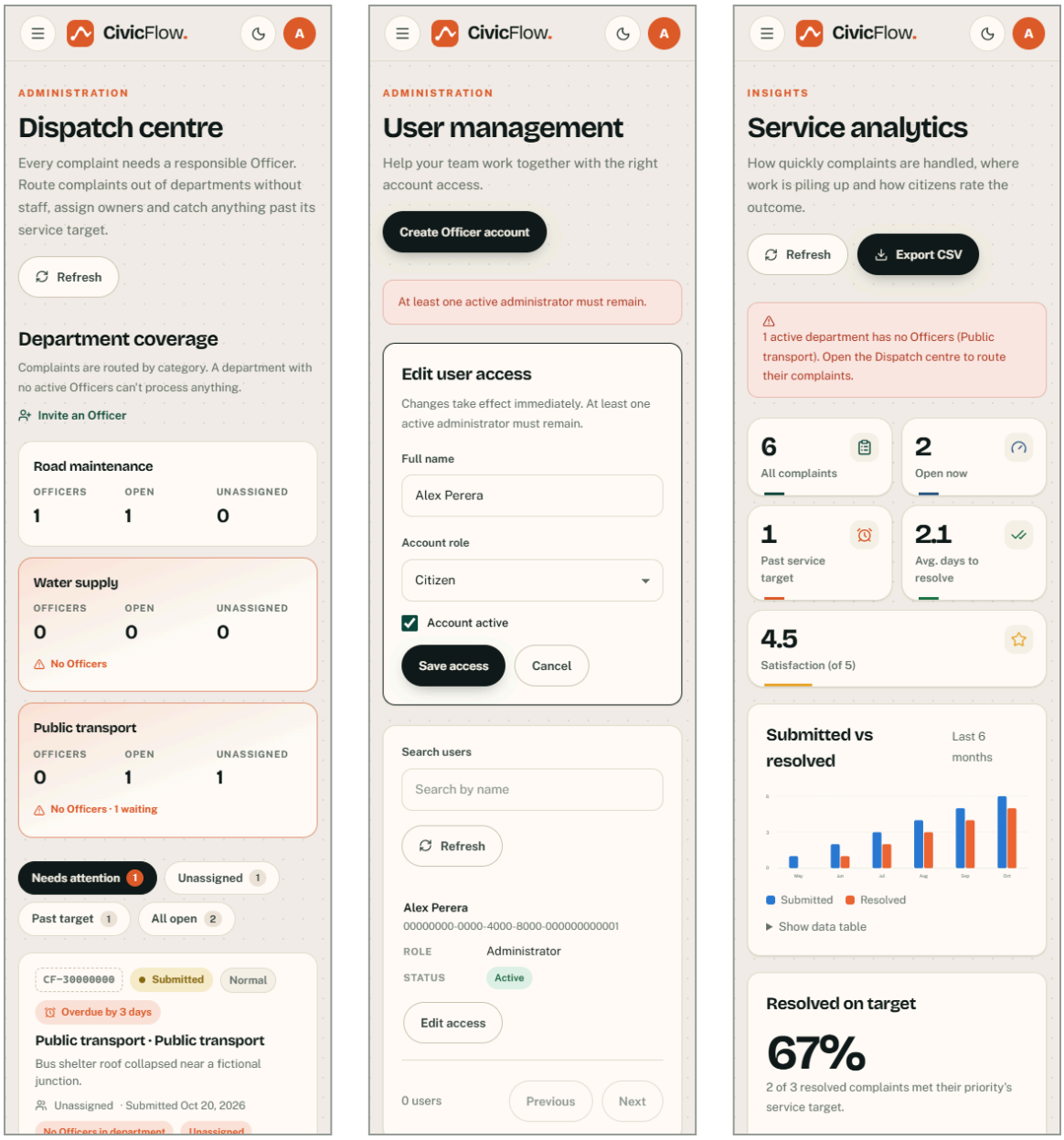



**Figure 9.13:** Dispatch centre, user management and analytics on a 360 px wide phone

## 9.4 Input validation

Every input is checked twice. Zod schemas in the browser give the user immediate feedback, and CHECK constraints and function checks in the database make the final decision. Our tests check that both layers use the same limits.

**Table 9.3:** Input validation in two layers

| Field | Client rule (Zod) | Database rule |
|---|---|---|
| Complaint description | trimmed, 10–5,000 | CHECK + RPC check |
| Location text | trimmed, 1–300 | CHECK + RPC check |
| Coordinates | finite, lat −90…90, lon −180…180 | CHECK + RPC check |
| Withdrawal reason | trimmed, 10–2,000, confirmation | RPC check + table constraint |

| Field | Client rule (Zod) | Database rule |
|---|---|---|
| Progress remark / resolution | 1–2,000 / 10–5,000 | RPC checks + CHECK |
| Evidence | JPEG/PNG/PDF magic bytes, ≤ 5 MiB, ≤ 5 files | Declared type/size CHECK, bucket MIME/size limits, quota under lock, stored metadata verified at finalize |
| Feedback | integer 1–5, comment ≤ 2,000 | RPC check + CHECK |
| Category | name 1–100, description ≤ 500, department required | RPC check + CHECK + unique name |
| Password | ≥ 12 characters at registration | Supabase Auth minimum length 12 |

## 9.5 Error handling

- Database functions return fixed error codes such as not_authorized, invalid_input, stale_version and feedback_exists. Each feature turns these codes into plain messages. Unknown errors show a general message and never reveal server details.
- If someone else changed the complaint first (HTTP 409), the user's input is kept and they are asked to refresh. This request is never retried automatically.
- Creating and withdrawing a complaint use idempotency keys, so the request can be sent again safely if the response is lost.
- When only part of an action succeeds, the user is told exactly what happened, for example "Your complaint is saved, but the file has not uploaded", with a Retry button.
- If a database migration is missing, the user sees an explanation instead of a crash. Authorisation errors are still treated as refusals.

## 9.6 Coding standards and version control

- Strict TypeScript with no any; database types generated from the migrations and checked for drift in CI.
- ESLint with zero warnings; snake_case in SQL, camelCase/PascalCase in TypeScript; feature folders named after the domain.
- SQL functions are SECURITY DEFINER with an empty search_path, fully qualified names, explicit EXECUTE grants and stable error codes.
- No string-built HTML: all user text is rendered as text by React.
- Git history on GitHub (https://github.com/razim2003/civic-flow) with feature branches per increment (for example sprint-4/feedback-inbox-categories, sprint-5/dispatch-and-experience, ui/responsive-layout) merged to main only when CI is green. Applied migrations are never edited; fixes are new migrations.

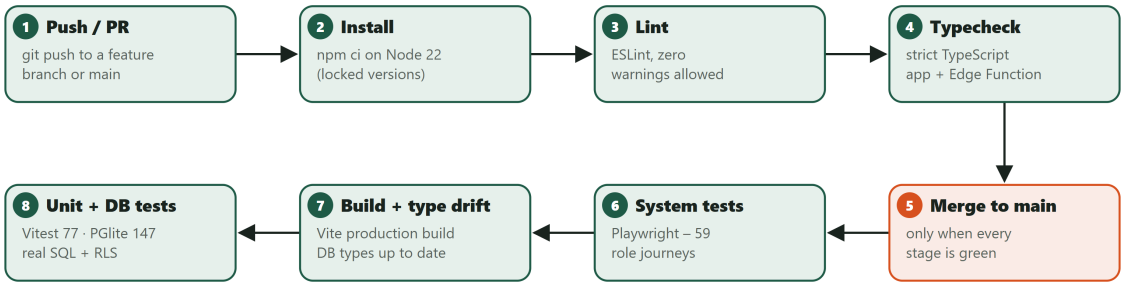



**Figure 9.14:** Continuous-integration pipeline that runs on every push

# 10 Testing and Quality Assurance

## 10.1 Test plan

Testing had three aims: to show that every requirement works as specified, that the backend itself refuses unauthorised users (hiding a button is not enough), and that every defect is fixed and tested again. The plan was based on risk. The biggest risks were data leaking between citizens or departments, skipping steps in the workflow, lost updates and invalid data. We tested these in the database with real SQL run as ordinary users without special privileges, and again through the user interface.

**Table 10.1:** Test levels

| Level | What is tested | Tool | Environment |
|---|---|---|---|
| Unit | Pure validation rules, error translation, service-level targets, analytics maths, session handling, Edge Function handler | Vitest | Node / jsdom |
| Integration (database) | All 15 migrations, grants, RLS, RPCs, constraints, rollbacks and concurrency, run as anon / authenticated users with fixture identities | Vitest + PGlite (PostgreSQL 17 in-process) | Fresh disposable database per test file |
| Integration (HTTP) | Real Supabase Auth JWTs + PostgREST + PostgreSQL in containers; concurrent race conditions | Node script + Docker | Disposable local containers |
| System | Complete role journeys through the real UI in a browser, with every backend response controlled | Playwright (Microsoft Edge / Chromium) | Local dev server |
| Acceptance | Acceptance criteria of the major user stories, checked by mapped system tests, plus a smoke test on the deployed system | Playwright + manual checklist | Local and deployed URL |

Entry criteria: the code compiles and lint passes. Exit criteria: all automated suites pass, no high-severity defect is open, and every Must requirement has at least one passing positive test and one passing negative test. All test data is fictional, for example "Alex Perera", "Sample Road" and @example.test e-mail addresses.

## 10.2 Execution results

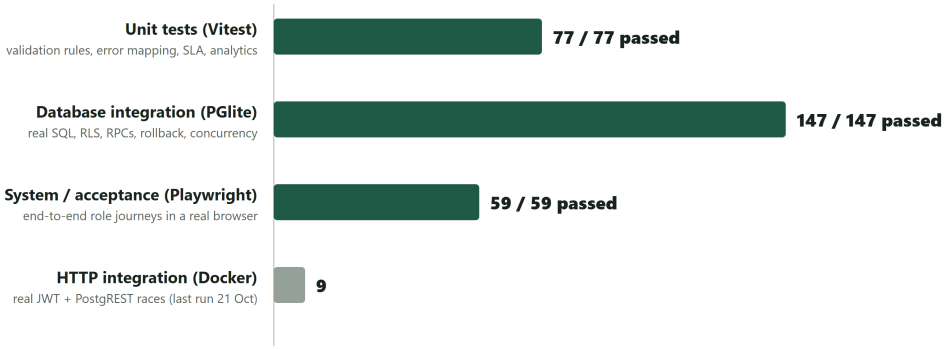



**Figure 10.1:** Automated test results by level (latest run, 30 October 2026)

**Table 10.2:** Execution summary

| Suite | Files | Tests | Passed | Failed |
|---|---|---|---|---|
| Lint / typecheck / production build | — | — | ✓ | — |
| Unit (Vitest) | 13 | 77 | 77 | 0 |
| Database integration (PGlite, 15 migrations + demo data) | 9 | 147 | 147 | 0 |
| System (Playwright, Edge) | 5 | 59 | 59 | 0 |
| HTTP integration (Docker) | 1 | 9 | 9 (21 Oct) | 0 |
| Total executed in the latest run | 27 | 283 | 283 | 0 |

The documented test cases contain 139 entries: 57 of them include positive checks and 89 include negative checks (misuse or invalid input). We also ran a layout audit that rendered 24 key screens at 360, 414, 768 and 1280 px and checked automatically for sideways scrolling and content spilling out of its box. After the Sprint 5 fixes it found no problems.

## 10.3 Representative test cases

The table below shows a selection of test cases from all levels. Appendix B has the complete list with preconditions, expected results and actual results.

**Table 10.3:** Representative test cases

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-AUTH-02 | CF-01 | +/− | Submit an empty form, then valid data | Field errors first; signup request carries only display_name (no role); "Check your email" shown | As expected | Pass |
| TC-AUTH-04 | CF-03 | − | Citizen types /admin/users | "Administrator access required"; no admin data requested | As expected | Pass |
| TC-DB-FND-02 | CF-03, NFR-05 | + | Inspect catalog | RLS enabled on every application table | As expected | Pass |

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-SUB-01 | CF-04, 06, 16, 18 | + | Citizen submits a valid complaint; first upload fails, retry; download; open detail | Reference shown; request has no citizen_id/status; upload retry reuses the same key; private download; history shows "Supporting evidence added" | As expected | Pass |
| TC-SUB-02 | CF-04 | − | Blank coordinates; map tiles unavailable | Validation error; map click selects; numeric entry still works | As expected | Pass |
| TC-TRK-03 | CF-07, NFR-11 | − | Open another citizen's reference; description containing HTML | "Complaint unavailable"; HTML shown as text | As expected | Pass |
| TC-DB-EV-03 | CF-16 | − | Declared size 0 / > 5 MiB; MIME text/html | Rejected | As expected | Pass |
| TC-OFF-01 | CF-09–12 | + | Officer reviews, assigns, prioritises, starts, adds progress, resolves with confirmation | Each step "Complaint updated"; expected versions 1→6 sent; complaint closed with resolution text | As expected | Pass |
| TC-OFF-03 | NFR-06 | − | Another user changed the complaint (409) | Remark kept; "Refresh and review"; retry after refresh uses the new version | As expected | Pass |
| TC-DB-OFF-05 | CF-11, NFR-06 | − | Skip/repeat transitions; stale version; no-ops | invalid_transition; PT409 stale_version; no-op adds no event | As expected | Pass |
| TC-DB-OFF-08 | NFR-06 | − | Notification insert forced to fail during review | Status/version and history unchanged | As expected | Pass |
| TC-FB-02 | CF-19 | − | Server answers feedback_exists | "Feedback was already submitted…"; form stays; no false success | As expected | Pass |
| TC-NTF-01 | CF-13 | + | Two unread, one read update; mark one; filter unread; mark all | Badge "2 unread" → "1 unread" → none; HTML remark shown as text; exact RPC payloads | As expected | Pass |
| TC-CAT-02 | CF-20 | − | Server answers category_in_use | Explanation shown; input kept | As expected | Pass |
| TC-DSP-01 | CF-24 | +/− | Admin opens the Dispatch centre with a complaint in an unstaffed department; dispatches with no reason, then with a reason | Coverage card shows "No Officers"; row flagged unstaffed, unassigned and past target; reason error with no request; then exactly one RPC with version 1 and a success message | As expected | Pass |
| TC-UX-03 | NFR-03, NFR-04 | + | 1280 px, then 768 px and 360 px. Open the menu, press Escape, then open each admin page from the menu | Desktop shows the sidebar with no menu button. Tablet/phone hide the sidebar links behind Open menu (aria-expanded). Focus moves to Close menu and returns to the button on Escape. Choosing a page closes the menu. No page is wider than the screen | As expected | Pass |
| TC-DB-DSP-07 | CF-24 | − | Short reason, mismatched route, foreign/citizen assignee, stale/null version, withdrawn | reason_required, category_unavailable, invalid_assignee, PT409, invalid_input, complaint_closed | As expected | Pass |

## 10.4 Acceptance testing

**Table 10.4:** Acceptance test summary

| Story | Acceptance criteria | Result |
|---|---|---|
| US-04/05/06 Submit complaint | AC1–AC4 | Pass (TC-SUB-01…03, TC-DB-SUB-01…04) |
| US-07 Track complaint | AC1–AC3 | Pass |

| Story | Acceptance criteria | Result |
|---|---|---|
| US-11 Update status | AC1–AC4 | Pass |
| US-12 Resolve | AC1–AC4 | Pass |
| US-13 Updates inbox | AC1–AC3 | Pass |
| US-14 Manage users | AC1–AC4 | Pass |
| US-18 Feedback | AC1–AC3 | Pass |
| US-19 Categories | AC1–AC4 | Pass |
| US-20 Withdrawal | AC1–AC2 | Pass |
| Manual UAT on the deployed system | Smoke checklist (Section 11.6) | S1–S3 passed on the live URL (30 Oct 2026); signed-in checks S4–S12 are run by the team before the demonstration |

## 10.5 Defects identified and corrected

We recorded 19 defects. For each one we found the root cause, fixed it and tested it again. The most important were DEF-01, an endless retry loop when two users edited the same complaint (found by the HTTP integration test); DEF-13, complaints in departments without Officers that nobody could handle (found while using the hosted system and fixed through change request CR-006); and DEF-16 to DEF-19, layout problems such as feedback options spilling out of their boxes and the sidebar squeezing the page on tablets (found by the layout audit).

**Table 10.5:** Defect log

| ID | Found | How found | Severity | Description | Root cause | Correction | Re-test |
|---|---|---|---|---|---|---|---|
| DEF-01 | Sprint 3 (2026-10-21) | HTTP integration test (concurrent review) | High | A stale-version conflict kept the request in an endless retry loop | PostgREST 14 automatically retries SQLSTATE 40001 | Raise PT409 (HTTP 409). A new corrective migration 20261021000100 replaced the withdrawal function; Officer helpers use PT409 | TC-INT-03/04, TC-DB-OFF-05 pass |
| DEF-02 | Sprint 2 (2026-10-19) | Hosted preflight / manual demo | High | Citizen category dropdown was empty | Category migration 006 not yet applied to the hosted DB | Applied the reviewed migration; added an explicit "missing setup" state instead of a silent empty list | TC-SUB-05 pass |
| DEF-03 | Sprint 2 | Playwright | Medium | Duplicate React sibling key warning in the history list | Two events rendered with the same key | Keys use stable event IDs | Browser suite pass |
| DEF-04 | Sprint 2 | Playwright | Medium | Race between refreshing a complaint and submitting a withdrawal could use an old version | Refresh and submit not serialised | Controls disabled while refreshing; in-flight guard | TC-WD-02 pass |
| DEF-05 | Sprint 1/3 | Operator review | Medium | supabase config push reset hosted Auth settings (TOTP, OTP length, email frequency) to defaults | CLI pushes the whole local config | Values restored immediately; local config.toml now records them | Verified by later push (no drift) |

| ID | Found | How found | Severity | Description | Root cause | Correction | Re-test |
|---|---|---|---|---|---|---|---|
| DEF-06 | Sprint 3 | Playwright | Low (test) | Officer return-navigation test clicked before the detail page loaded | Test timing | Test waits for the detail return link | TC-OFF-09 pass |
| DEF-07 | Sprint 3 | Playwright (cold start) | Low (test) | First redirect assertion exceeded 5 s on a cold Vite start | Test timeout too short for first compile | Wait for login readiness (15 s bound) | TC-AUTH-01 pass |
| DEF-08 | Sprint 3 | Unit run on Windows | Low (tooling) | Unit worker start-up timeouts | Too many jsdom workers | Pure tests run in Node; worker count bounded | Unit suite pass |
| DEF-09 | Sprint 4 (2026-10-29) | Production build warning | Medium | The app shell imported the notifications module both statically and lazily, pulling complaint pages into the first download | The unread-badge hook was in the same module as the inbox page | Separate lightweight notifications/badge entry and zod-free unread API | Build with no warnings; TC-NTF-01 pass |
| DEF-10 | Sprint 4 | Database regression suite | n/a (expected) | foundation.test.ts failed: feedback and notifications became readable | The guard intentionally asserted "closed until policies ship" | Guard updated to check authorised reads with writes still denied | 119/119 pass |
| DEF-11 | Sprint 4 | Local set-up | Low (environment) | vitest could not start: missing Rolldown native binding | Dependencies installed with Node 20 (project requires ≥ 22.18) | Reinstalled with Node 22; requirement documented | All suites pass |
| DEF-12 | Sprint 4 | Screenshot review | Low (test) | Inbox fixture showed "Date unavailable" | Malformed fixture timestamp T010:00 | Fixture fixed. The app's graceful fallback for bad dates is correct behaviour. | TC-NTF-01 pass |
| DEF-13 | Sprint 5 (2026-10-30) | Hosted use / read-only data check | High | 4 of 5 open complaints sat in departments with no Officers and could never be handled; the Administrator could not act | Routing assumed every department is staffed; no assignment at submission; Administrator was read-only | CR-006: auto-assignment, Administrator dispatch, coverage flags (migration 015) | TC-DB-DSP-01… 10, TC-DSP-01 pass |
| DEF-14 | Sprint 5 | Screenshot review | Medium | Analytics bars below the fold stayed empty until scrolled into view, so printouts and screenshots under-reported data | Bars animated on scroll-into-view | Bars animate on mount; scroll reveals kept only for decorative sections | TC-DSP-02 pass; screenshots re-reviewed |
| DEF-15 | Sprint 5 | Screenshot review | Low | Sign-in route illustration clipped its end labels; ticket notches misaligned with the perforation | Coordinates too close to the viewBox edge; notch tied to card bottom | Route moved inward; notches anchored to the perforated line | Visual re-check |
| DEF-16 | Sprint 5 (2026-10-30) | User report + layout audit | Medium | Feedback rating options spilled out of their boxes at every width above 414 px | Rating cards had a 150 px minimum, narrower than the radio, five stars and label combined | One aligned option per row (radio, label, stars); stars drop below the label on very small phones | TC-FB-01, TC-UX-04 |
| DEF-17 | Sprint 5 | Layout audit | Medium | On tablets the full sidebar stayed open, squeezing pages into ~500 px: two thin detail columns, stat cards cut off, long words breaking | The sidebar only collapsed below 760 px, and then into wrapped link chips | Below 1100 px the sidebar is an accessible slide-out menu opened from a sticky header. Detail pages use one column below 1180 px, and stat cards auto-fit | TC-UX-03, TC-UX-04 |

| ID | Found | How found | Severity | Description | Root cause | Correction | Re-test |
|---|---|---|---|---|---|---|---|
| DEF-18 | Sprint 5 | Layout audit | Medium | User management and Service categories scrolled sideways on phones (up to 134 px) | Wide no-wrap tables | Admin tables become stacked, labelled rows on phones | TC-UX-03, TC-UX-04 |
| DEF-19 | Sprint 5 | Layout audit | Low | Ticket notches extended outside the card; heading links ("Invite an Officer") were squeezed into a thin column on phones | Notches drawn as full circles and clipped; heading rows could not wrap | Half-circle notches drawn inside the card; heading rows wrap and their links do not | TC-UX-04 |

## 10.6 Limitations of the test evidence

- The browser tests replace the backend with controlled responses, so they check the interface and the exact requests it sends. Authorisation is checked separately by the database suite, using real SQL as anonymous and signed-in users.
- PGlite uses fixture Auth and Storage schemas; the Storage HTTP service (actual file bytes) was not integration-tested.
- The Docker HTTP suite last ran on 21 October 2026, before the Sprint 4–5 SQL; the same locking pattern is covered by the database suite.
- We did not test with real citizens or Officers; usability was judged by the team.

# 11 Deployment

CivicFlow is live at https://civic-flow-rho.vercel.app. The frontend is a static build served from Vercel's CDN, and the backend is the Supabase project xapqfnumcpjckhpdyuxz (PostgreSQL 17, Auth, Storage and one Edge Function).

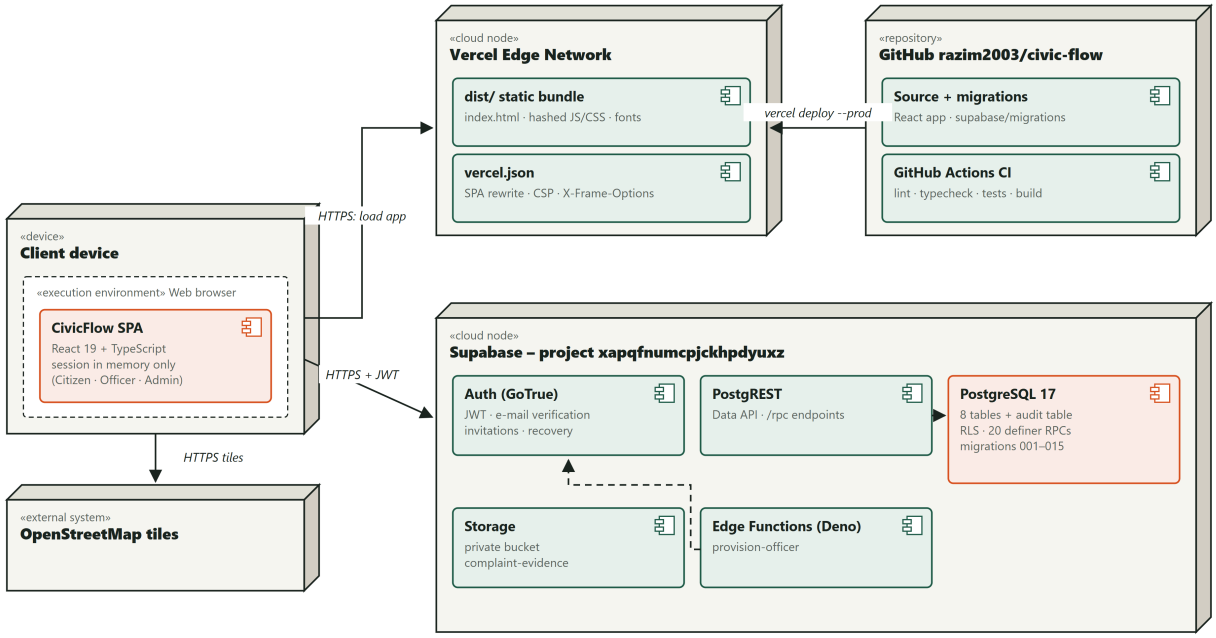



**Figure 11.1:** UML deployment diagram of the production system

## 11.1 Deployed architecture and database configuration

- Frontend: a static bundle with hashed file names, served by Vercel. No server code runs on Vercel and the browser holds no secrets. vercel.json adds rewrites for deep links and the security headers Content-Security-Policy, X-Frame-Options DENY, nosniff and Referrer-Policy.
- Backend: Supabase. The database schema comes only from the 15 migrations in the repository. Before applying each migration to the hosted database we reviewed the target, the SQL, the expected effects and the validation and recovery steps, and we compared data checksums before and after.
- The browser needs only two public values, VITE_SUPABASE_URL and VITE_SUPABASE_PUBLISHABLE_KEY. They are safe to publish because access is controlled by Row Level Security.
- The Officer-invitation Edge Function is the only place that uses the privileged key. It runs on the server and uses the CIVICFLOW_APP_URL secret so that invitation links open the live site.

## 11.2 System requirements

**Table 11.1:** System requirements

| Component | Requirement |
|---|---|
| End users | A current Chrome, Edge, Firefox or Safari; JavaScript enabled; internet access. Works on phones (≥ 375 px wide). |
| Developer machine | Node.js ≥ 22.18 with npm 10; Git; optional Docker Desktop (for test:integration) and the Supabase CLI |
| Hosting | A Vercel account linked to the GitHub repository; a Supabase project |

## 11.3 Configuration

**Table 11.2:** Configuration settings

| Setting | Where | Value / note |
|---|---|---|
| VITE_SUPABASE_URL | .env.local (local) / Vercel Environment Variables | https://xapqfnumcpjckhpdyuxz.supabase.co |
| VITE_SUPABASE_PUBLISHABLE_KEY | same | The project's publishable key (sb_publishable_…). Never use the secret or service-role key. |
| Auth → URL configuration | Supabase dashboard | Site URL = the deployed URL. Redirect allow-list: <deployed>/auth/callback, <deployed>/reset-password, <deployed>/accept-invite, plus the local 127.0.0.1:5173 equivalents. |
| Auth → providers | Supabase dashboard | Email with confirmation on; minimum password length 12; anonymous sign-ins off |
| CIVICFLOW_APP_URL | Supabase Edge Function secrets | The deployed origin, so Officer invitation links point to it |
| SMTP | Supabase Auth | Custom SMTP is required to invite Officers at arbitrary email addresses. The default sender only delivers to project-team addresses. |
| Security headers | vercel.json (already in the repo) | CSP, X-Frame-Options: DENY, nosniff, Referrer-Policy |

## 11.4 Installation and deployment steps

1. Local installation: git clone the repository, run npm ci (Node.js 22.18 or later), copy .env.example to .env.local and set the two public values, then npm run dev and open http://127.0.0.1:5173.
2. Verification: npm run lint, npm run typecheck, npm run test, npm run test:db, npm run build and npx playwright test.
3. Database: review each new migration, take a read-only baseline, apply it to the Supabase project only after approval, then verify grants, RLS, anonymous refusal and unchanged checksums.
4. Frontend: create the Vercel project, add the two public environment variables for Production and Preview, and run vercel deploy --prod (or connect the GitHub repository for automatic deployments).
5. Post-deployment: set the Supabase Auth Site URL and redirect URLs (/auth/callback, /reset-password, /accept-invite) and the CIVICFLOW_APP_URL secret to the production URL, then run the smoke test.

## 11.5 Deployment verification (smoke test)

**Table 11.3:** Smoke-test and acceptance checklist on the deployed URL

| # | Check | Expected | Result |
|---|---|---|---|
| S1 | Open / and /workspace signed out | Redirect to sign-in | Pass 2026-10-30: / shows the landing page; /workspace redirects to /login |
| S2 | Deep link /complaints/new reloads | SPA loads (no 404) and asks to sign in | Pass 2026-10-30: HTTP 200, then redirected to sign-in; no console errors |
| S3 | Response headers | CSP, X-Frame-Options: DENY, nosniff present | Pass 2026-10-30: CSP, X-Frame-Options: DENY, nosniff and Referrer-Policy present |
| S4 | Register a demo Citizen; confirm email; sign in | Citizen workspace | Run by the team with the fictional demo accounts before the demonstration |
| S5 | Submit a complaint with a map location and a small PNG | Reference shown; file downloadable | Run by the team with the fictional demo accounts before the demonstration |
| S6 | Officer (demo) reviews, assigns, starts, adds progress, resolves | Each step in history | Run by the team with the fictional demo accounts before the demonstration |
| S7 | Citizen sees the Updates badge, opens the inbox, marks it read | Badge clears | Run by the team with the fictional demo accounts before the demonstration |
| S8 | Citizen gives feedback (rating 4) | Read-only feedback shown; Officer can see it | Run by the team with the fictional demo accounts before the demonstration |
| S9 | Admin creates a category; Citizen form shows it | Category appears | Run by the team with the fictional demo accounts before the demonstration |
| S10 | Admin overview counts | Match the demo data | Run by the team with the fictional demo accounts before the demonstration |
| S11 | Citizen opens another user's complaint URL | "Complaint unavailable" | Run by the team with the fictional demo accounts before the demonstration |
| S12 | Page load timing (DevTools, broadband) | NFR-02 targets met | Run by the team with the fictional demo accounts before the demonstration |

For the demonstration we loaded a fictional Sri Lankan data set into the hosted database: ten citizens, Officers for the departments that had none, and 36 complaints from places such as Nugegoda, Kandy, Jaffna, Galle and Trincomalee. The data was created through the normal workflow functions, so every screen shows realistic content. The demo accounts cannot sign in, and a tested script removes the data.

# 12 Maintenance and Change Request

Once the first version was working, we treated every change as a maintenance task and followed the same steps each time: record the request, analyse its impact, classify and estimate it, get approval, implement it on a branch with a new migration, run the full regression suite, update every affected document, and apply it to the hosted system only after explicit approval.

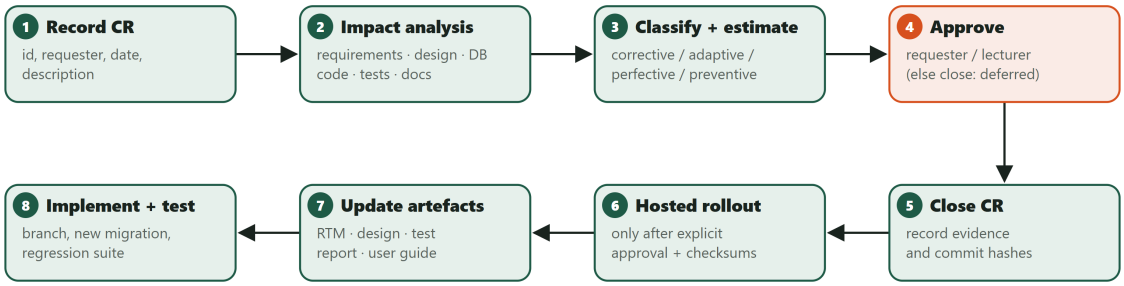



**Figure 12.1:** Maintenance process used for every change request

## 12.1 Change request register

**Table 12.1:** Change request register

| CR | Title | Requested by / date | Type | Status |
|---|---|---|---|---|
| CR-001 | Citizen can withdraw an unresolved complaint (CF-22) | Project owner, 2026-10-19 | Perfective | Implemented, deployed to DB, tested |
| CR-002 | Stale-version conflicts must not loop (PT409) | Found by testing, 2026-10-21 | Corrective | Implemented, applied, tested |
| CR-003 | Administrator-created Officer accounts (CF-14) | Project owner, 2026-10-20 | Perfective (+ preventive security) | Implemented, applied, tested |
| CR-004 | "Use my current location" and closer default map (CF-04) | Project owner, 2026-10-21 | Perfective (usability) | Implemented, tested |
| CR-005 | Lecturer-provided change request | Lecturer, date | to classify | Awaiting request |
| CR-006 | Every complaint must reach a responsible Officer (auto-assignment, dispatch, service targets) and product completion/redesign | Project owner after hosted testing, 2026-10-30 | Corrective (coverage defect) + perfective | Implemented, applied to the hosted database, tested and deployed |

**Lecturer change request**

The lecturer-provided change request (CR-005) had not been received when this report was prepared. When it arrives it will be handled with the same process and template: description, affected requirements, impact analysis across requirements, design, database, code, tests and documentation, an estimate, classification and the implementation and test evidence.

## 12.2 Main change request CR-006: every complaint must reach a responsible Officer

**Table 12.2:** CR-006 change request

| Field | Entry |
|---|---|
| Requested by / date | Project owner after hosted testing, 30 October 2026 |
| Description of the change | Complaints were routed to a department by category and only that department's Officers could act. Departments without Officers silently collected complaints: the hosted data showed 4 of 5 open complaints stuck this way, and the Administrator could see them but had no permission to act. Every complaint must reach a responsible Officer. |
| Classification | Corrective for the coverage defect (the system failed its own objective O3, "enable officers to manage … complaints"); perfective for the service targets, analytics and dispatch tools added with it |
| Affected requirements | CF-09, CF-13, CF-21; new CF-23 (automatic assignment), CF-24 (Administrator dispatch), CF-25 (service-level targets); NFR-03 |
| Affected stories and use cases | New US-22, US-23, US-24; new UC-15 Dispatch complaint and UC-16 Auto-assign Officer; UC-03 changed |

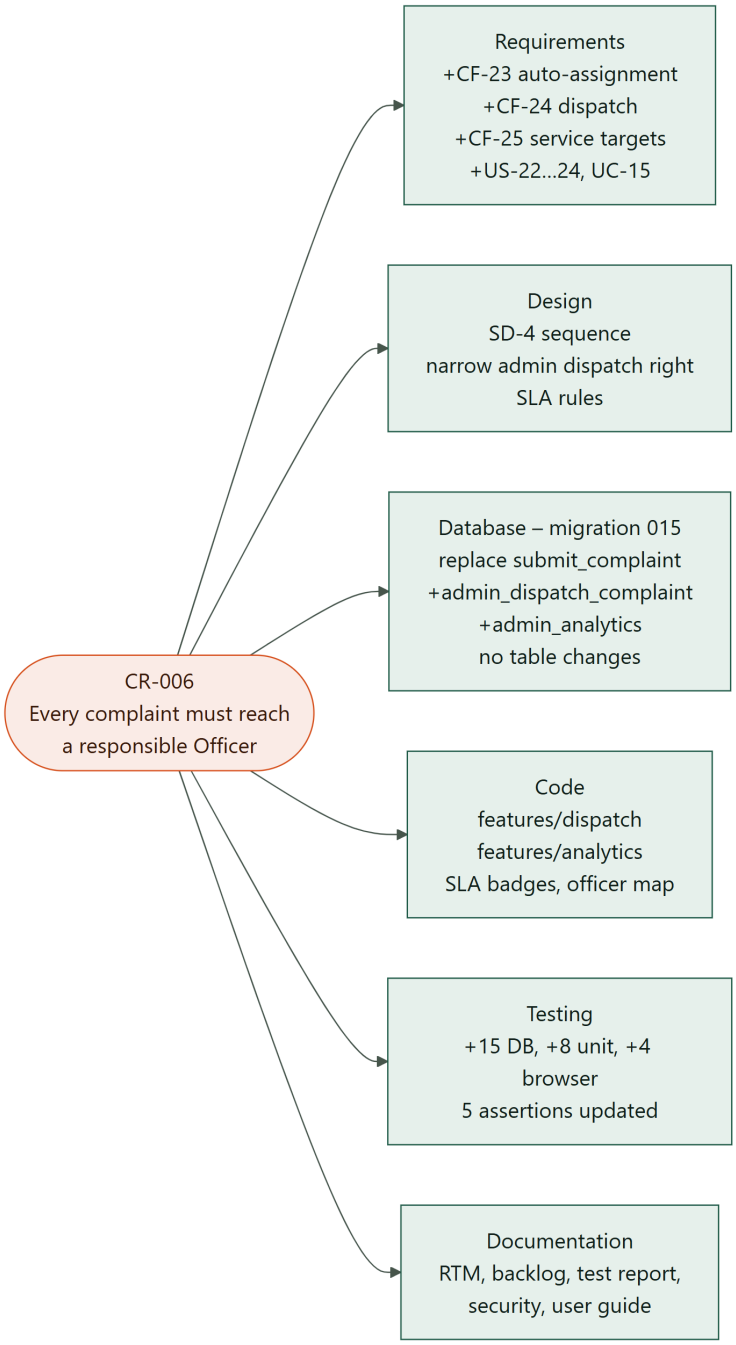



**Figure 12.2:** CR-006 impact across SDLC artefacts

**Table 12.3:** CR-006 impact analysis and estimate

| Area | Affected items | Change made | Effort (pts) |
|---|---|---|---|
| Requirements | Backlog, FR list, user stories, use cases, RTM | 3 new functional requirements, 3 user stories, UC-15/16; RTM rows added | 2 |
| Design | Architecture, sequence diagrams, UI | SD-4 sequence; Administrator gets a narrow, audited dispatch right (no processing rights); SLA rules; dispatch centre and analytics screens | 3 |

| Area | Affected items | Change made | Effort (pts) |
|---|---|---|---|
| Database | submit_complaint, record_officer_event | Migration 015 (new, never editing applied ones): workload-balanced assignment, admin_dispatch_complaint, admin_analytics, community_impact; no table changes and no data changed at install | 6 |
| Code | complaints, officer, admin | New dispatch and analytics features, SLA badges, Officer map | 6 |
| Testing | Database, unit, browser suites | +15 database, +8 unit, +4 browser tests; 5 assertions updated because the assigned Officer is now also notified | 3 |
| Documentation & security | Test report, security, user guide | Threat T11 and requirement SR-12 added; user guide and deployment record updated | 1 |
| Total estimate | | About one sprint for a pair of developers | 21 |

### 12.2.1 Modification plan and verification

1. Create branch sprint-5/dispatch-and-experience.
2. Write migration 015 and its 15 database tests first (test-first for the security-sensitive SQL).
3. Implement the dispatch and analytics features, SLA badges and Officer notifications.
4. Run the full regression suite: 77 unit, 138 database and 58 browser tests passed at that point, with CI green.
5. Apply migration 015 to the hosted database after explicit approval. All existing data checksums were identical before and after, anonymous calls to the new functions were refused (HTTP 401), and the live analytics confirmed the three unstaffed departments.
6. The four stuck complaints were not moved automatically: the Administrator dispatches them with a reason, so each citizen sees why their complaint moved.

### 12.2.2 Other change requests handled

- CR-001 Citizen withdrawal (perfective): a citizen can withdraw an unresolved complaint with a recorded reason; records, evidence and history are kept.
- CR-002 Non-retrying conflict code (corrective): the stale-version error was changed from SQLSTATE 40001 to PT409 (HTTP 409) after testing showed PostgREST retried 40001 indefinitely.
- CR-003 Administrator-created Officer accounts (perfective and preventive): Officers can no longer self-register; an audited invitation flow was added.
- CR-004 Location usability (perfective): an optional "Use my current location" button and a closer default map view.
- Responsive-layout maintenance (corrective, DEF-16–19): a slide-out menu for tablets and phones, an aligned feedback form and stacked admin tables.

# 13 Security

CivicFlow stores personal data (names, e-mail addresses, complaint locations and descriptions, and evidence files) and has three privilege levels, so we planned security from Sprint 1 instead of adding it at the end.

**Table 13.1:** Assets and their sensitivity

| Asset | Sensitivity |
|---|---|
| Citizen identity (name, email) and complaint location/description | Personal data: high |
| Evidence files (photos, PDFs) | Personal data, may show people or homes: high |
| Workflow integrity (status, assignment, resolution, history) | Accountability: high |
| Administrator/Officer privileges | Privilege: critical |
| Service configuration (categories, departments) | Medium |

## 13.1 Threats and risk assessment

11 threats were identified using the STRIDE categories. Likelihood (L) and impact (I) are rated from 1 (low) to 3 (high); risk = L × I. The residual risk is the rating after the implemented controls.

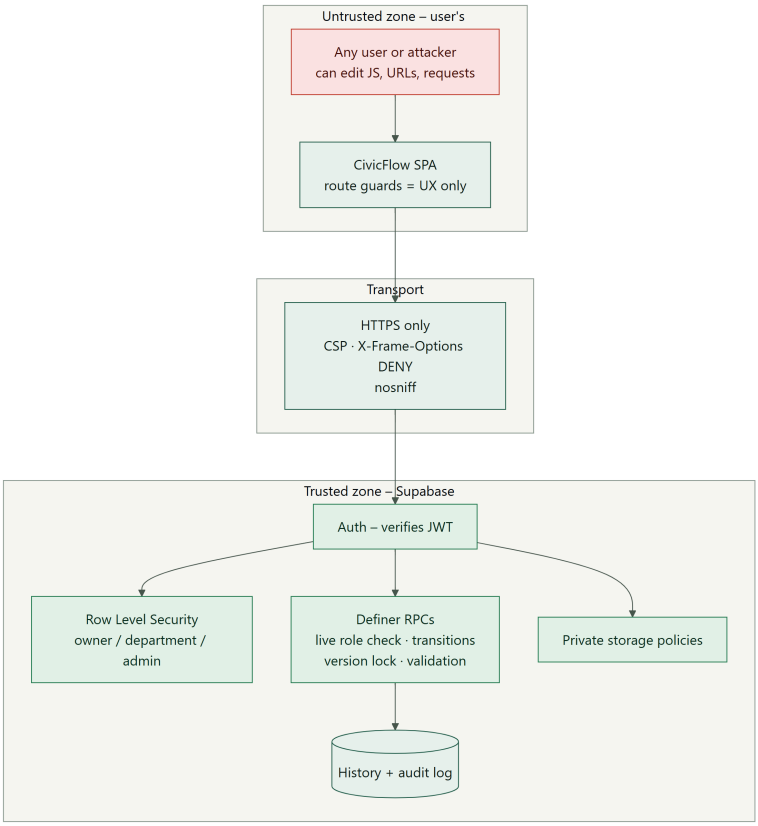



**Figure 13.1:** Trust boundary: every decision is made on the server side

**Table 13.2:** Threats and risk assessment

| ID | Threat (STRIDE) | Scenario | L | I | Risk | Controls | Residual |
|---|---|---|---|---|---|---|---|
| T1 | Elevation of privilege | A Citizen sends role: administrator in signup metadata or runs UPDATE profiles SET role=… from the browser console | 3 | 3 | 9 | Signup trigger hard-codes Citizen; no browser UPDATE grant; role changes only through the audited admin RPC, which checks the live Administrator | 1 |
| T2 | Information disclosure (IDOR) | A Citizen changes the complaint ID in the URL or REST query to read another person's complaint, history, evidence or feedback | 3 | 3 | 9 | RLS on every table via can_read_complaint (owner / same-department Officer / Admin); child tables inherit; Storage policies check the parent | 1 |
| T3 | Tampering (workflow bypass) | An Officer or Citizen calls PostgREST directly to set status='resolved', skip steps or edit another department's complaint | 2 | 3 | 6 | Direct DML revoked; narrow RPCs with transition rules, department/assignee checks, row locks and expected version | 1 |
| T4 | Spoofing / stale privilege | A deactivated or demoted user keeps using an unexpired JWT | 2 | 3 | 6 | Every policy and RPC re-reads the live profile (is_active, role, department); no authorisation from JWT claims | 1 |
| T5 | Tampering / XSS | A malicious description, remark, filename or comment contains <script> or onerror to steal sessions | 2 | 3 | 6 | React text rendering only; no dangerouslySetInnerHTML; no string-built map popups; CSP script-src 'self'; sessions kept in memory | 1 |
| T6 | Malicious upload / storage abuse | Upload of HTML/SVG/executables, huge files, overwriting someone else's file, or uploading to another path | 2 | 2 | 4 | Private bucket; MIME allow-list (JPEG/PNG/PDF) and 5 MiB limit in the bucket and RPC; server-reserved UUID paths; no UPDATE/DELETE policy; 5-file quota; stored metadata verified at finalize | 2 (content isn't malware-scanned) |
| T7 | Weak authentication | Short or reused passwords; account enumeration via recovery | 2 | 2 | 4 | 12-character minimum; email verification; generic recovery response; Supabase Auth rate limits | 2 |
| T8 | Secret leakage | A service-role key or DB password bundled into the frontend or committed to Git | 1 | 3 | 3 | Only VITE_ public URL and publishable key; the env validator rejects sb_secret_ keys; .env* git-ignored; the service key only inside the Edge Function runtime | 1 |
| T9 | Denial of service / spam | Mass complaint submission or repeated requests | 2 | 2 | 4 | Auth required for all writes; bounded inputs and pages; idempotent submission; platform rate limits | 3 (no app-level rate limit) |
| T11 | Tampering (CSV injection) / disclosure via analytics | A category name like =HYPERLINK(...) runs when an exported CSV is opened; non-admins call the analytics RPC | 1 | 2 | 2 | CSV cells are quoted and formula prefixes neutralised; admin_analytics checks a live active Administrator; community_impact returns aggregates only | 1 |
| T10 | Repudiation | An Officer or Admin denies making a change | 2 | 2 | 4 | Append-only complaint_events with actor and time; private admin_audit_events with before/after values | 1 |

## 13.2 Security requirements

**Table 13.3:** Security requirements

| ID | Requirement | Threats | Verified by |
|---|---|---|---|
| SR-01 | Every protected read and write requires an authenticated and active profile. | T4 | TC-DB-FND-03, TC-DB-ADM-04, TC-DB-NTF-04 |
| SR-02 | Roles can only be changed by an active Administrator through an audited operation. Signup always creates a Citizen. | T1 | TC-DB-FND-01, TC-DB-FND-04, TC-DB-ADM-01, TC-INT-02 |
| SR-03 | Citizens can read only their own complaints and children. Officers can read only their department's. | T2 | TC-DB-SUB-06, TC-TRK-03, TC-DB-FB-05, TC-INT-01 |
| SR-04 | Browsers have no direct INSERT/UPDATE/DELETE on application tables. Business changes go only through validated RPCs. | T3 | TC-DB-OFF-03, TC-DB-FB-05, TC-DB-CAT-06 |
| SR-05 | Workflow changes enforce valid transitions, assignee rules and optimistic locking. | T3 | TC-DB-OFF-02, TC-DB-OFF-05, TC-INT-04 |
| SR-06 | User-supplied text is never interpreted as HTML. | T5 | TC-TRK-03, TC-NTF-01 |
| SR-07 | Evidence is private, type- and size-limited, stored at server-chosen paths and never overwritten. | T6 | TC-DB-EV-01…06 |
| SR-08 | Passwords are at least 12 characters, and email ownership is verified. | T7 | TC-U-AUTH-01, Auth configuration |
| SR-09 | No secret credential is present in source, the build or browser configuration. | T8 | TC-U-AUTH-01 ("rejects a secret key …"), .gitignore |
| SR-10 | All administrative and workflow changes are attributable. | T10 | TC-DB-ADM-02, TC-DB-CAT-01, TC-DB-OFF-01 |
| SR-11 | The last active Administrator cannot be removed, even concurrently. | T1 | TC-DB-ADM-03, TC-INT-08 |
| SR-12 | Dispatch is limited to active Administrators, is version-checked, audited and notified; analytics are Administrator-only; aggregates never identify citizens. | T1, T3, T11 | TC-DB-DSP-04…12, TC-U-ANL-01 |

## 13.3 Implemented security controls

The guidelines ask for at least three implemented controls. We implemented 8, and each one can be demonstrated.

**Table 13.4:** Implemented controls and how to demonstrate them

| # | Control | Where | How to demonstrate |
|---|---|---|---|
| C1 | Row Level Security (role, ownership and department authorisation) | supabase/migrations/…0200, …0400, …200100 (can_read_complaint, read policies) | Sign in as Citizen B and open Citizen A's complaint URL: "Complaint unavailable". Run npm run test:db and show "isolates Citizens and departments…" passing. |

| # | Control | Where | How to demonstrate |
|---|---|---|---|
| C2 | Protected workflow RPCs (no direct DML; transitions, assignee, version) | … 20261020000100_officer_processing .sql, …0200_officer_resolution.sql | Send PATCH /rest/v1/complaints?id=eq.<id> with a signed-in Officer's token (DevTools → Network → copy as cURL, change it to PATCH {"status":"resolved"}): 403 permission denied (also asserted by the HTTP integration test). A stale tab shows "This complaint changed while you were viewing it". |
| C3 | Private evidence storage | … 0500_private_complaint_evidence.s ql | A public object URL (/storage/v1/object/public/complaint-evidence/…) is refused because the bucket isn't public. The download works only when signed in as an authorised user. |
| C4 | Fail-closed live authorisation | private.active_role, checks in every RPC | Deactivate an Officer in User management: their next action is refused without signing them out. |
| C5 | Input validation in two layers | Zod model/*.ts + SQL CHECK/RPC | Submit a 5-character description: client error. The same request sent directly gets invalid_input. |
| C6 | XSS defences | React text rendering, CSP in vercel.json | A complaint containing <img src=x onerror=alert(1)> displays as text (TC-TRK-03, TC-NTF-01). |
| C7 | Secure session handling | shared/lib/supabase/client.ts (persistSession:false), AuthProvider | Reloading the page signs you out; no token is stored in localStorage (TC-AUTH-06). |
| C8 | Audit trails | complaint_events, private.admin_audit_events | Every step appears in the complaint history. Category changes write audit rows (TC-DB-CAT-01). |

## 13.4 Security decisions

- Authorisation happens in the database. The user controls the browser, so a check that exists only in React can be bypassed. All checks are in PostgreSQL and the interface only reflects them.
- SECURITY DEFINER functions bypass Row Level Security, so we hardened them: an empty search_path, fully qualified names, no dynamic SQL, EXECUTE granted only to signed-in users, and an explicit check of the caller's current role.
- The privileged service key is used in exactly one place, the invitation Edge Function, and only after it has verified that the caller is a live Administrator.
- Sessions are kept in memory only. Users must sign in again after refreshing the page, but their tokens are less exposed on shared computers.
- Accepted residual risks for the prototype: no malware scanning of uploads, no application-level rate limiting beyond the platform's, and the default Supabase e-mail sender. Enabling Supabase leaked-password protection is recommended before public use.

# 14 Project Management and Individual Contributions

## 14.1 Team coordination

- Roles: each member owned one module end to end (Section 6.5) and reviewed another member's work at the end of each sprint.
- Ceremonies: sprint planning at the start of each sprint, short progress check-ins, and a sprint review and retrospective whose outcomes were recorded in the sprint documents (docs/sprint-*.md).
- Integration: modules integrate only through shared database contracts and small public feature interfaces, so members could work in parallel without breaking each other's code; CI protected main.
- Change control: database migrations and hosted rollouts required explicit approval; scope changes were recorded as change requests.

## 14.2 Individual contributions

As Section 13 of the guidelines requires, every member worked on requirements, design, implementation and testing, and not only on documentation or the presentation.

**Table 14.1:** Contribution matrix by SDLC activity

| Member | Requirements | Design | Implementation | Testing |
|---|---|---|---|---|
| TG/2023/1761 M.S.M. Sharfan | Admin, data and coverage requirements (CF-14, 18, 20, 21, 23, 24); NFR-05/06 | ER and database design, access matrix, RLS policies, architecture, SD-4 | Migrations and protected functions; user and category management; admin dashboard and analytics; dispatch centre (CR-006); hosted database operations | Database integration suites (foundation, dispatch, sprint 4, demo data) |
| TG/2023/1731 H.M. Amadi Nethma | Stakeholder analysis; citizen FRs; user stories and acceptance criteria | Use case diagram; Citizen screens and wireframes | Registration, login, profile; complaint form with map; My complaints; withdrawal; Updates inbox and feedback UI | Authentication and citizen browser tests; validation unit tests |
| TG/2023/1710 M.F. Rishma | Category, priority, evidence and service-target requirements (CF-05, 06, 09, 10, 16, 25) | Class / domain model; SD-1 submit sequence; evidence storage design | Category routing; complaint details; private evidence upload (reserve → upload → finalize); assignment and priority rules; SLA badges | Complaint and evidence database tests; SLA unit tests |
| TG/2023/1757 N.M. Razim Mohamed | Officer use cases UC-05–07 and stories US-08–12 | Complaint state machine; activity diagram; SD-2 resolve sequence; Officer screens | Officer queues and filters; review, assign, priority, start, progress, transfer and resolve; Officer map; project repository set-up | Officer database tests; Officer browser journeys |
| TG/2023/1776 V. Kayaththiry | Non-functional and security requirements (NFR-01–11, SR-01–12) | Test plan; threat model and trust boundary; deployment architecture | Error handling and messages; security headers and environment validation; CI pipeline; Vercel deployment configuration | System and HTTP integration testing; defect log; responsive-layout audit; smoke test; CR impact analysis |

## 14.3 Contribution log

**Table 14.2:** Contribution log (major tasks)

| Sprint | Member | SDLC stage | Task completed | Evidence |
|---|---|---|---|---|
| Planning | All | Definition, Agile, requirements | Project proposal: problem, stakeholders, backlog, user stories, FR/NFR, use cases, RTM | grp15 proposal (Milestones 1–3) |
| 1 | 1761, 1731 | Design / implementation | Database foundation with RLS; registration, login, roles; admin user management | Migrations 001–003; features/auth, features/admin |
| 1 | 1776 | Implementation / testing | CI workflow, environment validation, test harness | .github/workflows/ci.yml; supabase/tests/harness.ts |
| 2 | 1731, 1710 | Implementation / testing | Complaint form with map, categories and routing, idempotent submission, private evidence, withdrawal | Migrations 004–008, 011; features/complaints, features/evidence |
| 3 | 1757 | Design / implementation / testing | Officer queues and the complete processing workflow with stale-write protection | Migrations 009–010; features/officer; officer.test.ts |
| 3 | 1776 | Testing / maintenance | Found the PostgREST retry-loop defect with HTTP integration tests; drove CR-002 | scripts/test-postgres-http.mjs; DEF-01 |
| 4 | 1761, 1731 | Implementation / testing | Feedback, Updates inbox, category management, monitoring | Migrations 012–014; sprint4.test.ts; sprint4.spec.ts |
| 5 | 1761, 1710 | Maintenance / implementation | CR-006 auto-assignment, dispatch centre, service targets, analytics | Migration 015; features/dispatch, features/analytics; dispatch.test.ts |
| 5 | 1757, 1731 | Design / implementation | Redesign with motion and dark mode; Officer map; landing page | src/shared/ui, features/landing, sprint5.spec.ts |
| 5 | 1776, 1757 | Testing / deployment / security | Responsive audit and fixes; security assessment; Vercel production deployment and smoke test | tests/e2e/capture.ts; vercel.json; DEF-16–19 |

## 14.4 Version-control evidence

All work is in the GitHub repository https://github.com/razim2003/civic-flow. The history shows the increments of every sprint, merged through feature branches (for example sprint-4/feedback-inbox-categories, sprint-5/dispatch-and-experience, demo/sri-lanka-data, ui/responsive-layout and release/vercel-production) with CI passing before each merge. A Git identity shows who pushed a commit, not necessarily who designed or wrote each part; where members worked together on one machine, each member's part is recorded in the contribution log above.

# 15 Conclusion

CivicFlow meets the goals we set in the proposal. A Citizen can report a public service problem once and follow it until it is fixed. Officers have a clear workflow with service targets, and the system stops anyone from skipping its steps. Administrators can manage access and services, make sure every complaint has an owner, and see how well each department performs.

The project also showed us how the output of one SDLC stage feeds the next. The stakeholder needs became objectives and prioritised requirements. The requirements became user stories with acceptance criteria and a traceability matrix. The design turned them into an architecture, UML models and a database whose security rules enforce the requirements, and the implementation was checked by 283 automated tests that trace back to them. Problems found during real use, such as complaints stuck in departments without staff and layout faults on tablets, were handled as maintenance through change requests and defect fixes, and every affected document was updated.

## 15.1 Lessons learned

- Keeping authorisation in the database and testing it as real users meant we could show that the security rules work, instead of assuming that they do.
- Short sprints that ended with a demonstration showed problems early: the empty category list in Sprint 2, the retry loop in Sprint 3 and the coverage gap in Sprint 5.
- Testing the hosted system with realistic data found a defect (fixed by CR-006) that unit tests could not find, because it depended on how the departments were staffed.
- Automated checks help with layout too: the layout audit found problems that we had missed when checking by hand.

## 15.2 Future work

- Validate the system with real citizens and Officers, and measure page-load performance on the deployed system (NFR-02).
- Add e-mail or SMS notifications through a custom SMTP or SMS provider, and application-level rate limiting.
- Malware scanning of uploaded evidence and persistent "remember me" sessions after a security review.
- A public, privacy-preserving map of resolved issues and multilingual support (Sinhala and Tamil).

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# 17 Appendices

## 17.1 Appendix A: User guide (summary)

**Table 17.1:** User guide by role

| Role | How to use CivicFlow |
|---|---|
| Citizen | Create an account (name, e-mail, password of at least 12 characters) and confirm the e-mail. Choose Make a complaint, pick a category, describe the issue, mark the location on the map or type coordinates, and submit; attach photos or PDFs if needed. Follow progress in My complaints and the Updates inbox. Withdraw an unresolved complaint with a reason, or rate the service once it is resolved. |
| Officer | Accept the invitation e-mail and set a password. Use Department complaints and Assigned to me with filters. Open a complaint and use Process complaint: begin review, assign, set priority, start work, add progress and resolve. Due and overdue badges show the service target; the department map shows locations. |
| Administrator | Dispatch centre: see departments with no Officers and complaints that need attention, then dispatch with a reason. Analytics: service targets, coverage, workload, satisfaction and the six-month trend, with CSV export. User management: edit access or create Officer accounts. Service categories: add, edit, route and deactivate categories. |
| All users | On phones and tablets the menu is behind the ☰ button. You stay signed in only while the page is open. Use the moon/sun button to switch between light and dark themes. |

## 17.2 Appendix B: Complete test cases

Legend: + positive test, − negative test (misuse or invalid input). The actual results come from the automated runs recorded in the repository test report (docs/sdlc/07-test-report.md), and each case can be re-run with the automated test named in that report.

**Table 17.2:** Authentication and access (CF-01…03, CF-15)

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-AUTH-01 | CF-03 | − | No session; open /admin/users | Redirected to sign-in; no admin navigation | As expected | Pass |
| TC-AUTH-02 | CF-01 | +/− | Submit an empty form, then valid data | Field errors first; signup request carries only display_name (no role); "Check your email" shown | As expected | Pass |
| TC-AUTH-03 | CF-02, 03 | + | Sign in as Citizen, Officer, Administrator; sign out | Each lands in its own workspace; sign-out clears the workspace | As expected (3 cases) | Pass |
| TC-AUTH-04 | CF-03 | − | Citizen types /admin/users | "Administrator access required"; no admin data requested | As expected | Pass |
| TC-AUTH-05 | CF-03 | − | Signed-in account with no profile row | Fails closed with retry/sign-out; no workspace | As expected | Pass |
| TC-AUTH-06 | CF-02 | + | Sign in, reload page | Must sign in again (session in memory only) | As expected | Pass |
| TC-AUTH-07 | CF-02 | − | Password recovery; expired reset link | Generic response (no account enumeration); expired link can't set a password | As expected | Pass |

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-AUTH-08 | CF-15 | + | Change display name | Name updates; role unchanged | As expected | Pass |
| TC-U-AUTH-01 | CF-01, 02 | − | Short passwords, blank names, bad email, secret key as public key, insecure URL | Rejected by client rules | As expected | Pass |
| TC-U-AUTH-02 | CF-02 | − | A profile response arrives after sign-out or account switch | Late data discarded; fails closed on error | As expected | Pass |
| TC-DB-FND-01 | CF-01, 03 | − | Signup with metadata role=administrator | Profile created as active Citizen; metadata ignored | As expected | Pass |
| TC-DB-FND-02 | CF-03, NFR-05 | + | Inspect catalog | RLS enabled on every application table | As expected | Pass |
| TC-DB-FND-03 | NFR-04 | − | anon role reads tables / updates profiles | Permission denied | As expected | Pass |
| TC-DB-FND-04 | CF-03 | − | Citizen runs UPDATE profiles SET role='administrator', deletes, reads audit | Permission denied | As expected | Pass |
| TC-RBAC-01 | CF-03 | − | Citizen opens /admin/categories | "Administrator access required"; no nav link | As expected | Pass |

**Table 17.3:** Complaint submission and tracking (CF-04…07, 16…18)

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-SUB-01 | CF-04, 06, 16, 18 | + | Citizen submits a valid complaint; first upload fails, retry; download; open detail | Reference shown; request has no citizen_id/status; upload retry reuses the same key; private download; history shows "Supporting evidence added" | As expected | Pass |
| TC-SUB-02 | CF-04 | − | Blank coordinates; map tiles unavailable | Validation error; map click selects; numeric entry still works | As expected | Pass |
| TC-SUB-03 | CF-06 | − | First submit response lost (503) | Form kept; retry sends the same request ID; one complaint | As expected | Pass |
| TC-SUB-04 | CF-05 | + | Open category dropdown | Six categories, Other last; real category UUID submitted | As expected | Pass |
| TC-SUB-05 | CF-05 | − | No categories configured | Explained; submission blocked; no fabricated IDs | As expected | Pass |
| TC-LOC-01 | CF-04 | + | Open the form; click "Use my current location" | Map opens over Southern Province with no pin; opt-in fix sets an editable pin and zooms in | As expected | Pass |
| TC-LOC-02 | CF-04 | − | Geolocation denied / unavailable / timeout / unsupported / cancelled / late | Chosen point preserved; submission allowed; late results ignored | As expected (6 cases) | Pass |
| TC-TRK-01 | CF-07, 17, NFR-03 | + | 11 complaints; paginate, search, filter; mobile width | Correct pages and filters; only own complaints; fits 375 px | As expected | Pass |
| TC-TRK-02 | CF-07, 12 | + | Open own resolved complaint | Progress steps, resolution text and date, history | As expected | Pass |
| TC-TRK-03 | CF-07, NFR-11 | − | Open another citizen's reference; description containing HTML | "Complaint unavailable"; HTML shown as text | As expected | Pass |

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-U-CMP-01 | CF-04 | – | Invalid field values; zero coordinates | Invalid values rejected; 0,0 accepted as a value | As expected | Pass |
| TC-U-EV-01 | CF-16 | – | Spoofed MIME, unsupported, empty, oversized files | Rejected client-side (usability, not malware safety) | As expected | Pass |
| TC-DB-SUB-01 | CF-04–06, 18 | + | Citizen calls submit_complaint | Trimmed values, routed department, status Submitted, version 1, event and notification created | As expected | Pass |
| TC-DB-SUB-02 | CF-06 | +/– | Same request ID twice; same ID with changed input | Same row returned; changed input → request_reused | As expected | Pass |
| TC-DB-SUB-03 | CF-04, NFR-07 | – | Description < 10 or > 5,000 characters; invalid/NaN/∞ coordinates | invalid_input | As expected | Pass |
| TC-DB-SUB-04 | CF-03, 04 | – | Officer, Administrator, inactive Citizen, anonymous | not_authorized / permission denied | As expected | Pass |
| TC-DB-SUB-05 | NFR-06 | – | Notification insert forced to fail | Complaint and event rolled back | As expected | Pass |
| TC-DB-SUB-06 | CF-07, 08 | – | Citizen B and a foreign-department Officer read Citizen A's complaint/history | 0 rows | As expected | Pass |
| TC-DB-SUB-07 | CF-05 | – | Inactive category or department | category_unavailable; historic reference still readable | As expected | Pass |
| TC-DB-EV-01 | CF-16, NFR-11 | + | Inspect bucket; reserve | Private bucket, 5 MiB, JPEG/PNG/PDF; server path citizen/complaint/evidence.ext | As expected | Pass |
| TC-DB-EV-02 | CF-16 | – | Foreign Citizen, Officer, Administrator, inactive reserve | Denied | As expected | Pass |
| TC-DB-EV-03 | CF-16 | – | Declared size 0 / > 5 MiB; MIME text/html | Rejected | As expected | Pass |
| TC-DB-EV-04 | CF-16 | – | Storage writes to foreign, unreserved or spoofed-owner paths | Denied by Storage policy | As expected | Pass |
| TC-DB-EV-05 | CF-16 | – | Sixth file | evidence_limit; expired reservations free slots | As expected | Pass |
| TC-DB-EV-06 | CF-16 | – | Upload after review began | Reserve, upload and finalize all refused | As expected | Pass |

**Table 17.4:** Officer processing and resolution (CF-08…12)

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-OFF-01 | CF-09–12 | + | Officer reviews, assigns, prioritises, starts, adds progress, resolves with confirmation | Each step "Complaint updated"; expected versions 1→6 sent; complaint closed with resolution text | As expected | Pass |
| TC-OFF-02 | CF-08, 17 | + | "Assigned to me" with a priority filter | Scoped to the department and the Officer; filter narrows the list | As expected | Pass |
| TC-OFF-03 | NFR-06 | – | Another user changed the complaint (409) | Remark kept; "Refresh and review"; retry after refresh uses the new version | As expected | Pass |
| TC-OFF-04 | CF-09 | + | Transfer to Water supply | Officer returned to queue; old department loses access | As expected | Pass |

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-OFF-05 | CF-11 | – | Complaint assigned to another Officer | Start/progress/resolve not offered | As expected | Pass |
| TC-OFF-06 | CF-22, NFR-03 | – | Withdrawn complaint, mobile | Readable; all processing disabled | As expected | Pass |
| TC-OFF-07 | CF-08 | – | Workflow RPC missing vs. authorisation denied | Missing: read-only notice, activates after setup. Denied: fails closed. | As expected (2 cases) | Pass |
| TC-OFF-08 | CF-10 | + | Set an unchanged priority | "No change was needed"; no new history claimed | As expected | Pass |
| TC-OFF-09 | CF-17 | + | Search and filter, open a detail, go back | Filters preserved; Clear resets all | As expected | Pass |
| TC-U-OFF-01 | CF-09–12 | – | Missing reasons, short remarks, tampered URL filters | Rejected or bounded client-side | As expected | Pass |
| TC-DB-OFF-01 | CF-11, 12, 18 | + | Full lifecycle via RPCs | Versions 1→7; 7 events; Citizen notifications; last is resolution | As expected | Pass |
| TC-DB-OFF-02 | CF-03, 08 | – | Citizen, other Citizen, foreign Officer, inactive Officer, Administrator call 7 RPCs | not_authorized for all | As expected (5×7) | Pass |
| TC-DB-OFF-03 | NFR-05 | – | Anonymous RPC; direct UPDATE complaints; call private helper | Permission denied | As expected | Pass |
| TC-DB-OFF-04 | CF-09 | – | Assign to Citizen/foreign/inactive/Admin/null; roster contents | invalid_assignee; roster shows only active same-department IDs and names | As expected | Pass |
| TC-DB-OFF-05 | CF-11, NFR-06 | – | Skip/repeat transitions; stale version; no-ops | invalid_transition; PT409 stale_version; no-op adds no event | As expected | Pass |
| TC-DB-OFF-06 | CF-11, 12 | – | Empty/long progress; resolution < 10 or > 5,000 | progress_required / resolution_required; version unchanged | As expected | Pass |
| TC-DB-OFF-07 | CF-12, 22 | – | Any workflow RPC on a Resolved or Withdrawn complaint | complaint_closed | As expected | Pass |
| TC-DB-OFF-08 | NFR-06 | – | Notification insert forced to fail during review | Status/version and history unchanged | As expected | Pass |
| TC-DB-OFF-09 | CF-09 | +/– | Transfer with mismatched category, short reason, valid; after start | category_unavailable / reason_required / success with access moved / invalid_transition | As expected | Pass |

**Table 17.5:** Withdrawal (CF-22)

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-WD-01 | CF-22 | + | Owner gives a reason and confirms | Status Withdrawn; reason in history; still searchable | As expected | Pass |
| TC-WD-02 | CF-22 | – | Stale page | Reason kept; refresh required | As expected | Pass |
| TC-WD-03 | CF-22 | – | Response lost | Same intent retried safely | As expected | Pass |
| TC-WD-04 | CF-22 | – | Resolved / withdrawn complaint | Withdraw not offered | As expected | Pass |

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-U-WD-01 | CF-22 | – | No reason / no confirmation / final states | Rejected | As expected | Pass |
| TC-DB-WD-01 | CF-22 | + | Withdraw from each open state | Reason, actor and time recorded; assignment released; event and notification atomic | As expected (3 states) | Pass |
| TC-DB-WD-02 | CF-22 | – | Foreign / non-Citizen / inactive / anonymous | Denied without revealing whether the complaint exists | As expected | Pass |
| TC-DB-WD-03 | CF-22 | – | Short reason; invalid/stale version; resolved | Rejected; records unchanged | As expected | Pass |
| TC-DB-WD-04 | CF-22 | + | Identical retry after a lost response | Returns the existing withdrawal; no duplicate history | As expected | Pass |
| TC-DB-WD-05 | NFR-06 | – | Notification failure | All withdrawal effects rolled back | As expected | Pass |

**Table 17.6:** Sprint 4: feedback, inbox, categories, monitoring (CF-13, 19, 20, 21)

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-FB-01 | CF-19 | +/– | Submit with no rating; then "Good" + padded comment | "Choose a rating" and no request; then one call {rating 4, comment trimmed}; panel becomes read-only | As expected | Pass |
| TC-FB-02 | CF-19 | – | Server answers feedback_exists | "Feedback was already submitted…"; form stays; no false success | As expected | Pass |
| TC-U-FB-01 | CF-19 | – | Ratings 0, 6, 2.5, missing; comment 2,001 characters | Rejected; 1–5 and 2,000 accepted | As expected | Pass |
| TC-DB-FB-01 | CF-19 | +/– | Owner rates a resolved complaint; identical retry; different second attempt | Stored once (trimmed); retry returns it; second → feedback_exists | As expected | Pass |
| TC-DB-FB-02 | CF-19, NFR-07 | – | Ratings 0, 6, null; comment 2,001 characters; whitespace comment | invalid_feedback; whitespace stored as null | As expected | Pass |
| TC-DB-FB-03 | CF-19 | – | Complaint not resolved; withdrawn | feedback_closed | As expected | Pass |
| TC-DB-FB-04 | CF-19, CF-03 | – | Other Citizen, Officer, Administrator, inactive owner | not_authorized | As expected (4) | Pass |
| TC-DB-FB-05 | NFR-05 | – | Anonymous RPC; direct UPDATE/DELETE/INSERT; reads by each role | Denied; readable by owner, department Officer and Admin only | As expected | Pass |
| TC-NTF-01 | CF-13 | + | Two unread, one read update; mark one; filter unread; mark all | Badge "2 unread" → "1 unread" → none; HTML remark shown as text; exact RPC payloads | As expected | Pass |
| TC-U-NTF-01 | CF-13 | +/– | Event label and remark; missing event; unknown kind | Label used; fallback title; invalid kind rejected | As expected | Pass |
| TC-DB-NTF-01 | CF-13, NFR-11 | +/– | Owner and others list notifications; mark one | Owner sees 5 kinds in order; others see 0; one marked; re-marking is a no-op | As expected | Pass |
| TC-DB-NTF-02 | CF-13 | – | Mark another user's notification (by ID and "all") | 0 rows changed; target stays unread | As expected | Pass |
| TC-DB-NTF-03 | CF-13 | + | Mark all | All of the caller's unread rows set read | As expected | Pass |

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-DB-NTF-04 | CF-13 | − | Null/empty/101 IDs; null flag; inactive; anonymous; direct UPDATE/DELETE | invalid_input / not_authorized / permission denied | As expected | Pass |
| TC-DB-NTF-05 | CF-13 | − | Inactive recipient reads | 0 rows | As expected | Pass |
| TC-CAT-01 | CF-20 | +/− | Admin opens categories; creates with no name, then valid; edits an inactive one | Validation error; created and listed; inactive departments offered only for the existing route; exact RPC payloads | As expected | Pass |
| TC-CAT-02 | CF-20 | − | Server answers category_in_use | Explanation shown; input kept | As expected | Pass |
| TC-U-CAT-01 | CF-20 | − | Empty/long name, long description, missing department | Rejected | As expected | Pass |
| TC-DB-CAT-01 | CF-20 | + | Create, no-op, update | Two audit events (created, updated) with before/after; no-op not audited | As expected | Pass |
| TC-DB-CAT-02 | CF-20, NFR-10 | − | Inactive category | Hidden from Citizens; submission → category_unavailable | As expected | Pass |
| TC-DB-CAT-03 | CF-20 | − | Duplicate name (different case); bounds; unknown/inactive department | category_name_taken, invalid_input, department_unavailable | As expected | Pass |
| TC-DB-CAT-04 | CF-20 | − | Change department of a referenced category | category_in_use; rename still allowed | As expected | Pass |
| TC-DB-CAT-05 | CF-03 | − | Citizen, Officer, inactive Admin, anonymous | not_authorized / permission denied | As expected (4) | Pass |
| TC-DB-CAT-06 | NFR-05 | − | Direct UPDATE categories; read audit table | Permission denied | As expected | Pass |
| TC-MON-01 | CF-21 | + | Admin overview with 1 resolved complaint, ratings 5 and 4 | All 1, Open 0, Average 4.5; no processing buttons | As expected | Pass |
| TC-U-MON-01 | CF-21 | + | Counts and ratings | Open excludes final states; average to 1 decimal; none when no feedback | As expected | Pass |
| TC-UX-01 | NFR-03 | + | 375×812 viewport; detail and inbox | No horizontal overflow | As expected | Pass |

**Table 17.7:** Dispatch, coverage and service targets (CR-006: CF-23, 24, 25, 21)

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-DSP-01 | CF-24 | +/− | Admin opens the Dispatch centre with a complaint in an unstaffed department; dispatches with no reason, then with a reason | Coverage card shows "No Officers"; row flagged unstaffed, unassigned and past target; reason error with no request; then exactly one RPC with version 1 and a success message | As expected | Pass |
| TC-DSP-02 | CF-21, 25 | + | Admin opens Analytics; exports CSV | Unstaffed warning; 67% on target; trend chart; CSV downloaded with dated name | As expected | Pass |
| TC-DSP-03 | CF-13, 23, 25 | + | Officer with an auto-assigned complaint | Updates badge "1 unread"; inbox link goes to the Officer detail path; map shows 1 marker; queue shows a "Due in" badge | As expected | Pass |
| TC-UX-02 | NFR-03 | + | Landing page; toggle theme | Hero and CTA shown; theme switches; no horizontal overflow | As expected | Pass |

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-UX-03 | NFR-03, NFR-04 | + | 1280 px, then 768 px and 360 px. Open the menu, press Escape, then open each admin page from the menu | Desktop shows the sidebar with no menu button. Tablet/phone hide the sidebar links behind Open menu (aria-expanded). Focus moves to Close menu and returns to the button on Escape. Choosing a page closes the menu. No page is wider than the screen | As expected | Pass |
| TC-UX-04 | NFR-03 | + | Layout audit of 24 screens × 4 widths | No horizontal scroll and no overflowing content (see 7.2) | As expected | Pass |
| TC-U-SLA-01 | CF-25 | +/− | Targets and states at fixed times | 24/72/168/336 h; on track / due soon / overdue; closed and invalid dates ignored | As expected | Pass |
| TC-U-DSP-01 | CF-24 | +/− | Attention flags; input bounds; messages | Unstaffed/unassigned/overdue flags; reason 10–2,000; refusals explained | As expected | Pass |
| TC-U-ANL-01 | CF-21, SR-12 | +/− | Summary maths; CSV with =HYPERLINK | Totals and rates correct; cells quoted; formula neutralised | As expected | Pass |
| TC-DB-DSP-01 | CF-23 | + | Two Officers; two submissions | One complaint each (balanced); assignee notified; event auto_assigned | As expected | Pass |
| TC-DB-DSP-02 | CF-23 | − | Only Officer inactive | Unassigned; citizen notified only | As expected | Pass |
| TC-DB-DSP-03 | CF-23 | + | Identical retry | Same row, same assignee, version 1 | As expected | Pass |
| TC-DB-DSP-04 | CF-13 | + | Officer assigns a teammate, then themselves | Teammate notified; self-assignment not notified | As expected | Pass |
| TC-DB-DSP-05 | CF-24 | + | Dispatch stuck complaint to Water supply with an Officer | Route, assignee and version 2; transferred event by admin with reason; citizen and Officer notified; Officer can read it | As expected | Pass |
| TC-DB-DSP-06 | CF-24 | + | Same-department reassignment; repeat | assigned event; repeat is a no-op | As expected | Pass |
| TC-DB-DSP-07 | CF-24 | − | Short reason, mismatched route, foreign/citizen assignee, stale/null version, withdrawn | reason_required, category_unavailable, invalid_assignee, PT409, invalid_input, complaint_closed | As expected | Pass |
| TC-DB-DSP-08 | CF-24 | − | In-progress complaint moved or unassigned | invalid_transition; Officer swap within the department allowed | As expected | Pass |
| TC-DB-DSP-09 | CF-24, CF-03 | − | Citizen, Officer, inactive Admin, anonymous | not_authorized / permission denied | As expected (4) | Pass |
| TC-DB-DSP-10 | NFR-06 | − | Notification failure during dispatch | Route and version unchanged | As expected | Pass |
| TC-DB-DSP-11 | CF-21, 25 | +/− | Analytics as admin; as citizen, Officer, inactive admin | Coverage, overdue, workload, 6-month trend; others refused | As expected | Pass |
| TC-DB-DSP-12 | CF-21, NFR-11 | +/− | Community impact as citizen, inactive, anonymous; private helper | Only 6 aggregate keys; inactive/anonymous refused; helper not executable | As expected | Pass |

**Table 17.8:** Sri Lankan demonstration data (supabase/demo)

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-DB-DEMO-01 | NFR-11 | + | Empty database; run the seed | 10 Citizens and 6 Officers who cannot sign in (.test e-mail, no password); Other public services left unstaffed | As expected | Pass |

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-DB-DEMO-02 | CF-04… 12, 19, 22, 23 | + | Seed through the workflow RPCs | 36 complaints: 21 resolved, 4 in progress, 3 under review, 5 submitted, 3 withdrawn; 1 transfer; 33 auto-assigned; 19 ratings; 3 unassigned in the unstaffed department | As expected | Pass |
| TC-DB-DEMO-03 | NFR-06 | + | Backdated history | Events 1..version with no gaps, in time order; first and last events match created/resolved/withdrawn times; every event notifies the citizen; times within 180 days; coordinates inside Sri Lanka; feedback after resolution | As expected | Pass |
| TC-DB-DEMO-04 | CF-21, 25 | + | Analytics after seeding; Officer reviews a seeded complaint | 6 months of submissions, overdue > 0, some late resolutions; seeded complaint still works with the normal RPC | As expected | Pass |
| TC-DB-DEMO-05 | NFR-06 | +/− | Existing Road Officer; re-run; remove while a real complaint is assigned to a demo Officer; remove again | Existing Officer used (no demo Road Officer); re-run changes nothing; removal refused with demo_data_in_use; then removes only demo rows and keeps real ones | As expected | Pass |

**Table 17.9:** Administration (CF-14)

| ID | Req | ± | Preconditions / input | Expected result | Actual | Status |
|---|---|---|---|---|---|---|
| TC-ADM-01 | CF-14 | − | Citizen opens Officer / Officer-provisioning routes | Refused | As expected | Pass |
| TC-ADM-02 | CF-14 | − | Demote the last admin | Backend refusal shown | As expected | Pass |
| TC-ADM-03 | CF-14 | +/− | Invite an Officer; email delivery refused | Department required; no password/role fields; refusal shown without false success or retry | As expected (2) | Pass |
| TC-U-INV-01 | CF-14 | − | Edge Function: no session, non-admin, foreign origin, bad input, inactive department, provider failures | Refused before any invitation; no promotion on partial failure | As expected (9) | Pass |
| TC-DB-ADM-01 | CF-14 | − | Non-admin calls admin_update_user | not_authorized | As expected | Pass |
| TC-DB-ADM-02 | CF-14 | + | Promote to Officer without and with a department | Requires an active department; audit row written | As expected | Pass |
| TC-DB-ADM-03 | CF-14 | − | Demote the last admin | last_administrator; change rolled back | As expected | Pass |
| TC-DB-ADM-04 | NFR-04 | − | Deactivate a user who has a valid claim | Reads and writes refused immediately | As expected | Pass |

**Table 17.10:** HTTP integration with real JWTs (last executed 2026-10-21)

| ID | Scenario (from the Officer verification record) | Result 2026-10-21 |
|---|---|---|
| TC-INT-01 | Owner / department / inactive / role denial through PostgREST with real user JWTs | Pass |
| TC-INT-02 | Signup metadata privilege escalation through Auth | Pass |
| TC-INT-03 | Stale withdrawal returns HTTP 409 immediately (no retry loop) | Pass |
| TC-INT-04 | Concurrent review by two Officers: one succeeds, one gets 409 | Pass |
| TC-INT-05 | Resolution racing withdrawal: exactly one final state | Pass |

| ID | Scenario (from the Officer verification record) | Result 2026-10-21 |
|---|---|---|
| TC-INT-06 | Transaction rollback through HTTP | Pass |
| TC-INT-07 | Transfer removes old-department access | Pass |
| TC-INT-08 | Two Administrators demote each other concurrently: at least one remains | Pass |
| TC-INT-09 | Only the assignee can start (403 for teammate); short resolution rejected (400); submission and owner reads work over HTTP | Pass |

## 17.3 Appendix C: Repository, deployment and demonstration

**Table 17.11:** Key links and commands

| Item | Value |
|---|---|
| Source code | https://github.com/razim2003/civic-flow |
| Deployed application | https://civic-flow-rho.vercel.app |
| SDLC evidence pack | docs/sdlc/ (README, 01–11, user guide, screenshots) |
| Run locally | npm ci, then npm run dev (Node.js ≥ 22.18) |
| Run all checks | npm run lint, npm run typecheck, npm test, npm run test:db, npm run build, npx playwright test |
| Layout audit | LAYOUT_AUDIT=<folder> npx playwright test |
| Demo data | supabase/demo/sri-lanka-demo.sql (remove with remove-sri-lanka-demo.sql) |

**Table 17.12:** Demonstration script for the final evaluation

| Step | What to show |
|---|---|
| 1 | Problem and stakeholders (Chapters 2–3); backlog and one user story with acceptance criteria (Chapter 6) |
| 2 | Scrum process and sprint history; architecture and sequence diagram SD-1 (Chapter 8) |
| 3 | Live demo on the deployed URL: a Citizen submits a complaint, the automatically assigned Officer processes and resolves it, the Citizen sees the update and rates it, and the Administrator views the analytics and dispatches a stuck complaint |
| 4 | Run npm run test:db and show a database test refusing a direct UPDATE (Chapter 10) |
| 5 | Change request CR-006 and its impact analysis (Chapter 12) |
| 6 | Security: open another citizen's complaint URL and show the refusal; show a <script> description rendered as text (Chapter 13) |